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60-1: Bat Diets

One of the most dramatic measures of the diversity of bats is the variety of food they consume. Although some 600 species eat insects as the main dietary staple, others live on fruit, nectar and pollen, fish, frogs, birds, small mammals, blood, and even other bats. Most of this diversity occurs in the tropics, although many bats from temperate regions vary their diets by eating a wide range of insects.

What any bat eats is determined by two important factors: the need for enough energy to keep the body going and the need for essential chemicals to maintain it. Just as an automobile requires fuel to move and lubricants to keep the engine running smoothly, animals require protein, carbohydrates, or fat for energy, along with vitamins and minerals to stay healthy. Since it is essential for bats to consume enough calories to make flight, reproduction, growth, and other bodily functions possible, it is easy to see why such variety is necessary in their diets. Species that feed mainly on plant products require protein, which can be obtained by eating animals (insects), pollen (which provides only moderate amounts of protein), or large quantities of fruit (a very limited source of protein). Bats feeding on insects appear to obtain a balanced diet from this source of nutrition.

To obtain the energy and chemicals they need, bats consume vast quantities of food. Small insectivorous species (species that eat mainly insects) eat at least 30 percent of their body weight each night they are active, and in nursing mothers, the amount may exceed 50 percent. These figures appear to apply to all bats, regardless of diet. It may not be impressive to learn that a little brown bat eats three grams of insects on a summer night, but to find 150 mosquitoes in its stomach certainly is, especially when you realize they are not an entire night's ration and the bat probably caught them in less than fifteen minutes. The most impressive statistic of insect consumption comes from Texas, where it is estimated that a local population of Mexican free-tailed bats eats slightly more than 6,000 tons of insects each summer.

The basic design of a bat imposes certain restrictions on the range of food available to it. Bats determine the direction and distance of objects in their environment by emitting

high-pitched sounds and interpreting their echoes to find their location-the term for this is echolocation. Because they can echolocate, New World leaf-nosed bats that feed mainly on fruit are able to catch insects to supplement their diet with protein. Flying foxes (a species of bat found mainly in Indonesia and Malaysia) and their relatives, however, do not echolocate and must obtain their protein from something other than insects. Recent studies in the Ivory Coast by the Canadian biologist Donald Thomas suggest that several species of the herbivorous bat family pteropodidae get their protein from the fruit that composes the main part of their diet. The levels of protein in the fruit are low, but this is countered by the consumption of large quantities of fruit and by enzymes in the digestive tract that efficiently extract what little protein is available. Size is also a factor in determining the diet of a bat; a 3-gram butterfly bat has fewer prey species from which to choose than a 40-gram, large slit-faced bat. As a rule, smaller bats are almost entirely insectivorous, and larger species include larger prey in their diet, readily switching to small vertebrates such as fish, birds, and frogs. The smallest bats that feed on plant material are nectar and pollen feeders. Larger species more often feed on fruit but may also supplement their diet with nectar and pollen.

The food selected by bats also depends on where they feed. Flying bats chasing flying insects will not catch scorpions that do not fly, but they often snatch spiders ballooning on pieces of web. Bats feeding on stationary or terrestrial prey often catch resting insects that are able to fly. Bats concentrating their feeding activity over water catch more aquatic insects than those feeding high over the forest, and species hunting over water have opportunities to catch fish not available to high-flying species that visit the water only to drink.

60-2: Continuous Script and Oral Culture in Europe

Today people commonly read in silence and in private, but that was not the case in Europe during the periods of ancient Greece (800-146 B.C.E.), ancient Rome (753 B.C.E. -476 C.E.), and the centuries that followed. In ancient Greek times, reading was an oral (spoken) practice--one reflected in writing itself. Greek texts were written in a continuous script (without spaces between words) and with minimal punctuation; this both required and rewarded sounding them aloud. A continuous script could not have developed without the Greek introduction of letters for vowels, which allowed readers to identify syllables and hold them in memory as the eye moved across the text.

Though it seems awkward to us now, the continuous script was not a natural construction, but a choice, as demonstrated by the fact that the Romans discarded their own punctuation in the first century in favor of the Greek model. It established literacy (ability to read and write) as the domain of a cultured elite, who either studied from a young age to master the skills appropriate for reading each individual text or employed a professional reader, or lector, for the task. It also facilitated a culture of shared inquiry, in which challenging texts were read aloud in groups as an incentive for debate. In ancient Greece, literature was primarily a social activity, with audiences gathering for performances of epic poetry and drama. Epic poems bear the hallmarks of this orality: they rely on repetition, formulaic images, meter, and rhyme as mnemonic (memory) aids to the performer. The term used to describe performances of such works, rhapsody, means "to stitch together" -suggesting the extent to which oral composition relies on weaving familiar lines.

The great thinkers of ancient Greece, in fact, mistrusted writing as a technology that would destroy the oral arts of debate and storytelling on which they based their sense of the world, philosophy, and time and space. In Plato's dramatic play *Phaedrus*, the philosopher Socrates looks down on the written word for separating ideas from their source, citing Egyptian king Thamus as the first to voice this concern when he received the gift of writing from the god Thoth. Transcription (writing speech down), Socrates fears, is an aid that will both interfere with memory and trap philosophical thought in ambiguity, leaving interpretation in the hands of the reader. Texts, after all, can circulate without their author, thus preventing one from explaining or defending them. Despite these fears, the very writing Plato used to record his works proved instrumental in the development of ancient Greek oratory (the art of speech-making). As scholar Walter Ong

points out in *Orality and Literacy*, his study of the ways writing technologies restructure consciousness, the written word enabled Greek scholars to transcribe and codify effective rhetorical (speaking) strategies. It also vastly increased human vocabulary, since we no longer had to rely on memory to hold all of the language for immediate use. Writing, in fact, allowed rhetoric to flourish.

For the kind of silent reading we now experience to take hold, reading would have to change its context and text its form. It would have to become a more private experience, which means literacy would have to extend beyond the elite and monastic (religious) communities. Texts, too, would need to become more legible, with standardized punctuation and word spaces so that the mumbling of readers sounding out text, common through the sixth century, could disappear. And libraries designed for quiet, contemplative reading could then develop to serve this new readership.

British scribes, like those who crafted the Book of Kells (around 800 C.E.), played a key role in making the text more accessible. They wrote in Latin, and because it was a second language, and one more challenging to sound out in continuous script, they introduced several changes to improve its legibility, including word separation (around 675 C.E.), additional punctuation, and simplified letterforms.

Still, it took nearly four hundred years for these small innovations to spread. The translation of Arabic scientific writing into Latin in tenth-century Europe likely played a vital role in solidifying word separation, since it was inherent in the language (because, unlike Greek and Latin, it is written in consonants). Translators kept Arabic word separation when rendering these texts in Latin, in part because it made the complex technical prose significantly more comprehensible.



61-1: Tern Hunting of Fish

Terns are a group of bird species that hunt fish close to the water's surface. Fish have adaptations to avoid being eaten, including cryptic coloration (most fish are dark along the back, even if they are silvery on the sides), habitually swimming at depths below those at which they are in danger from above, and taking evasive action if they see a bird diving toward them. Terns have their own cryptic coloration, most having white or light grey underparts so that they are difficult to see from below the surface, and they dive vertically toward the fish at the highest possible speeds.

Terns also have to learn to adjust for parallax: sunlight changes direction at the air-water interface, which makes fish always appear to be at shallower depths than they really are. This is an especially difficult problem when the water surface is wavy, which makes the image of the fish move around horizontally as well as vertically—as one can see by looking down from a bridge or dock into rippled water. Terns must learn by trial and error to compensate for parallax and dive accurately toward a target that appears to be moving unpredictably. It has been shown that terns' success in catching fish declines with increasing wind speed, presumably because increasing waviness of the surface makes the fish increasingly hard to locate. At the other extreme, terns' fishing success declines again at very low wind speeds, either because the terns have more difficulty hovering to locate the fish in still air, or because the fish can see the terns more easily when the surface is smooth. Except in the most favorable circumstances, terns' fishing success is usually quite low—typically only one fish caught for every three or four dives, with many attempts broken off even before the tern hits the water.

At most times and in most places, terns are unable to catch fish because the fish are swimming deeper below the surface than the terns can dive. Terns' ability to catch fish usually depends on factors that bring the fish toward the surface. Many fishes follow vertical movements of their invertebrate prey, or zooplankton (tiny water-based animals),

but these prey habitually come to the surface at night when terns are usually unable to catch fish. Sand eels, one of the main prey species for terns, come to the surface to spawn (lay eggs), and several other fishes spawn in shallow water near the shore. These spawning events provide good opportunities for terns to catch fish, but they are usually localized and transitory. Fish larvae and juvenile fish are more likely to swim near the surface than adult fish, but these prey are often too small to be good food sources for terns.

An important factor that brings larger fish close to the surface is the vertical movements of the water caused by currents running over obstacles or through narrow passages. Terns regularly concentrate over tide "races" such as that at Dungeness, England, where currents running around narrow strips of land generate turbulent eddies (circular currents) that bring fish to the surface, or at gaps in barrier beaches such as those of the north Norfolk coast (also in England), where tidal currents running in and out twice each day produce similar eddies. Other sites suitable for tern feeding are where tidal currents run around islands or over shallow rocks, reefs, or sandbars, bringing fish close to the surface at predictable times of the tidal cycle. Roseate terns, in particular, feed regularly over such places, and in some areas, travel long distances to feed at sites where fish can be caught predictably.

The most widespread factor making fish come toward the surface, however, is predatory fish chasing them from below. In many coastal areas, flocks of terns gather over schools (groups) of predatory fish and follow them as they feed, diving frantically in the short intervals when the prey fish are forced close to the surface by the predators pursuing them from below. Many tropical terns habitually follow tuna, bonito, or other predatory fish that hunt in schools. The sooty tern is thought to be dependent for much of the year on tuna and the dolphins that accompany them, ranging widely over tropical oceans and coming together wherever the tuna come near the surface to feed.

61-2: Chaco Canyon

Archaeologists have discovered that the Anasazi civilization of present-day southwestern United States flourished even after environmental problems had reduced crop production and virtually eliminated timber supplies in Chaco Canyon, the center of the Anasazi population. Despite these problems, or because of the solutions the Anasazi found for them, the canyon's population continued to increase, particularly during a big spurt (sudden and brief increase) of construction that began in A.D. 1029. Such spurts went on, especially during wet decades when more rain meant more food, more people, and more need for buildings. A dense population is attested not only by the famous great houses (such as Pueblo Bonito) spaced about a mile apart on the north side of Chaco Canyon, but also by holes drilled into the northern cliff face to support roof beams, indicating a continuous line of residences at the base of the cliffs between the great houses, and by the remains of hundreds of small settlements on the south side of the canyon. The size of the canyon's total population remains unknown and much debated. Many archaeologists think that it was less than 5,000 and that those enormous buildings had few permanent occupants except priests and were just visited seasonally by peasants at the time of rituals. Other archaeologists note that Pueblo Bonito, which is just one of the large houses at Chaco Canyon, by itself was a building of 600 rooms and that all those post holes suggest dwellings for much of the length of the canyon, thus implying a population much greater than 5,000. Such debates about estimated population sizes arise frequently in archaeology.

Whatever the number, this dense population could no longer support itself but was subsidized by outlying satellite settlements constructed in similar architectural styles and joined to Chaco Canyon by a radiating regional network of hundreds of miles of roads that are still visible today. Those outliers had dams to catch rain, which fell unpredictably and very patchily; a thunderstorm might produce abundant rain in one desert area and no rain in another area just a mile away. The dams meant that when a particular area was fortunate enough to receive a rainstorm, much of the rainwater became stored behind the dam, and people living there could quickly plant crops, irrigate, and grow a huge surplus of food in that area in that year. The surplus could then feed people living at all the other outliers that did not happen to receive rain then.

Chaco Canyon became a black hole into which goods were imported but from which nothing tangible was exported. Into Chaco Canyon came tens of thousands of big trees for construction; pottery (all late-period pottery in Chaco Canyon was imported, probably

because exhaustion of local firewood supplies precluded firing pots within the canyon itself); stone of good quality for making stone tools; turquoise from other areas of New Mexico for making ornaments; and macaws (parrots), shell jewelry, and copper bells from the Hohokam and from Mexico, as luxury goods. Even food had to be imported from places as far as 50 to 60 miles away, as shown by a recent study tracing the origins of corn cobs excavated from Pueblo Bonito.

Chaco society turned into a mini-empire, divided between a well-fed elite living in luxury and a less well-fed peasantry doing the work and raising the food. The road system and the regional extent of standardized architecture testify to the large size of the area over which the economy and culture of Chaco and its outliers were regionally integrated.

Why would outlying settlements have supported the Chaco center, dutifully delivering timber, pottery, stone, turquoise, and food without receiving anything material in return? The answer is probably the same as the reason why outlying areas of Italy and Britain today support cities such as Rome and London, which also produce no timber or food but serve as political and religious centers. Like the modern Italians and British, Chacoans were now irreversibly committed to living in a complex, interdependent society. They could no longer revert to their original condition of self-supporting mobile little groups, because the trees in the canyon were gone, irrigation was impossible, and the growing population had filled up the region and left no unoccupied suitable areas to which to move.

62-1: Energy Distribution in Plants

Annuals are plants that go through their entire life cycle within a single year. Annuals begin their life cycles in the spring when seeds that survived the winter germinate (begin to grow). In regions with distinct dry and wet seasons, germination occurs with the onset of the rainy season. Because it has only one growing season, an annual has to distribute its photosynthates (energy-rich molecules) first to leaves. Leaves, in turn, become involved in photosynthesis, which replenishes the supply of photosynthates and increases overall plant biomass. At the time of flowering, the plant decreases the amount of energy distributed to leaves and diverts most of its photosynthate to reproduction. For example, in the sunflower, the biomass of leaves declines from approximately 60 percent of the total plant weight during the period of growth to 10 to 20 percent by the time the seeds are ripe. When in bloom, the sunflower distributes 90 percent of its photosynthate to the flower head and the remainder to the leaves, stem, and roots.

Perennial plants maintain a vegetative structure over several years. Once established, they distribute their energy in a very different manner from annuals. Before perennials expend any energy on reproduction, they divert photosynthate to the roots. This distribution to roots is in excess of that required for the development of roots for the uptake of nutrients and water from the soil. In some species, such as the skunk cabbage, the roots develop into large storage organs. Energy stored in the roots makes up a reserve upon which the plants draw when they begin growth the following growing season. When they are ready to flower, perennials divert energy from storage to the production of flowers and fruit. As the flowers fade and the fruits ripen, the plant once more sends photosynthate to the roots to build up the reserves it will need for the following spring.

Trees and woody shrubs live a long time, which greatly influences the manner in which they distribute energy. Early in life, leaves make up more than one-half of their biomass; however, as trees age, they accumulate more woody growth. Trunks and stems become thicker and heavier, and the ratio of leaves to woody tissue changes. Eventually, leaves account for only 1 to 5 percent of the total mass of the tree. The production system (the leaf mass) that supplies the energy is considerably less than the rest of the biomass it supports. Thus, as the woody plant grows, much of the energy goes into support and maintenance, which increases as the plant ages.

When deciduous trees (trees that lose their foliage in winter) produce leaves again in the spring, they expend up to one-third of their reserve energy on the growth and expansion of leaves. This expenditure is repaid as the leaves carry out photosynthesis during the spring and summer. After leaves, trees give preference to flowers; then tissues that transport nutrients and water, new leaf buds, deposits of starch in roots and bark, and finally, new flower buds.

Evergreen trees have a somewhat different approach. Many have fine, sharp-pointed leaves called "needles." Because the photosynthetic tissues in these needles can function year-round when temperature and moisture conditions permit, they do not need to draw on root reserves for new growth in the spring. They can afford to wait until later in the growing season to produce new growth. Then evergreens can draw upon energy built up earlier in the spring. For the same reason, new growth develops rapidly and matures within a few weeks.

Reproduction and vegetative growth compete for energy allotments. If photosynthesis is limited, vegetative growth gets the first claim. Because the energy that reproduction demands is high-up to 15 percent in pines, 20 percent in deciduous trees, and 35 percent or more in fruit trees-trees can afford an abundance of fruit only periodically, once every two to three years in deciduous trees and two to six years in evergreens.

The proportionate distribution of net production to above-ground and below-ground biomass tells much about different ecosystems. Low light conditions favor the distribution of energy to the production of leaves and stems at the expense of roots. A reduction in water or nutrient availability favors the distribution of energy to the roots.

62-2: The Early American Economy

In the sixteenth century, settlers from England, a part of Britain, began to colonize North America. These colonies remained under British control until the American Revolution, which occurred at the end of the eighteenth century. The economic theory of mercantilism that was dominant from the sixteenth to the eighteenth century, held that colonies should benefit the mother country by helping her become self-sufficient and wealthy. Beginning in the 1650s, the British Parliament passed a series of laws relating to the American colonies that were designed to advance those goals. Colonists were required to buy goods only from English merchants using English ships operated by English crews. The laws stated that certain products, such as tobacco, rice, and indigo, could be sold only in England and nowhere else. On the other hand, colonial agricultural products that competed with those produced by English farmers, such as wheat or meat, could not be sold in England.

The mercantile system was designed to profit the colonizers-not the colonies-and Americans grumbled about some aspects of it and evaded some of its regulations. But on balance, the colonies probably benefited from it. For one thing, it gave colonial producers of protected goods a monopoly in the British market, and English economic energy and strength maintained a vigorous resale trade in American commodities on the European continent. Also, the producers of some commodities received the inducement of a cash subsidy, called a bounty, that profited them immensely. Finally, all American trade benefited from a strong British commercial and financial infrastructure and from the physical protection provided by the powerful British navy, which controlled the North Atlantic. The British connection was especially helpful in the second third of the eighteenth century, when English and European prosperity and population growth led to increased demand for American products. The British even lowered their barriers to American foodstuffs, creating a new market for colonial farmers.

This prosperity had a generally positive, if uneven, effect on American standards of living. As productive people enjoying extensive use of a rich resource base and taking advantage of growing market opportunities, the colonists probably lived as well as any people anywhere. For the colonies as a whole, per capita wealth on the eve of the Revolution was only slightly lower than in England, and in some places—Pennsylvania, for example—was actually higher.

As a consequence, Americans were progressively better fed, clothed, and housed. They drank coffee, tea, and rum, flavored their food with sugar and spices, and wore clothes constructed from imported textiles. They enjoyed luxuries their parents and grandparents had been unable to acquire, such as glass windows, china, silver candlesticks, and lace. Especially attractive to colonial consumers were luxury foods for the table, an indication that they were entertaining guests and perhaps that women were playing an important role in purchasing decisions.

Prosperity fueled expectations among colonists and prospective colonists. Immigrants poured in to share the American bounty, not only from England but from Ireland, Scotland, and Germany. Farmers bought and cleared more land and hired labor. Planters went into debt for luxuries, assuming that the upward economic trend minimized their risk.

The problem with discussing broad trends in economies or the economic behavior of people in the aggregate is that it leads us to lose sight of important qualifications and distinctions. The first and perhaps the most important qualification that must be made involves farmers' behavior and intent. Virtually all colonial farmers-including even the most substantial planters-practiced a safety-first agriculture that was modified only slowly and incompletely by growing commercial opportunities. The essential goal of most farmers was not to fill a market demand but to provide sufficiency for their families and dependents. It was their first priority to feed their families from their own flocks and fields, clothe their families with thread spun and cloth woven at home, and warm their families from their own woodlots. Self-sufficiency is not the same as subsistence, however, and most farmers produced some sort of surplus for the market most of the time. Market involvement was necessary because few farmers could produce enough of everything their families consumed. Hence, one might sell surplus wheat to purchase another's surplus apples or to buy a cooking pot. Moreover, farmers needed a surplus to generate profits to pay taxes, and they wanted a surplus in order to buy those luxuries that would make their lives more attractive.



63-1: The Mycenaean Collapse

The Mycenaean culture of ancient Greece, with major centers at Mycenae and Pylos on the Greek mainland and Knossos on the island of Crete in the Aegean Sea, was at its height from 1400 B.C.E. to 1200 B.C.E., with growing cities, thriving trade, and a prosperous economy. Around 1200 B.C.E., however, the Mycenaean world showed signs of great trouble, and by 1100 B.C.E., it was gone. Its palaces were destroyed, many of its cities were abandoned, and its art, patterns of life, and system of writing were buried and forgotten.

What happened? Some recent scholars, noting evidence that the Aegean island of Thira (modern Santorini) suffered a massive volcanic explosion in the middle to late second millennium B.C.E., have suggested that this natural disaster was responsible. According to one version of the theory, the explosion occurred around 1400 B.C.E., blackening and poisoning the air for many miles around and sending a monstrous tidal wave that destroyed the great palace at Knossos and, with it, Minoan culture. According to another version, the explosion took place about 1200 B.C.E., destroying the Bronze Age culture throughout the Aegean Sea region. This second version conveniently accounts for the end of both the Mycenaean and Minoan civilizations in a single blow, but the evidence does not support it. The Mycenaean towns were not destroyed all at once; many fell around 1200 B.C.E., but some flourished for another century, and the Athens of the period was never destroyed or abandoned. No theory of natural disaster can account for this pattern, leaving us to seek less dramatic explanations for the end of Mycenaean civilization.

Some scholars have suggested that sea raiders destroyed Pylos and, perhaps, other sites on the mainland. The Greeks themselves believed in a legend that told of the Dorians, a rude people from the north who spoke a Greek dialect different from that of the Mycenaean peoples. According to the legend, the Dorians joined with one of the Greek tribes, the Heraclidae, in an attack on the southern Greek peninsula of Peloponnese, which was repulsed. One hundred years later, they returned and gained full control. Recent historians have identified this legend of the return of the Heraclidae with a Dorian

invasion.

Archaeology has not provided material evidence of whether there was a single Dorian invasion or a series of them, and it is impossible to say yet with any certainty what happened at the end of the Bronze Age in the Aegean Sea region. The chances are good, however, that Mycenaean civilization ended gradually over the century between 1200 B.C.E. and 1100 B.C.E. Its end may have been the result of internal conflicts among the Mycenaean kings combined with continuous pressure from outsiders, who raided, infiltrated, and eventually dominated Greece and its neighboring islands. There is reason to believe that Mycenaean society suffered internal weaknesses because of its organization around the centralized control of military force and agricultural production. This rigid organization may have deprived it of flexibility and vitality, leaving it vulnerable to outside challengers.

The immediate effects of the Dorian invasion were disastrous for the inhabitants of the Mycenaean world. The palaces and the kings and bureaucrats who managed them were destroyed. The wealth and organization that had supported the artists and merchants were, likewise, swept away by a barbarous people who did not have the knowledge or social organization to maintain them. Many villages were abandoned and never resettled. Some of their inhabitants probably turned to a nomadic life, and many perished.

Another result of the invasion was the spread of Greek people eastward from the mainland to the Aegean islands and the coast of Asia Minor. The Dorians themselves, after occupying most of the Peloponnese, swept across the Aegean Sea region to occupy the southern islands and the southern part of the Anatolian coast. These migrations made the Aegean Sea a Greek lake. Trade with the old civilizations of the Near East, however, was virtually ended by the fall of the Minoan and Mycenaean civilizations, nor was there much internal trade among the different parts of Greece. The Greeks were forced to turn inward, and each community was left largely to maintain itself.

63-2: Examining the Diets of Prehistoric People

One of the most important pieces of information about prehistoric people is the nature of their diet. This information is especially relevant when trying to answer questions related to the origins and development of agriculture. Archaeologists can approach diet in a number of ways. They can study diet indirectly by figuring out what people might have eaten based on what was available in their natural environment. For example, deer, duck, turkey, and fish are known to have been available in what is now the northeastern United States for about 7,000 years. Hunting-and-gathering people in this area are likely to have used such resources at one time or another. The weakness of this method of reasoning is that we cannot know with certainty which of the available foods were most important in the diet.

On the other hand, archaeologists can approach the question of diet more directly if there has been good preservation. In many instances, food remains themselves are still present in archaeological sites. Archaeologists can study the fireplaces and garbage heaps of the people who lived at a site and recover food material if it has been preserved. Such remains as bone, seeds, and nuts are often fragile and fragmentary, however, making it difficult to get them out of the ground and back to the lab for identification and analysis. In many cases, an archaeologist takes the entire feature, including all soil, back to the laboratory, instead of attempting to separate the dry soil matrix from the fragile archaeological remains in the field. In the lab, through a number of different procedures collectively called flotation, the archaeologist uses liquid to do the job of separation, taking advantage of the fact that soil and rock will not float in some liquids, whereas organic remains will.

The next task in the reconstruction of a prehistoric diet is the identification of the species of plant or animal represented by the remains. This task can be difficult because of the fragmentary nature of such remains and the changes in them as they decay. In some cases, no precise identification can be made-the piece of bone is too small or the seed too broken up to tell what it is with any degree of confidence. But by comparing the fragment to items in a "library" of bones, nuts, and seeds, archaeologists can often identify many of the dietary remains found at a site.

The examination of the animal remains found at a site is called faunal analysis. Here, the species represented, their sex, ages at death, health, and physical characteristics are identified. Knowing the species of the remains as well as their age, sex, and health status,

can provide insight into the hunting practices of prehistoric people. Were the people hunting large numbers of herd animals in group hunts, or were their prey solitary creatures that could be hunted by individual hunters? Were the ancient people able to kill large animals in their prime or only those that were very young, very old, or sick? Also, because many animals give birth to their young seasonally, knowing the age at death of a young animal helps determine the season of the hunt. For example, the North American white-tailed deer usually gives birth to its young in May or June. If the bones of a deer found at an archaeological site indicate an age of nine months, the animal must have been killed, and the site was most probably occupied, in February or March.

Information about diet can sometimes be gathered from an animal's teeth and bones. For example, in mammals, teeth provide clues to overall food sources. Carnivores such as dogs and cats, for example, have slicing teeth adapted to meat eating. Humans have a more generalized dentition, built for chewing a variety of foods. Certain wear patterns on the teeth, examined microscopically, can reveal whether the diet was made up of soft foods such as fruits or more abrasive foods such as roots and tubers. Prehistoric bones can be analyzed to determine what proportion of their chemical composition is strontium and calcium; this information can be used to determine whether plants or meat made up the bulk of the diet of certain populations.

64-1: The Development of Factories

One of the most enduring images of the eighteenth and nineteenth-century British Industrial Revolution is that of the large factory, filled with workers laboring amid massive machinery driven by either water or steam power. Mechanized factory production evolved out of forms of industry that had emerged only during the early modern period (1500-1750). In the Middle Ages, virtually all industry in Europe was undertaken by skilled craftsmen belonging to urban guilds, and these artisans, working either by themselves or with the assistance of apprentices or journeymen, produced everything from candlesticks and hats to oxcarts and beds. During the early modern period, the urban craftsman's shop gave way to two different types of industrial workplaces, the rural cottage and the large handicraft workshop. Both of these served as halfway houses, or transitional stages, to the large factory.

Beginning in the sixteenth century, entrepreneurs began employing families in the countryside to spin and weave cloth and make nails and cutlery. By locating industry in the countryside, the entrepreneurs were able to escape the regulations imposed by the guilds regarding employment and the price of finished products. They also paid lower wages, because the rural workers, who also received an income from farming, were willing to work for less than the residents of towns. Another attraction of rural industry was that all the members of the family, including children, participated in the process. In this "domestic system," a capitalist entrepreneur provided the workers with the raw materials and sometimes the tools they needed. He later paid them a fixed rate for each finished product. The entrepreneur was also responsible for having the finished cloth dyed and for marketing the commodities in regional towns.

The rural household industry was widespread not only in certain regions of Britain but also in most European countries. In the late eighteenth century, it gradually gave way to the factory system. The great attraction of factory production was mechanization, which became cost-efficient only when it was introduced in a central industrial workplace. In factories, moreover, the entrepreneur could reduce the cost of labor and transportation, exercise tighter control over the quality of goods, and increase productivity by concentrating workers in one location. Temporary labor shortages sometimes made the transition from rural industry to factory production imperative.

The second type of industrial workplace that emerged during the early modern period was the large handicraft workshop. Usually located in the towns and cities rather than in the countryside, these workshops employed relatively small numbers of people with different skills who worked collectively on the manufacture of a variety of items, such as pottery and munitions. The owner of the workshop supplied the raw materials, paid the workers' wages, and gained a profit from selling the finished products.

The large handicraft workshop made possible a division of labor- the assignment of one stage of production to each worker or group of workers. The effect of the division of labor on productivity was evident even in the manufacture of simple items such as buttons and pins. In *The Wealth of Nations* (1776), the economist Adam Smith (1723-1790) used a pin factory in London to illustrate how the division of labor could increase per capita productivity from no more than twenty pins a day to the astonishing total of 4,800.

Like the cottages engaged in rural industry, the large handicraft workshop eventually gave way to the mechanized factory. The main difference between the workshop and the factory was that the factory did not require a body of skilled workers. When production became mechanized, the worker's job was simply to tend to the machinery. The only skill factory workers needed was manual dexterity to operate the machinery. Only those workers who made industrial machinery remained craftsmen or skilled workers in the traditional sense of the word.

With the advent of mechanization, factory owners gained much tighter control over the entire production process. Indeed, they began to enforce unprecedented discipline among their workers, who had to accommodate themselves to the boredom of repetitive work and a timetable set by the machines. Craftsmen who had been accustomed to working at their own pace now had to adjust to an entirely new and demanding schedule.

64-2: Causes of Glacial Ages

Any theory that attempts to explain the causes of glacial ages must successfully address two basic questions. First, what causes the onset of glacial conditions? For ice sheets to have formed, average temperatures must have been somewhat lower than at present and perhaps substantially lower than throughout much of geologic time. For that reason, a successful explanation would have to account for the gradual cooling that finally leads to glacial conditions. The second question is: What caused the alternation of glacial and interglacial stages that have been documented for the Pleistocene epoch (1.6 million to 10,000 years ago)? Whereas the first question deals with long-term trends in temperature that occur on a scale of millions of years, this second question relates to much shorter-term changes.

Probably the most attractive proposal for explaining why extensive glaciations have occurred only a few times in the geologic past comes from the theory of plate tectonics. Not only does this theory provide geologists with explanations about many previously misunderstood processes and features, it also provides a possible explanation for some previously unexplainable climatic changes, including the onset of glacial conditions. Since glaciers can form only on land, we know that landmasses must exist somewhere in the higher latitudes before an ice age can commence. Many believe that ice ages have only occurred when Earth's shifting crustal plates carried the continents from tropical latitudes to more poleward positions.

Glacial features in present-day Africa, Australia, South America, and India indicate that these regions experienced an ice age near the end of the Paleozoic era, about 250 million years ago. For many years, this puzzled scientists. Was the climate in these relatively tropical latitudes once like it is today in Greenland and Antarctica? Why did glaciers not form in North America and Eurasia? Until the plate tectonics theory was formulated and proven, there was no reasonable explanation. Today, scientists realize that the areas containing these ancient glacial features were joined together as a single supercontinent called Pangaea, that was located at high latitudes far to the south of their present positions. Later, this landmass broke apart, and its pieces, each moving on a different plate, drifted toward their present locations. It is now believed that during the geologic past, plate movements accounted for many dramatic climatic changes as landmasses shifted in relation to one another and moved to different latitudinal positions. Changes in oceanic circulation also must have occurred, altering the transport of heat and moisture

and, consequently, the climate as well. Since the rate of plate movement is very slow, on the order of a few centimeters per year, appreciable changes in the positions of the continents occur only over great spans of geologic time.

Since climatic changes brought about by moving plates are extremely gradual, the plate tectonics theory cannot explain the alternation of glacial and interglacial climates that occurred during the Pleistocene epoch. Therefore, we must look to some other triggering mechanism that may cause climatic change on a scale of thousands rather than millions of years. Today, many scientists believe, or strongly suspect, that the climatic oscillations that characterized the Pleistocene may be linked to variations in Earth's orbit. This hypothesis was first developed and advocated by the Yugoslavian scientist Milutin Milankovitch and is based on the premise that variation in incoming solar radiation is a principal factor in controlling Earth's climate. Milankovitch formulated a comprehensive mathematical model based on the following elements: variations in the shape (eccentricity) of Earth's orbit around the Sun; changes in obliquity, that is, changes in the angle that Earth's axis makes with the plane of Earth's orbit; and the unsteady wobbling of Earth's axis, called precession.

Using these three factors, Milankovitch calculated variations in the seasonal timing of the receipt of solar energy and the corresponding surface temperature of Earth over a long period of time in an attempt to correlate these changes with the climatic fluctuations of the Pleistocene. In explaining climatic changes that result from these three variables, it should be noted that they cause little or no variation in the total amount of solar energy annually reaching the ground. Instead, their impact is felt because they change the degree of contrast between seasons. Somewhat milder winters in the middle to high latitudes mean greater snowfall totals, while cooler summers would bring a reduction in snowmelt.

65-1: Imitation in Child Development

When does imitation become possible? How important is this ability in the learning of infants and children? Andrew Meltzoff and M. Keith Moore argue that even newborns and very young infants imitate a variety of responses, including tongue protrusion, mouth opening, and possibly even facial expressions portraying such emotions as happiness, sadness, and surprise. Although some investigators have been unable to replicate these results, many others, including the authors of one study involving infants from Nepal, report a high degree of imitative competence in newborns.

Even more controversial is what imitative behaviors mean. The influential Swiss psychologist Jean Piaget, for example, claimed infants younger than eight to twelve months could imitate someone else's behavior only when able to see themselves making these responses. Because babies do not ordinarily view their own faces, imitative facial gestures would be impossible, according to Piaget, until after about a year of age, when symbolic capacities (abilities to use a symbol, an object, or a word to stand for something) emerge. From this perspective, then, the facial gestures that infants younger than a year make in response to a model's facial expression are stereotyped, rigid responses set off by or linked to limited forms of stimulation rather than imitations of what the child has seen. For example, perhaps tongue protrusion by a model arouses the infant, which in turn promotes a sucking response that naturally invokes tongue protrusion from the infant. If this is the case, infants could be responding to just a few types of stimuli and producing a kind of reflexive (automatic) motor activity that is not really a form of imitation.

Meltzoff and Moore counter that very young infants imitate a variety of responses, modify their imitations to increasingly match the modeled behavior over time, and exhibit their imitations primarily to other people and not to inanimate objects. These arguments contradict the view that such behaviors are simply a fixed pattern of reflexive actions. They propose instead that infants imitate in order to prolong interaction with others. In fact, babies as young as six weeks will imitate the behaviors of a model up to twenty-four hours later. Such imitation is a way to continue the earlier imitative communication with the model. If this interpretation is correct, imitation has an important social-communicative function and is one of the earliest games babies play with other people.

Between six and twelve months of age, infants display far more frequent and precise imitations, matching a wide range of modeled behaviors. Piaget and others believed that deferred imitation, the ability to imitate well after some activity has been demonstrated, is not possible until about eighteen to twenty-four months of age. Piaget believed deferred imitation, along with pretend play and the emergence of language, marks an important transition from one stage of thinking to the next and provides one of the first major pieces of evidence for symbolic capacities. However, as we have already seen, Meltzoff and Moore claim infants as young as six weeks can reproduce a model's behavior a day after seeing it.

Deferred imitation involving actions associated with objects can also be observed far earlier than Piaget claimed. For example, six-month-olds will remove a mitten from a puppet's hand, shake it, and try to put it back on the puppet after observing this sequence of actions performed by a model twenty-four hours earlier. Moreover, toddlers as young as fourteen months who see a peer pulling, pushing, poking, and inserting toys in the laboratory or at a daycare center will reproduce the behaviors in their own homes as much as two days later when given the same toys. The capacity for deferred imitation, then, appears to exist much sooner than previously assumed. In fact, the results are consistent with research on memory, showing that infants younger than one year can recognize stimuli hours and even days later. The observances of imitation at very young ages are important from a social learning perspective, providing clear and compelling evidence that infants, as well as older children, learn many new behaviors by observing others.

65-2: Birds and Food Shortage

The term "food shortage" has been used for food that is inadequate in quantity, quality, or availability. Whatever the abundance of food in the environment, if it is for some reason unavailable, bird species in either inland or coastal areas may starve. Food sources like intertidal invertebrates retreat deeper in the mud in cold weather, putting them beyond the reach of shorebirds, while many soil-dwelling invertebrates move deeper into the ground during drought. Such behaviors illustrate the general points that only a proportion of potential food items may be accessible to birds at any one time, and that this proportion may change through time in response to other conditions. Secondly, if food is plentiful but of low nutrient content or digestibility, a bird may be unable to process enough each day to maintain its weight. This is particularly true of herbivores (plant eaters) eating fibrous, or coarse vegetation, and food shortage may arise partly from the structure of a bird's inner parts, or gut, which limits the rate at which food can be processed. In general, among birds, species differences in gut structure correspond with dietary differences, and in species whose diet changes seasonally, the gut structure may change accordingly. The need for change in gut structure can make some species vulnerable at times of rapid shift in diet, as when snowfall forces birds suddenly from one food type to another.

Many bird species cushion the effects of food shortage by storage, either of body fat or food itself. In small birds, fat stores are accumulated mainly for use overnight, and few species can last more than a day or two without feeding. Larger birds, with their relatively lower metabolic (energy) demands, can store relatively more energy per unit of body weight. As a result, larger birds can rely on internal reserves for longer than small ones, more than two weeks in large waterfowl. Many birds store some body fat and protein in preparation for breeding, extreme examples being Arctic-nesting geese, in which the females produce and incubate their eggs almost entirely on reserves accumulated in wintering or migration areas. For some weeks, then, they are protected against the need to feed on their breeding grounds. Most migratory birds accumulate body fat for the journey, the amount depending on the duration of the unbroken flight.

Species that store food items outside the body may retrieve them hours, days, or even months later, depending partly on the type of food. Some meat eaters cache (store) prey for consumption later in the day or in the following days, while some seed eaters may store food in autumn to last through the winter or even year-round. Long-term storage is

frequent among various birds, such as tits, nuthatches, and woodpeckers, reaching its extreme in certain jays and nutcrackers. In some such species, stored food can provide more than 80 percent of the winter diet, and can also feed the adults into the next breeding season, influencing the numbers of young raised. In good seed years, huge amounts of seed can be stored. Crucial aspects of hoarding behavior are that individuals should guard their stores or should hide them (to reduce theft) and remember where they are. Some species store items individually, each in a separate place, and in experiments, have shown incredible feats of memory. But none of this behavior alters the dependence of such birds on their food supplies; it simply enables them to use their foods more effectively, taking advantage of temporary gluts (oversupply) to lessen later shortages. It does not prevent them from being limited by shortages in years of poor seed crops.

Yet other birds can cushion themselves against temporary food shortages by becoming torpid, meaning they achieve an inactive state in which body temperature and heart, respiratory, and metabolic rates are greatly depressed, thereby conserving energy. In some small species, this happens to some degree every night, but it is especially marked in some small hummingbirds, such as Anna's hummingbird of western North America. Some other species, such as the common swift, can remain torpid for several days at a time when cold weather renders their insect prey inactive, and at least one species, the common poorwill, can remain torpid for several weeks, in a state akin to hibernation in mammals. These adaptations help the species concerned to last through difficult periods, whether hours, days, or weeks, but they cannot eliminate altogether the risk of starvation.



66-1: Cotton Ginning and Interchangeable Parts: The Legacy of Eli Whitney

Cotton is a soft fiber that grows around the seeds of the cotton plant. It was first cultivated in the Indus Valley about 6,000 years ago and then spread to Egypt, Persia, China, and the Mediterranean. For thousands of years, the cultivation of cotton was very labor-intensive. Many steps were involved in preparing cotton to be used in a garment, and many machines were invented to make cotton more valuable as a general-purpose textile material.

The process of separating cotton fibers from seeds is called ginning, which is depicted as far back as the fifth century in paintings in the Ajanta Caves in India. Egyptian cotton has the longest fibers and is used to make premium fabrics. In the American South, the high-quality long-staple cotton can be grown only in a narrow band along the Carolina and Georgia coast and is called Sea Island cotton. The interior areas of the American South can only grow Upland short-staple cotton. The shorter-length fibers mean reduced yarn and cloth quality, and they have "fuzzy" seeds because the fibers are tightly attached to the entire seed surface. This strong attachment of fibers to seeds makes it difficult to remove the fibers without damage, which made the process expensive as it was time-consuming and labor-intensive. These time-consuming tasks were done by slaves in the American southern states during the 1800s.

The modern cotton gin was invented by the American Eli Whitney (1765-1825), who graduated from Yale University in 1789 and is known for two great inventions: the cotton gin, which transformed the southern economy, and interchangeable parts in machinery,

which led to the mass-production method of manufacturing. When he graduated, he was short of funds and went to Georgia to seek his fortune in plantations. He was introduced to the problem of finding a better way to gin the Upland short-fiber cotton, which led him to invent the cotton gin in 1794. Whitney's cotton gin made the mechanical separation of fiber from seeds much more efficient. His machine had spiked sawteeth mounted on a revolving cylinder in a box, which was turned by a crank. The teeth pulled the cotton fiber through small slotted openings that were too small for the seeds to pass; thus, the lint was separated from the seeds. Then, a rotating brush removed the fibrous lint from the teeth, and the seeds fell into a hopper. Whitney's machine could separate up to 50 pounds of cleaned cotton daily, making cotton production profitable for the southern states.

Unfortunately for Whitney, his cotton gin was extremely simple and easy to copy, so he failed to profit from his invention. Soon after, imitations of his machine appeared, and his 1794 patent for the cotton gin could not be upheld in court until 1807. Instead, he decided to make money by going into the ginning business, and manufacturing and installing cotton gins throughout Georgia and the southern states. However, his ginning plants were resented by the planters, who thought that he charged too much, and hence, found ways to circumvent his patent.

In later years, Eli Whitney cultivated social and political connections through his status as a Yale alumnus and through his marriage. Although he had never made a gun in his life, through those connections, he won a contract from the War Department in 1798 to make and deliver 10,000 muskets (firearms used by soldiers before the invention of the rifle) for the army in 1800. Treasury Secretary Wolcott sent him a "foreign pamphlet" on arms manufacturing techniques, and Whitney began to talk about interchangeable parts. This concept was a departure from the usual method of having an individual master machinist make each part of the musket and fit them together, which is slow and painstaking. Whitney's method was to have many less-skilled machinists specialize in making only one part per person to highly specified dimensions, so that the parts would be interchangeable and could then be assembled. This is the heart of the mass-production method, later made more famous by Henry Ford in the manufacture of the Model T automobile. Whitney ultimately delivered on the army contract in 1809, and he spent the rest of his life promoting interchangeability.



66-2: Interpreting Prehistoric Cave Art

The Upper Paleolithic period began about 45,000 B.C. It is from this period, in several caves located in Europe, that archaeologists have discovered remarkable examples of prehistoric art. One of the early interpretations of cave art was that it was “art for the sake of art,” much as we today might go to a museum to see the skills of artists. This is a Western-culture-centered interpretation, however, and some would argue that there might be other, context-specific interpretations that are more suitable. The location of this art deep within dark caves, in which small flickering lamps would only reveal small portions of painted walls, also does not seem to fit with an interpretation of prehistoric “art galleries.”

Another early interpretation was based on the fact that most of the images are animals, and the majority of these, such as horses and bison, were hunted for food. This viewpoint became known as the “hunting magic” explanation. The drawing of the animals was interpreted as a way to magically ensure that an upcoming hunt would be successful. The hunting-magic interpretation has much appeal because we know that hunting was an important part of daily life during this period. And, in direct contrast to the art-for-art's-sake explanation, the practice of these “rituals” deep within caves suggests that not everyone participated. There are many nonhunting images in the caves, however, that do not support this explanation, such as geometric shapes and human figures.

In an effort to include all the types of images in Upper Paleolithic cave art in a comprehensive interpretation, some researchers turned to aspects of how the human mind works during altered states of consciousness and what images the mind “sees” during different phases of altered states. This explanation is called “entoptic phenomena,” and it argues that all modern human brains experience the same sets of visual images in the same progression. For example, during the first stage, it is common to see geometric patterns; during the second stage, the brain begins to associate various

geometric designs with real objects; and during the third stage, the brain sees actual animals, people, and monsters. Altered states can be achieved in many ways—drugs, intense dancing, sitting in absolute darkness as in a deep cave—and the entoptic phenomena interpretation argues that cave-art images represent the "visions" seen during these experiences.

Other explanations focus more on the use of cave art as a form of communication. That is, its presence and the types of images were used to establish social identities and perhaps as territorial markers. Communication, as an explanation, is based on identifying different styles that represent different groups of people and is a key element at aggregation sites, places where these people came together at certain times. Communication also is seen as adaptive because it enhances the survival of groups using social networks or alliances. The abundance of cave art in France and Spain, particularly during the Late Upper Paleolithic, is thought to be one outcome of the dense packing of people as they moved south to escape the harshest conditions of the extreme cold of the glacial maximum of the last ice age. Art was used to form alliances and thus resolve disputes about resources between groups who could not easily move away because of the close presence of many groups and the inhospitable nature of more northern areas in Europe.

In recent years, accumulating evidence suggests that Upper Paleolithic people created wall art not only in relatively inaccessible caves, but also in the rock shelters where they lived. Like Later Stone Age rock shelters in Africa, those of the Upper Paleolithic in Europe were exposed to sun, rain, and other weather, and most of the art present in these locales has long since disappeared. The traces we do have, however, suggest that wall art was a much more common feature of daily life than the art present deep in caves might suggest. And, if wall art was much more widespread and was typical of people's living sites, then single-cause explanations for wall art, especially those that are based, in part, on its inaccessibility in deep caves, seem less likely to be accurate models for all Upper Paleolithic wall art.



67-1: Sea Ice

Sea ice covers an area of almost 16 million square kilometers of the Arctic seas during its maximum extent in winter. Each summer, the pack ice (Arctic ice cover) melts and shrinks in the area and, at a minimum, extends over about 9 million square kilometers. The sea ice is an important part of the climate of the Arctic, and indeed of the whole world, because it has a large effect on the way the surface of the Arctic is heated. Sea ice is highly reflective; we see this by the need to wear sunglasses when walking over snow or ice. Water, on the other hand, appears rather dark by comparison. Where sea ice covers the Arctic seas, much radiation from the Sun is reflected back into the atmosphere. However, where open water is present, most of that radiation is absorbed and acts to heat the water. As the water heats up, more ice can be melted, and so on. This is called a positive feedback effect and is one of the key reasons why scientists are monitoring both the extent and the thickness of Arctic sea ice today. If sea ice covers a progressively smaller part of the Arctic seas in a world that may be affected by greenhouse gases, then this positive feedback could be critical to the rate at which our planet warms up.

Another important aspect of the formation of sea ice is that when water freezes, a large proportion of its salt content is rejected (forced out into the surrounding water). Indeed, it is possible to melt old sea ice and drink the water. However, the seawater produced in areas where sea ice forms is both very cold and also of high salinity (salt content). The presence of so much salt produces some of the densest water found in the world's oceans, and this water sinks to the seafloor as a result. In fact, the very dense waters produced by sea ice freezing in the Arctic are a vital factor in driving the circulation of the oceans because these waters flow to the deepest parts of the ocean floor. Some areas of the Arctic seas are places of intense sea ice production, where upwelling (the movement of above-freezing water from the lower depths of the ocean to the surface) or strong winds consistently keep the sea open year-round. One example is the North Water at the head of Baffin Bay. New sea ice can, therefore, continue to form throughout the winter in such open-water areas, which are known as polynyas, from the Russian language.

Sea ice also presents a number of hazards to human activity in the waters around the Arctic islands. In the heroic era of nineteenth-and early-twentieth-century exploration of the Arctic archipelagos (groups of islands), the presence of sea ice in many passages between islands made travel by sea difficult and dangerous. The British navy, for example, sent several tens of ships to the Canadian Arctic during the first half of the nineteenth century in an attempt to find a navigable passage connecting the North Atlantic and Pacific Oceans, referred to as the Northwest Passage, as a route for trade with Asia. However, this period of exploration coincided with some of the coldest weather of the last few thousand years, which is known as the Little Ice Age. The ice conditions encountered by many of these ships were appalling. A number of vessels became trapped in sea ice as the short summers came to an end, and several were crushed when pressure built up in the sea ice as it was driven by strong winds. The great dangers inherent in these early explorations of the islands and passages in, for example, the Canadian north are exemplified by the remarkable story of Sir John Franklin and his crews of the ships *Erebus* and *Terror*, who disappeared in the late 1840s and whose fate has never been finally established. By contrast, the Norwegian explorer Roald Amundsen was the first to navigate a ship through the Northwest Passage, and he did so only in 1903-06 when the climate had warmed up after the Little Ice Age and sea ice conditions had become less severe.



67-2: The Early Cyclades

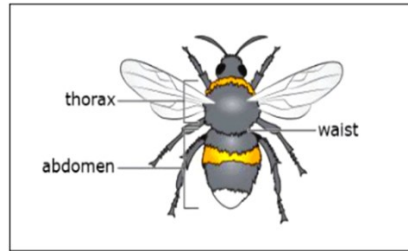
The Cyclades, a group of islands in the Aegean Sea to the east of mainland Greece, eventually played an important part in Greek civilization, though at first they were sparsely populated. Radiocarbon dates from excavations throughout the islands tell us that farmers had settled on Naxos as early as the fifth millennium B.C.E. Between 4000 and 3000 B.C.E., farming communities flourished on Andros, relatively close to the mainland, as far out as Paros, and possibly on Mykonos, in the heart of the Cyclades. As far as one can tell, the early settlers avoided smaller islands, with their tiny patches of arable soil and often sparse water supplies. Milos was rich in obsidian, a volcanic glass used for tools and jewelry, and the island was certainly visited for fishing and fine-grained stone, but the first date for permanent occupation remains an open question. Throughout the Cyclades, the founding settlers seem to have preferred larger and medium-sized islands, even if they were far from the mainland. This speaks volumes about the seaworthiness of what by now must have been fairly large boats. Quite apart from anything else, such vessels would have had to transport sheep and goats across open water for as long as two days.

Why did people colonize islands far out in the Aegean? One theory argues that regular visits in search of obsidian, or for tuna fishing in otherwise inaccessible waters, might have led in due time to permanent settlement. Archaeological finds on the mainland do not support the obsidian hypothesis, for early obsidian finds are merely fragments, apparently carried back haphazardly. It was only much later, some centuries before 3000 B.C.E., that much greater quantities of obsidian arrived on the mainland, deliberately prepared in a variety of ways, arriving as cargoes that included carefully hewed blocks of the volcanic glass. Whether greater loads resulted from colonization or simply from an increase in the tempo of exploitation and visitation is a matter for discussion, nor is fishing a particularly compelling reason for colonization. Fish were never more than a dietary supplement in these waters, where marine life was far from abundant. Tuna migrations

move rapidly and unpredictably, usually heading south in the fall. Laborious voyages in search of such an unreliable food source were not worth the trouble or the risk, given the difficulty of predicting them.

Living permanently on the Cyclades was never easy, for survival depended on diversifying one's crops among carefully selected microenvironments. Barley, wheat, and pulses (a plant group including peas and beans) were staples, while goats and sheep thrived on rugged island slopes. Olives and vines became important crops later, when they were widely traded commodities, but almost certainly not in earlier millennia. The uncertainty of the Aegean agricultural year meant that storage of foods of all kinds was of great importance, a process that must have involved sharing among neighboring communities and perhaps even between one island and another. Unlike mainland farmers, islanders could not fall back on a cushion of reliable wild plant foods or game like fallow deer. The realities of daily life meant that most island communities were small and widely dispersed over generally rugged island terrain. Populations were tiny, to the point that people in the Cyclades were a scarce resource. Few islands attained self-sufficient populations and marriage networks of about three hundred to five hundred people. There were many more people living in a single early Mesopotamian city, such as Uruk, one of the earliest urban settlements in the world in the fourth millennium B.C.E., than in the entire Cyclades. Inevitably, then, there was a high degree of interdependence among the islands, for exchange of food and other commodities and mobility were the only long-term survival strategies.

This constant movement touched every facet of island existence- marriage and kin ties, planting and harvest, food storage, the need for toolmaking stone and other commodities as prosaic as grinding stones, movements triggered by drought, excessive rainfall, and other short-and long-term climatic events, chance migrations of tuna shoals, and all the complex demands of ceremonial life and ritual observance. Seafaring operated here within ever-changing lattices of interaction that linked people over considerable distances and across often turbulent seas.



68-1: Bumblebee Heat Regulation

While flying, bumblebees have a body temperature that is generally well above that of the air around them and tends to be constant at about 35 C. Keeping warm is harder the smaller you are. Small creatures, like bees, have a vast surface area relative to their volume and lose heat incredibly quickly. Yet bumblebees can keep themselves warm even when the surrounding air is 30C cooler than their body temperature. Scientist Bernd Heinrich found that the answer to bees' ability to retain heat is in two parts: keeping heat in, and generating it in the first place. Keeping warm is helpful if you have insulation, and of course, bumblebees have furry coats. The vital thing for a bumblebee is to keep its thorax (the middle section) warm, because this is where the flight muscles are; unless the thorax is warm enough, the muscles cannot contract sufficiently fast, and the bee cannot take off. The temperature of the abdomen does not matter much in flight. The abdomen and thorax in a bumblebee are connected by a very narrow waist, and the front part of the abdomen contains a sac of air, so heat loss from the thorax to the abdomen is minimal (air is a poor conductor). A bumblebee's heat is generated by the contractions of the flight muscles. In flight, a bumblebee flaps its wings 200 times per second. This generates a lot of heat, but of course, it comes at a cost: bumblebee flight is enormously expensive in terms of the energy that it uses. Consequently, bumblebees have to eat almost continually to keep warm.

A bumblebee's dense furry coat has obvious advantages, but it can create problems when the weather is warm. Bumblebees cannot help but produce lots of heat when they fly, which can be difficult to get rid of if the air temperature is also high. This is probably why bumblebees are not common in Mediterranean countries. If their body temperature exceeds 44 C, they will die; as they approach this lethal limit, their metabolism collapses, and they become unable to fly. It is noticeable that those species of bumblebee that occur in warmer climates tend to have shorter fur, while those from high latitudes and altitudes tend to be very large and very furry. Hence, the huge size and long shaggy coat of the world's largest bumblebee, *Bombus dahlbomii*, which inhabits the high Andes of South America.

Bumblebees do have a trick to help them lose heat in hot weather. They normally keep their abdomen at a much lower temperature than their thorax, with the narrow waist connecting the two acting as a barrier. If the thorax starts to get too hot, the abdomen starts rhythmically to contract, sending surges of cool blood into the thorax and sucking back waves of hot blood. This heats the abdomen, and so raises the surface area from which heat can be lost. Nonetheless, on very hot days in summer, bumblebees will tend to stop seeking food as noon approaches and recommence in the cool of the evening.

Pumping heat from the thorax to the abdomen can also serve a very different purpose. The northernmost social insect in the world is a bumblebee known as *Bombus polaris*, which lives well within the Arctic Circle: large and unusually hairy, it can exist in regions where, even in the height of summer, air temperatures rarely exceed 5°C. Unlike other bumblebees, the *Bombus polaris* queens maintain a stable, high abdominal temperature (greater than 30°C) by pumping hot blood from their thorax to the abdomen. This enables them to develop eggs within their ovaries quickly, which is important in the very short Arctic summer. As the workers and males have no eggs to develop, their abdomens are substantially cooler.

The larger the bee, the easier it is for it to keep warm, but the more prone it is to overheating in hot weather. This may explain why queen bees are so much larger than workers, for they begin flying earlier, in the spring, when the weather is cold. It may also explain why the worker bees that leave the nest to gather food tend to be larger, on average, than those that stay in the nest to look after the brood.



Water travelling through a dam

68-2: Dams and Flood Control

There are both advantages and disadvantages to building dams on rivers. On the one hand, dams can be beneficial to human populations. They provide reserves of fresh water, a commodity crucial to life, both directly as drinking water and indirectly because irrigating farmland provides crops for food. Moreover, the energy of water flowing through dams can be harnessed to provide electricity. The location of dams on rivers also enables flooding to be controlled. Thus, dams can provide protection from the extremes of the local climate. Heavy rains in headwater areas, which might otherwise lead to downstream flooding, are impounded in the reservoir (water storage area) behind a dam. By the same token, during dry periods that might otherwise produce drought, water can be released from the reservoir to downstream areas.

On the other hand, introducing a dam into a river system will alter the structure of the river. Sediment will be deposited upstream of the dam and in the reservoir (reservoirs are inevitably only transient features and will ultimately fill up with sediment). Downstream, increased erosion will occur. Landscapes and ecosystems are submerged beneath the new reservoir. The dam itself represents a potential hazard should it ever fail. The collapse of the St. Francis Dam in southern California in 1928 was the consequence of building it against a weak rock support only two years earlier.

Government policies in the United States regarding the construction of dams have shifted considerably in the past century. The early part of the twentieth century saw an enormous boom in dam construction, particularly in the arid West. In recent years, this trend has been halted and even reversed in some cases, as plans to tear down dams and restore stream systems to their natural state are under consideration. Any discussion of the construction of a dam across a "wild" river usually pits conservationists against power companies and farmers. One side stresses the need to conserve for future generations; the other stresses the needs of the modern population and economy. The example of the

Trinity River Dam in northern California serves to illustrate many of the issues at work here.

Prior to the construction of a dam on the Trinity River near Mount Shasta in northern California, the river was a typical youthful mountain stream; its bed was rocky and strewn with boulders, and only minor brush (woody plants) grew close to the river's edge. It was a clear, cold stream with various game fish, including a large resident population of trout and annual spawning runs of steelhead and salmon. The stream bed was kept free of sediment and brush by vigorous floods derived from the spring thaw in the high mountains nearby. This mountainous region was a wilderness with little industry except logging. Downstream, however, was a growing population and economy. Energy needs of the area grew, and the spring floods became more than just a nuisance—they were a threat to safety. A dam was built to control the floods, conserve the water for later use in irrigation, and act as a hydroelectric facility.

After the construction of the dam, the spring floods were contained by the new Trinity Lake. Control of the river meant that regular water delivery could be provided, and irrigation for farms and ranches downstream was a boon to agriculture. Power generated from the hydroelectric plant is completely nonpolluting and provides a substantial proportion of the increasing energy needs of the area. But since there are no spring floods to clean the river of sediment, weeds, and brush, instead of a clear, fast-moving, cold stream, the Trinity River has become sediment-laden, slower-moving, warmer, and less hospitable to the prized game fish that formerly populated the river. There are no longer any salmon or steelhead runs up the Trinity River.

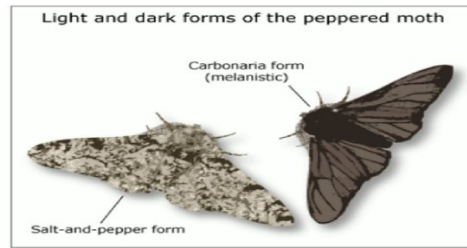
Trinity Lake, formed behind the dam, is a popular recreational area that attracts many people for boating, water skiing, and fishing. Agriculture and residents of the cities and towns downstream benefit from the reliable supply of water and power. The lake has an excellent resident population of trout, and because of the good fishing, the surrounding forest supports a growing population of osprey— a hawk that many conservationists consider on the brink of extinction.

69-1: The Development of Land Flora

Today's most primitive land plants, the liverworts and hornworts, have patterns in their DNA that indicate a connection to a group of complex green algae that are found only in freshwater, which suggests that land plants arose in freshwater lakes and streams or from river deltas-not from the ocean shore. But what might have caused a plant to adapt to the harshness of a land environment? The most reasonable scenario suggests that early land plants lived along streams or in ponds that dried up during part of the year. Their first adaptation was the spore, a reproductive cell that could survive this dry period. Carried by the wind, the microscopic spore could be scattered across wide areas, germinating (starting to grow) -if lucky- in a distant pool of quiet water or moist mud. Such spores are the first evidence of plants living on land.

Earlier, blue-green bacteria probably lived on damp soil and in shallow pools over hundreds of millions of years. They were probably joined by green algae and fungi to form puddles of green glop in moist depressions and along the edges of ponds and streams. These lineages (lines of descent) may have populated land surfaces for many millions of years before more complex land plants arose. And they, too, had to develop sporelike stages to survive drying out and achieve dispersal. Unfortunately, because they did not possess resistant tissues, these early forms of land flora left no trace in the fossil record. The first "true" land plants were probably flat liverwort-like plants with little leaves, only one cell layer thick. Again, these first land plants are remembered in the rocks only by their spores. Being able to withstand the physical stresses of drying and rehydration, the tough spore wall is also resistant to decomposition. Thus, spores became an important element in the fossil record. Algal and fungal spores are quite different from these. These early spores were first recorded in the rocks and dated at around 470 million years ago, and they are usually found in groups of four (called tetrads). By about 430 million years ago, spore tetrads declined in abundance, and separate individual spores with a Y-shaped (trilete) configuration on one side became common.

About 410 million years ago, with plant spores diversifying, the rocks preserved the first fragmentary evidence of larger plants. It was not until more complex land plants had evolved desiccation-resistant surfaces and a strong erect structure that a really three-dimensional terrestrial vegetation came into being. Waxy surfaces were a major innovation in the advent of larger terrestrial plants. In addition to reducing water loss through outer surfaces, a waxy layer protected against ultraviolet radiation, microbial attack, and corrosive chemicals. Unfortunately, such protection also presents a problem: how can a plant absorb carbon dioxide from the air if it's covered with wax? Moist interior cell surfaces must be exposed to air in order to absorb carbon dioxide for photosynthesis (the process through which plants obtain energy by using sunlight, carbon dioxide, and water). Inevitably, exposing moist cells to air will also result in water loss by evaporation from within the plant. And because carbon dioxide concentrations are low in air, a great deal of water will be lost in the process of absorbing carbon dioxide. (This is why plants require a lot more water to survive than do animals with a similar volume of active tissue.) The solution to this plant problem came in the form of special little pores called stomates on the surfaces of the plant: pores that could be opened for gas exchange and closed when conditions became too dry. The other major adaptation for life on land was a "plumbing system" to conduct water from the roots to the parts of the plant active in photosynthesis. The vascular/plumbing system includes thick-walled cells that quickly become dead, empty, and open at the ends. Laid out end-to-end, these empty cells resemble a system of pipes, facilitating the flow of water through the plant. These same dead, thick-walled cells would also serve as architectural support. With a vascular system and stomates in place, the earliest erect plants were able to draw water up to the aerial parts of the plant and provide strength in the face of wind and weather.



69-2: Evolution in Peppered Moths

Since Darwin proposed the theory of evolution, a vast body of literature has emerged that supports the precepts of the theory and increases our understanding of the mechanisms of evolution. One of the most striking cases of evolution in action is that of industrial melanism, or darkening of the body's tissues, in the peppered moth of England. Early in the nineteenth century, occasional dark (melanistic) specimens of the common peppered moth were collected. Over the next 100 years, this darker form, referred to as carbonaria, became increasingly common in forests near heavily industrialized regions of England, which is why the phenomenon is often referred to as industrial melanism. In places without factories and other heavy industry, the lighter-colored, salt-and-pepper form of the moth prevailed. This phenomenon aroused considerable interest among geneticists, who showed, by cross-mating light and dark forms, that melanism is an inherited trait determined by a single dominant gene. Because the melanistic trait is an inherited characteristic, its spread reflected genetic changes (evolution) in the population.

Peppered moths inhabit dense woods and rest on tree trunks during the day. Where melanistic individuals had become common, the environment must somehow have been altered so as to give dark forms a survival advantage over light forms. It seemed reasonable to suppose that natural selection- the process by which organisms best suited to survival in their environment achieve greater reproductive success, allowing advantageous genetic characteristics to pass on to future generations-had led to the replacement of the typical light individuals with darker carbonaria individuals. To test this hypothesis, the English biologist H.B.D. Kettlewell captured a sample of moths of both forms, marked each individual with a dot of cellulose, and then released them back into the woods. The mark was placed on the underside of the wing so that it would not attract the attention of predators to a moth resting on a tree trunk. Some days later, Kettlewell captured more moths by attracting them to a mercury-vapor lamp in the center of the woods or to caged virgin females at the edge of the woods. (Only males could be used in the study because females are attracted neither to lights nor to virgin females.) This type of study is referred to as a mark-recapture study.

In one such mark-recapture experiment, Kettlewell marked and released 201 lighter moths and 601 darker moths in a wooded area near industrial Birmingham, England. The results indicated that a greater percentage of the darker form survived over the course of the experiment. A similar experiment in a nonindustrial area revealed higher survival by the lighter salt-and-pepper form of the moth.

The specific agent of selection was easily identified. Kettlewell reasoned that in industrial areas, pollution had darkened the trunks of trees so much that typical moths stood out against them and were readily found by predators. Any unusual dark forms were better camouflaged against darkened tree trunks, and their coloration made them more likely to survive. Eventually, the different survival rates of dark and light forms would lead to changes in their relative frequency in the population. To test this idea, Kettlewell placed equal numbers of light and dark forms on tree trunks in polluted and unpolluted woods and watched them carefully from a concealed spot some distance away. He quickly discovered that several species of birds regularly searched tree trunks for moths and other insects, and that these birds more readily found a moth that contrasted with its background than one that resembled the bark it clung to. These data were consistent with the results of the mark-recapture experiments. Together, they clearly demonstrated the operation of natural selection, which, over a long period, resulted in genetic changes in populations of the peppered moth in polluted areas.

One of the most gratifying aspects of the peppered moth story is that with the advent of new smoke-control programs and the return of forests to a cleaner state, frequencies of melanistic moths have decreased. In the area around the industrial center of Kirby in northwestern England, for example, the darker carbonaria form decreased from more than 90 percent of the population to about 30 percent over a period of twenty years.

70-1: United States Literature and Art in the Nineteenth Century

At the beginning of the nineteenth century, the United States was a provincial culture, still looking to Britain for values, standards, literature, and art, despite all the rapid improvements in communication - for example, the great growth in the number of magazines and newspapers that occurred in the United States following independence from Great Britain. In a famous essay in the *Edinburgh Review* in 1820, Sydney Smith sarcastically inquired, "In the four quarters of the globe, who reads an American book? Or goes to an American play? Or looks at an American picture or statue?"

Yet, interest in the arts and the cultural institutions that sustain them was rapidly growing in America. To be sure, cultural institutions were difficult to create in the South, mainly because the population was so widely dispersed; and in the West, dominated in this era by pioneers, the emphasis was on the practical rather than on the literary or artistic. But many eastern seaboard cities were actively building the cultural foundation that would nurture distinctively American art and literature. During these years, Philadelphia's American Philosophical Society, founded by Benjamin Franklin in 1743, boasted a distinguished roster of scientists, including Thomas Jefferson, concurrently its president and president of the United States. Culturally, Boston ran a close second to Philadelphia, founding the Boston Athenaeum (1807), an impressive library and reading room containing "works of learning and science in all languages, particularly such rare and expensive publications as are not generally to be found in this country," The *North American Review*, which became the most important and long-lasting intellectual magazine in the country, was published in Boston. Devoted to keeping its readers in touch with European intellectual developments, it had a circulation of 3,000 in 1826, about the same as similar British journals.

Of the eastern cities, New York produced the first widely recognized American writers. In 1809, Washington Irving published America's first great book of comic literature - *A History of New York* - a humorous account of life in New York both in colonial times and in Irving's own day. James Fenimore Cooper's *Leatherstocking* novels (of which *The Last of the Mohicans*, published in 1826, is the best known) achieved wide success in both America and Europe. In these novels, Cooper established the American experience of westward expansion as a serious and distinctive American theme.

It was New England, however, that saw itself as the home of American cultural independence from Europe, a claim voiced in 1837 by Ralph Waldo Emerson in a lecture

at Harvard: "Our day of dependence, our long apprenticeship to the learning of other lands, draws to a close," Emerson announced. Rather than draw on British models, American artists, Emerson argued, should seek inspiration from the ordinary details of daily American life. Immensely popular, Emerson gave more than 1,500 lectures in 20 states between 1833 and 1860. "The American Scholar," his most famous lecture, carried a message of cultural self-sufficiency that Americans were eager to hear.

Artists in this period were as successful as writers in finding American themes. Thomas Cole, who came to the United States from England in 1818, found great inspiration in the American landscape. Cole founded the Hudson River School of American Painting, whose romantic renderings of New York landscapes were frankly nationalistic. Other painters - realists such as Karl Bodmer and George Catlin- drew on the dramatic landscapes and peoples of the American West. Their art was an important contribution to the American sense of the land and to the nation's identity. Catlin, driven by a need to document Native American life, spent eight years among the tribes along the upper Missouri River. Then, he assembled his collection of more than 500 paintings and toured the country in an effort to increase the public's awareness and appreciation of Native American lifeways. George Caleb Bingham, an accomplished genre painter, produced somewhat tidied-up scenes of real-life American workers, such as flatboat men on the Missouri River. All these painters found much to record and celebrate in American life. Ironically, the inspiration for the most prevalent theme, the American wilderness, was profoundly endangered by the rapid western settlement of which the nation was so proud. Despite such contradictions, the early nineteenth century was the period in which writers and painters found the national themes that defined the first distinctively American literature and art.

70-2: Development of Intensive Agriculture

Anthropologists and scientists in other fields have long been concerned with the origins of intensive farming and early civilization, with the social and economic structure of rural society, and with the strategies that have enabled diverse populations to adapt to environmental and other problems. Agricultural intensification involves increasing the yield from labor or land. This implies that due to the complex ways in which land and labor are interrelated, there is more than one route to intensification. Generally, the perspective of most American or European economists is to concentrate on labor. Agricultural history is usually described in terms of progress in labor-saving technology - the plow, seed drills, cultivators, and the like- because the economies of the Western world experienced labor shortages over much of their histories. Technology, of course, not only may increase food production by allowing the same labor force to cultivate more land but also may free up labor for other endeavors. Thus, Australia, Canada, and the United States produce massive amounts of food with a relatively small rural labor force.

Production can also be intensified by increasing the productivity of land without reducing the labor requirements; that is, expanding production by using an existing labor force or even a larger one. This can be important, for example, where there are few alternative sources for employment, as is often the case in densely populated developing countries. Irrigation and the introduction of new crop strains are well-known examples of this form of intensification. Water control may allow for multiple harvests of a particular crop; new plant strains may also increase productivity without new capital input needed to reduce labor requirements.

One of the most ancient (and still important) ways land productivity can be increased is through water management. So diverse are the ways in which moisture can be controlled that it is somewhat misleading to refer to them all as irrigation." Simply adding small stones to fields, as did some ancient peoples of North America, can enhance the field's ability to retain moisture. In this sense, irrigation, or, at least, "moisture control," is as early as agriculture itself. Archaeological evidence in the Middle East indicates that simple systems of water control predate the rise of large agrarian states with their concomitantly dense populations. Populations near rivers or marshes would simply capture or divert annual floodwaters or runoff from rains. The term "irrigation" refers to actually transporting water to the field and then managing its direct application and subsequent drainage, since drainage is important in order to maintain a salt-free soil base.

The earliest known large-scale system of irrigation in Egypt appears with the emergence of one government throughout that land in 5100 B.C. We know that even fairly complicated irrigation systems can be managed by local farmers, although the potential for interfamilial or intercommunity conflict is substantial. People who share a common water resource may have very different interests in its use. As a consequence, we see a widespread pattern of large-scale irrigation systems that come to be run by special managers, with a corresponding lessening of control by households or even by local communities. Centralized decision-making facilitates the mobilization of large workforces, allocation of water, conflict resolution, and storage of surpluses.

In the emerging prehistoric states of Mesopotamia, this managerial role was first assumed by religious leaders and only later by secular rulers. It is interesting to note a similar pattern in large-scale irrigation systems in the southwest United States that were managed by the Mormon Church (a religious organization) in the nineteenth century. Making water control feasible required centralized control by a committed bureaucracy willing to use resources to sustain building and rebuilding dams and canals beyond the ability of local communities.

The main impetus for irrigation, in most places, is simply the need to have water available in areas where rainfall is unpredictable -not necessarily a wish to increase the average yield of a unit of land. But with the advent of irrigation, slight differences in the productivity of different pieces of land become greatly magnified. Fields that lend themselves to irrigation -fields close to the water source or that drain well- produce far more than those less suited to irrigation.

71-1: Why Did Nonavian Dinosaurs Become Extinct?

Data from diverse sources indicate that the Late Cretaceous (100 to 65 million years ago) climate was milder than today's because of shallow seas covering the continents. At the end of the Cretaceous, the geological record shows that these seaways retreated from the continents back into the major ocean basins. No one knows why. Over a period of about 100,000 years, while the seas pulled back, climates around the world became dramatically more extreme: warmer days, cooler nights; hotter summers, colder winters. Perhaps nonavian dinosaurs (that is, all the dinosaurs except those belonging to the Aves or bird family) could not tolerate these extreme temperature changes and became extinct.

If true, though, why did cold-blooded animals, such as snakes, lizards, turtles, and crocodiles survive the freezing winters and torrid summers? Those animals are at the mercy of the climate to maintain a livable body temperature. It's hard to understand why they would not be affected, whereas nonavian dinosaurs were left too crippled to cope, especially if some dinosaurs were warm-blooded. Critics also point out that the shallow seaways had retreated from and advanced on the continents numerous times during the Mesozoic. So why did the nonavian dinosaurs survive the climatic changes associated with the earlier fluctuations but not with this one? Although initially appealing, the hypothesis of a simple climatic change related to sea levels is insufficient to explain all the data.

Volcanism has also been implicated in dinosaur extinction. The end of the Cretaceous coincided with a great increase in volcanic activity throughout the world. Lava flooded large areas of India, and explosive eruptions in the South Atlantic and the midwestern United States hurled ash over much of the globe. These eruptions could have spewed great quantities of poisonous gases into the atmosphere, causing acid rain and more acidic waters in the surface layers of the ocean. Over the short term, a cooling would result from the airborne volcanic ash, which would cut off sunlight. Climatic changes from increased volcanism may have caused the extinction of many nonavian dinosaurs, but they do not satisfactorily explain the selective patterns of extinction in the fossil record.

Dissatisfaction with conventional explanations for nonavian dinosaur extinctions eventually led to a key observation that, in turn, has fueled a decade of vigorous and often vitriolic debate. Many plants and animals disappear abruptly as one moves from older layers of rock, documenting the end of the Cretaceous up into younger rocks representing the beginning of the Cenozoic. Between the last layer representing the end of the

Cretaceous and the first layer representing the start of the Cenozoic, there is often a thin bed of clay. Scientists felt that they could get an idea of how long it took to deposit this 1 cm of clay by looking at the concentration of the element iridium (Ir) in it.

Iridium is no longer common on Earth's surface. Because it usually exists in a metallic state, it was preferentially incorporated into Earth's core as the planet cooled and consolidated. Iridium is found in high concentrations in some meteorites, in which the solar system's original chemical composition is preserved. Even today, microscopic meteorites continually bombard Earth, falling on both land and sea. By measuring how many of these meteorites fall to Earth over a given period of time, scientists can estimate how long it might have taken to deposit the observed amount of iridium in the boundary clay. These calculations suggest that a period of about one million years would have been required. On the basis of other evidence, however, scientists know the deposition of the boundary clay could not have lasted one million years, so the unusually high concentration of iridium seemed to require a special explanation.

Consequently, scientists hypothesized that a large asteroid, about 10 to 15 km across, collided with Earth, and the resulting fallout created the boundary clay. Their calculations show that the impact kicked up a dust cloud that cut off sunlight for several months, inhibiting photosynthesis in plants, decreasing surface temperatures on continents to below freezing, causing extreme episodes of acid rain, and significantly raising long-term global temperatures through the greenhouse effect. This disruption of the food chain and climate would have eradicated the nonavian dinosaurs and other organisms in less than 50 years.

71-2: Infant-Directed Speech

Infant-directed speech is a style of speech directed toward infants. This type of speech pattern was previously called motherese because it was assumed that it applied only to mothers. However, that assumption was wrong, and the gender-neutral term infant-directed speech is now used more frequently.

Infant-directed speech is characterized by short, simple sentences. Pitch (the highness or lowness of a sound) becomes generally higher, the range of frequencies (essentially, the difference between the highest and the lowest pitch used) increases, and intonation (the rise and fall in pitch) is more varied. There is also repetition of words, and topics are restricted to items that are assumed to be comprehensible to infants, such as concrete objects in the baby's environment. Sometimes infant-directed speech includes amusing sounds that are not even words, imitating the prelinguistic speech of infants. In other cases, it has little formal structure but is similar to the kind of telegraphic speech that infants use as they develop their own language skills.

Infant-directed speech changes as children become older. Around the end of the first year, infant-directed speech takes on more adultlike qualities. Sentences become longer and more complex, although individual words are still spoken slowly and deliberately. Pitch is also used to focus attention on particularly important words.

Infant-directed speech plays an important role in infants' acquisition of language. Newborns respond to such speech more readily than they do to regular language, a fact that suggests that they may be particularly receptive to it. Furthermore, some research suggests that unusually extensive exposure to infant-directed speech early in life is related to the comparatively early appearance of first words and earlier linguistic competence in other areas.

Is infant-directed speech similar across cultures? In some respects, it clearly is. According to a growing body of research, there are basic similarities across cultures in the nature of infant-directed speech. Consider, for instance, a comparison of the major characteristics of speech directed at infants used by native speakers of English and Spanish. Of the ten most frequent features, six are common to both: exaggerated intonation, high pitch,

lengthened vowels, repetition, lower volume, and instructional emphasis (that is, heavy stress on certain key words, such as emphasizing the word "ball" in the sentence, "No, that's not a ball."). Similarly, mothers in the United States, Sweden, and Russia all exaggerate and elongate the pronunciation of the three vowel sounds of "ee," "ah," and "oh" when speaking to infants in similar ways, despite differences in the languages in which the sounds are used. Even deaf mothers use a form of infant-directed speech: When communicating with their infants, deaf mothers use sign language at a significantly slower tempo than when communicating with adults, and they frequently repeat the signs.

The cross-cultural similarities in infant-directed speech are so great, in fact, that they appear in some facets of language specific to particular types of interactions. For instance, evidence comparing American English, German, and Mandarin Chinese speakers shows that in each of the languages, pitch rises when a mother is attempting to get an infant's attention or produce a response, while pitch falls when she is trying to calm an infant. Why do we find such similarities across very different languages? One hypothesis is that the characteristics of infant-directed speech activate innate responses in infants. As we have noted, infants seem to prefer infant-directed speech over adult-directed speech, suggesting that their perceptual systems may be more responsive to such characteristics. Another explanation is that infant-directed speech facilitates language development, providing cues as to the meaning of speech before infants have developed the capacity to understand the meanings of words.

Despite the similarities in the style of infant-directed speech across diverse cultures, there are some important cultural differences in the quantity of speech that infants hear from their parents. For example, although the Gusii of Kenya care for their infants in an extremely close, physical way, they speak to them less than American parents do.

72-1: A Mutualistic Fungus of Tall Fescue Grass

Mutualism is an interaction between two species in which both benefit. A classic example of this relationship is between tall fescue grass and a fungus named *Epichloë coenophiala*. That fungus is completely internal and grows intercellularly (between the cells) in the above-ground portion of the grass. The grass supplies all the nutritional needs of the fungus. Infected plants exhibit no external symptoms of the fungus and no disease symptoms. In contrast, pathogenic (disease-causing) fungi result in symptoms that include wilt, leaf spots, deformed plant structures, or death of plant tissue. Pathogenic fungi often invade plant host cells and absorb the cells' contents (thereby eventually killing the cells) or produce chemicals that kill them outright.

Tall fescue is extensively used as both a pasture grass and a turf or lawn grass. It is a hardy, vigorously growing grass but has been associated with various health problems in cattle and horses. Beef cattle often show reduced weight gains when grazing on tall fescue pastures, and dairy cattle experience lower milk yields. Female horses may be more likely to lose their offspring during pregnancy and may not produce milk following a live birth. Reports of such adverse effects resulted in a microscopic investigation of leaf and stem tissue that revealed the presence of the *Epichloe coenophiala* fungus. Surveys of infected pastures found a positive correlation between the percentage of infected plants and the severity of animal symptoms. The fungus in the grass was found to produce various alkaloids (toxic, nitrogen-containing, organic molecules). One type of alkaloid (ergot alkaloids) contributes to the livestock's symptoms and reduces grazing, whereas another type (loline alkaloids) makes the grass more resistant to insects. Interestingly, the insect-detering alkaloids are not produced when the fungus is grown in the laboratory apart from the plant. Because these chemicals defend the grass as well as the fungus from consumption by both livestock and insects, the mutualistic relationship has been termed defensive mutualism.

Because the *Epichloe coenophiala* fungus is not external to the plant, the usual mechanism of spread by fungi does not take place. Typical fungi form spores (reproductive units consisting of one or more cells that are similar to the seeds of plants). Fungal spores spread through the air to susceptible plants and usually require an extended period of moist conditions for their survival and germination (the beginning of the growth of a spore or seed) and for the subsequent infection of the host plant. The tall fescue fungus, in contrast, penetrates the flower head and infects the plant embryos in

the developing seeds. Therefore, seeds collected from infected plants result in infected seedlings following germination. As the plant grows, the fungus grows along with it by division and stretching of the hyphae (threadlike structures that make up the body of a fungus) at the base of the grass, where plant cell division also occurs. Thus, the growth of the fungus is concurrent with the growth of the plant, which ensures that new stems and leaves of the grass are infected with the fungus. Fungus-free tall fescue plants can result if infected seed is stored for several months before planting. The fungus usually dies out during most long-term storage conditions.

The alkaloids produced by the fungus also affect microorganisms in the immediate environment of the plant. Nondisease-causing bacteria, recovered from the surfaces of fungus-infected tall fescue plants, can use loline alkaloids, released on the leaves, as a source of nutrients. The population size of these bacteria is about eightfold greater on infected plants than on fungus-free tall fescue plants or on plants colonized by related fungi incapable of producing loline. Thus, many components of the fungus-infected tall fescue community are impacted by the fungus: the grass itself, animals, insects, and microbes.

Although the fungus of tall fescue is detrimental to livestock, it is considered beneficial when grass is grown for other purposes. Insect pests are numerous in turf (short, thick grass and the soil it grows in), and tall fescue's increased resistance to them is advantageous when grass is grown for house lawns or sports fields. It may be possible to develop pasture varieties of tall fescue that are infected with this fungus for the purpose of insect resistance, but with the deletion of the fungal genes for the toxins that harm livestock.

72-2: How Old is the Continental Crust?

Although we now know that the Earth formed roughly 4.5 billion years ago, the next 600 million years of Earth's history is essentially blank. The oldest rock samples yet identified on Earth (from the Northwest Territories in Canada) have an age slightly greater than 3.9 billion years. These rocks have undergone strong metamorphism (changes to the form of the existing rock due to heat and pressure), and it is difficult to tell much about their origin. However, they are not very different from many other typical continental rocks that are much younger. Thus, we know that there were at least some fragments of continental crust in existence 3.9 billion years ago.

The question of when the first continents formed is one that has long intrigued geologists, because it is evident that the continental crust has grown and evolved over geologic time. It is probable that there were small continents even before the creation of the 3.9-billion-year-old rocks. The clues that lead to this conclusion are rare, minute, and difficult to discover. Where would one look for such evidence? The answer provides a good example of how geologists often work: by using the present as a window into the past. We know that the products of erosion accumulate on the edge of continents today, and there is no reason to think that the situation was different in the past. Even the earliest continents must have had beaches. There is a chance that if some of those very old sediments have been preserved, they might contain mineral grains from even older continents.

Geologists have thus sifted through some of the oldest known sandstones, which were probably originally laid down along the shorelines of ancient continents, in search of mineral grains that are especially resistant to destruction during weathering and transport. One such search was in a 3.6-billion-year-old sandstone in Western Australia. Some of the grains in these rocks are much older than the sandstone itself and have apparently survived multiple cycles of erosion, deposition, consolidation into solid rock, uplift, and re-erosion. Geologists have found that a few grains of the weathering-resistant mineral zircon from these sandstones have ages in the range of 4.1 to almost 4.3 billion years.

Zircon crystals are small but common constituents of many rocks. The weathering and erosion processes that break down their parent rocks have little effect on the inert zircon crystals. Large, transparent zircon crystals are often sold as gemstones. But the ones that are most useful to geologists are the small zircon grains that are carried long distances in

streams or even by wind. These become tracers of the ultimate source for the sedimentary materials in which they now reside.

As their name implies, zircons are rich in the element zirconium. Fortunately, they also incorporate considerable amounts of uranium when they form, and the radioactive decay of uranium produces isotopes of lead that can be measured to determine the age of the grain. Modern techniques are so sensitive that even the minute amount of lead in a single zircon grain can be measured precisely and its age determined. Because these ancient zircons are single grains rather than rock fragments, it is difficult to say much about the types of rock from which they were originally eroded. However, zircons are common in continental rocks such as granite but virtually absent from the ubiquitous basalts of the ocean floor. The inference is that these grains must have come from continental rocks, and if this is indeed the case, the existence of continents can be extended back to almost 4.3 billion years, only a few hundred million years after the formation of the Earth.

Even if the Earth's crust began to form very early, there are several possible reasons why no fragments of continental rock are preserved from approximately the first 600 million years of our planet's existence. One is that for much of this period, the Earth was experiencing heavy bombardment from space as the residual material from the planet's formation process was gathered by the Earth's gravity. A second is that the early Earth was very hot. The vigorous circulatory motion of materials in the hot Earth may simply have destroyed much of the early formed crust.

73-1: Industry during the Roman Empire

Although agriculture was the economic mainstay of the Roman Empire, the development of industry under the Empire also made a significant contribution to its centuries of power and stability in Europe. The growth of industry was largely the result of economic and social factors. It grew because its markets increased with a rising population and the urbanization of the western provinces, such as Gaul (modern France) and Spain. Nevertheless, with only a handful of exceptions, it was usually concerned with supplying local requirements. To take one example, the products of even the more successful coarse pottery industries rarely traveled more than fifty or sixty kilometers from the production sites, and this was probably true of other industries as well. The main exporting industries were those that provided the raw materials without which Roman civilization could not have survived, such as foodstuffs, metals, timber, and cloth; or, much more rarely, were monopolies, in the way that the papyrus (paper) industry was a state monopoly in Egypt. Other exporting industries were largely dependent on being able to send their goods along routes and in ships that existed primarily to carry food and raw materials. It was for this reason that so many of the industrial centers of the Roman world lay on the coast or by one of the great navigable rivers.

The advantages of Roman manufacturers were immense when compared with those of their predecessors and successors. They worked within an efficient financial system, often in periods of prolonged peace when trade could flourish unhindered; they traded in what was probably the largest common market the world had ever seen, with only a few minor taxes to pay on crossing boundaries between provinces (i.e., between different territories of the Roman Empire); communications by both land and sea were better than they had ever been before or were to be again for many centuries; and they sold their goods in a world in which there was immense public and private wealth.

When all these factors are considered, it is necessary to ask why industry only expanded sufficiently to fill the increased demand rather than transforming itself and creating a true industrial revolution. The question has no certain answer, but it is possible to define a number of factors that would have had to be overcome if such a revolution were to have occurred. Perhaps the key one is the fact that industry was socially unacceptable to the upper classes, who effectively controlled the resources of the Empire. Only agriculture was a suitable occupation for them, and while they were not unwilling to obtain their share of the profits of trade and industry, they could not become openly involved in it.

Without their direct interest, change was far less likely, if not impossible. The Roman legal system made it difficult to create the type of limited-liability company with a multitude of shareholders (different people, each of whom owns a part of the company), which has proved particularly well suited to financing industrial concerns, and which ensures continuity through more than one generation. In consequence, most Roman industry remained in the hands of individuals or small groups. Raising funds to start an industrial business was usually a personal matter, for although banks existed, the main source of loans was wealthy individuals. Nor was industry a fashionable field of investment; land was always preferred.

Finally, there was an almost total lack of technical innovation and invention. The industries of the fourth century were those that had existed for centuries, and they were still using the same techniques and processes. What could happen when a revolutionary technique was applied was shown by the explosive growth of the glass industry after the invention of glassblowing, but this stands almost in isolation in all the centuries of Roman industrialization. The result was that in most industries, production could be increased only by multiplying the number of units produced, and not by changing to different processes that were better suited to large-scale production. Undoubtedly, the greatest such limitation was the almost total failure to utilize any sources of power other than those provided by animals and humans.

73-2: Origins of Life on Earth

The first experimental demonstration that complex molecules could have evolved naturally from simpler ingredients found on the primitive Earth came in 1953. Scientists Harold Urey and Stanley Miller took a mixture of the materials thought to be present on Earth long ago - water, methane, carbon dioxide, and ammonia - and energized it by passing an electrical discharge through it, simulating the effects of lightning. After a few days, they found their mixture contained many of the same amino acids found today in all living things on Earth. About a decade later, scientists succeeded in constructing nucleotide bases, carbon-based molecules that are the building blocks of life as we know it, in similar fashion.

None of these experiments has ever produced a living organism, and the molecules created are far less complex than a strand of DNA, which contains the nucleic acids that carry genetic information in cells and are associated with the formation of proteins. Nevertheless, they demonstrate conclusively that "biological" molecules can be synthesized by strictly nonbiological means, using raw materials available on the early Earth. More advanced experiments have fashioned protein-like blobs (tiny drops) that behave to some extent like true biological cells. They resist dissolution in water and tend to cluster into small droplets called microspheres -- a little like oil globules floating on the surface of water. The walls of these laboratory-made droplets permit the inward passage of small molecules, which then combine within the droplet to produce more complex molecules too large to pass back out through the walls. As the droplets "grow," they tend to "reproduce," forming smaller droplets.

Can we consider these protein-like microspheres to be alive? Almost certainly not. The microspheres contain many of the basic ingredients needed to form life, but are not life itself. They lack the hereditary molecule DNA. However, they do have similarities to ancient cells found in the fossil record. Thus, while no actual living cells have yet been created from scratch, many biochemists feel that many key steps in chemical evolution - the chain of events leading from simple nonbiological molecules to the building blocks of life itself - have now been demonstrated.

Recently, a dissenting view has emerged. Some scientists argue that there may not have been sufficient energy available to power the chemical reactions in Earth's primitive atmosphere, and there may not have been enough raw material for the reactions to occur at a significant rate in any case. Instead, they suggest, most of the organic material that combined to form the first living cells was produced in interstellar space and subsequently arrived on Earth in the form of interplanetary dust and meteors that did not burn up during their descent through the atmosphere.

To test this hypothesis, NASA researchers carried out their own version of the Urey-Miller experiment, exposing an icy mixture of water, methanol, ammonia, and carbon monoxide - representative of many interstellar grains - to ultraviolet radiation to simulate the energy from a nearby newborn star. When they later placed the irradiated ice in water and examined the results, they found that it had formed droplets, surrounded by membranes, containing complex organic molecules. No amino acids, proteins, or DNA were found in the mix, but the results, repeated numerous times, clearly show that even the harsh, cold vacuum of interstellar space can be a suitable medium for the formation of complex molecules and primitive cellular structures.

Interstellar molecular clouds are known to contain many complex molecules, and large amounts of organic material were detected on comets Halley and Hale-Bopp when they last visited the inner solar system. Many scientists think cometary impacts early in our planet's history were responsible for most of Earth's water. Perhaps those comets carried complex molecules too. In addition, a small fraction of meteorites that survive the plunge to Earth's surface have been shown to contain organic compounds. Such discoveries strongly suggest that complex molecules can form in an interplanetary or interstellar environment and could have reached Earth's surface unharmed after their fiery descent. The moderately large molecules found in meteorites and in interstellar clouds are our only evidence that chemical evolution has occurred elsewhere in the universe. However, most researchers regard this organic matter as prebiotic - matter that could eventually lead to life, but that has not yet done so. Nevertheless, the hypothesis that organic matter is constantly raining down on Earth from space in the form of interplanetary debris is quite plausible.

74-1: Towns in the High Middle Ages

Europe was fundamentally an agricultural society in the High Middle Ages (1000-1350). Powerful local lords controlled the countryside, and social and economic relationships were based on the realities of rural life; however, towns did exist during this period. One may call them towns because even though they varied enormously in numbers of inhabitants, and some were smaller than large villages in this respect, they shared certain characteristics that set them off from rural settlements.

Relatively speaking, there was a greater density of population in towns than villages, although the line separating the two is necessarily arbitrary. Relatively, too, there was a more highly developed specialization and greater diversity of labor, especially - though not exclusively - of craft labor, in urban settlements. To be called a town, a settlement with a relatively high density of population and specialization of labor also requires an economy based on money as opposed to barter (the simple exchange of goods) and one in which a sizeable proportion of income, again speaking relatively, comes from trade. The presence of a regulated periodic market (meeting every week, for example) did not define a town, for many a village had one, too. But a town without a periodic market is unthinkable for the Middle Ages, which is not the case for a village. Finally, towns were distinguished, if not fully distinct, from large villages by a concentration of diverse monumental buildings - large churches, bell towers, warehouses, permanent market halls, hospitals, town halls, and so on - although not every town had the full array of such buildings even by the end of the Middle Ages.

Some scholars would supplement the foregoing list of characteristics, for towns often had mints for making money, special legal status and modes of landholding for the inhabitants, as well as autonomous criminal and civil court systems. Because what distinguished towns from villages was not so much size as the density of economic and social activities, one writer has argued that an important criterion for a town is the existence of traffic jams - the hustle and bustle of oxcarts; long lines of wagons bringing fruits and vegetables, raw materials, and finished goods to markets; and the parade of men and women coming to shop, visit, or attend meetings.

Towns in Europe around the year 1000 were few in number and very small in size, yet by the early fourteenth century, the number and size of urban settlements had increased enormously. Central Europe, which had only the thinnest sprinkling of settlements that might be deemed urban in the year 1000, witnessed the creation of 1,500 new towns from the eleventh century up until 1250 and 1,500 more in the 50 years or so after that. The Rhine river valley, which had no more than eight towns well into the twelfth century, boasted more than 50 in the thirteenth. In southern Europe there was more continuity from earlier settlement patterns: growth came less from the creation of new towns than from the flow of immigrants to old settlements.

Most of the early eleventh-century towns were weak in relation to the lords that ruled the countryside. In the majority of cases in the north, the lords controlled or owned the towns even though they usually did not live in them. Within the towns, the agents of the lords' authority contended with bishops and other church officials for domination. The urban commercial element was ordinarily the weakest of the three contenders for power, but the relative weight of each party would change dramatically in favor of the merchants over the course of the High Middle Ages. In southern Europe (Italy), where traditions of urban oligarchic domination (control by a small group) had not been completely erased, merchants and senior craftsmen had a stronger say in town government. But here, too, there was a three-way struggle for power among the commercial oligarchy, the church, and the lay lords, who more often than in northern Europe actually had their principal residences in the towns. The outcome of these struggles would vary from region to region over time. Throughout Italy, but particularly in the northern region, these struggles would dominate the political culture of the entire period from 1000 to 1350.

74-2: Ice Age Sculpture

For many years, scientists debated whether prehistoric humans in Central Europe lived alongside the mammoth and whether they hunted it with their stone tools. The mammoth, an extinct species of the elephant family, roamed across Europe during Earth's last ice age - a period during which large parts of Earth were covered by ice. Protected from the cold by its hairy coat, it survived even the coldest period some 26,500 years ago, when Earth's ice cover was most extensive. Archeologists did not initially believe that humans could live in this inhospitable climate. However, in the late nineteenth century, the remains of ancient human settlements that included numerous burned mammoth bones were found at the Moravian Gate, a mountain pass in what is now the Czech Republic. This suggested that humans survived even in frozen regions near the margins of the ice sheets by using mammoths for food and for raw materials used in clothing, blankets, and shelter. These ice-age people left behind numerous small sculptures of mammoths and other animals, formed of ceramic or carved out of stone, bone, or tusks. While these works usually show animals in still positions, there are some that depict animals in motion.

One such animated sculpture is the Předmostí mammoth. The fragments from which it is restored were found by geologist Martin Kříž in 1895 when he took over the excavations started by mathematician Karel Maska in 1882 in the massive deposits of loess (accumulated windblown soil) on the Bečva River in Moravia. In the intervening period, Maska had exposed an area of about 200 square meters and found over 30,000 artifacts, human burials, and more than 1,000 kilograms of animal bones, most of which were those of mammoths sometimes discovered in organized piles. It was also Maska who joined the broken pieces of ivory to reveal the sculpture that was a unique and important find at that time. Made on a large, flat segment from a tusk of considerable size, the realism of the animal, obviously observed in life by the sculptor, immediately disproved the opinion of the Danish archaeologist Japetus Steenstrup, who had visited in 1888 and suggested that the occupants of the site had arrived much later than the mammoths whose bodies they had found frozen in the permafrost and utilized. The indisputably ancient context of the sculpture found among mammoth bones and stone tools was used to further confirm the great age of cave art when it was compared to mammoths known from the walls of caves such as Les Combarelles and Font de Gaume in southwest France, the authenticity of which had only recently been accepted after many years of doubt. However, Steenstrup's view that the massive accumulation of mammoth remains at Předmostí and

other Moravian sites was not due to hunting prevailed and still provokes questions about how we might interpret the sculpture.

Arguments against the hunting of mammoths in Moravia surfaced in the twentieth century when Wolfgang Soergel and later, Olga Soffer, suggested that humans had exploited the remains of mammoths that had died natural deaths rather than hunting such potentially dangerous prey. Herds of animals migrating through the Moravian Gate mountain pass stopped at mineral licks (exposed mineral deposits), which provided them with the salts essential to the diet, particularly of young animals and pregnant females during late spring-early summer, but weak, tired animals would also die there. Humans just moved in to take advantage of them. This picture is contested by Martin Oliva, who reiterates the archaeological evidence for hunting combined with some scavenging for available food and, in so doing, proposes that culture as well as nature made the mammoth important to society in ways over and above bare survival. He points out that some humanly constructed accumulations of mammoth remains discovered within these sites have no apparent utilitarian function and notes that mammoth bones and teeth were placed in human burials. Such behavioral gestures, now incomprehensible, may have been influenced by engagement with and dependence on the animal which consequently acquired special significance. Oliva also argues that people do not avoid hunting dangerous animals but take them on for the thrill of the competition in which they acquire social prestige and status.

75-1: The Cost of Venom

Venom, a poisonous substance that is produced by some animals and injected into an enemy or prey mainly by biting or stinging, is clearly an advantage for defense or hunting. But it is an expensive resource. Replenishing empty venom glands can greatly diminish an animal's energy supply. Even though the venom supply in snakes makes up only a tiny fraction of their weight, refilling venom glands after use can increase a snake's metabolic (energy use) rate by more than 20 percent. The cost of using venom is not restricted to the direct metabolic cost of replenishing it, however. Spending venom can result in temporary disarmament, so that, for instance, a predatory spider lacking its venom becomes defenseless prey. The speed of refueling venom glands varies considerably between groups. It can take theraphosid spiders - a group of mostly large and hairy spiders popularly known as tarantulas or bird-eating spiders - as long as 85 days to finish reloading their glands, while the tiny caterpillar-killing parasitoid wasp *Bracon brevicornis* can reload its glands in less than three hours. The North American centipede regenerates about 85 percent of its venom volume within two days.

The great cost of venom, direct or indirect, is also indicated by the phenomenon of "venom metering" (monitoring the amount of venom used). Snakes can control the amount of venom they release so they do not use more than is necessary. A larger, more dangerous, or more agile prey receives a larger dose of venom to ensure it is subdued effectively. Although venom metering has been poorly studied in spiders, researchers know that they are also able to make decisions about when and how much venom to deliver. The Central American hunting spider *Cupiennius salei* injects more venom into larger prey. When dealing with prey that have body armor, such as beetles, and the armor prevents the spider from injecting venom into an optimal location, the spider also increases the dose. Remarkably, this spider can also distinguish prey that are more or less sensitive to its venom (crickets and cockroaches, respectively) purely on the basis of smell, and choose the more vulnerable prey when its venom glands are depleted.

Venom is not just metered during prey capture. Snakes, scorpions, and spiders also show defensive venom metering. Females of the western black widow spider, for example, can adjust the amount of venom in their defensive bites depending on the threat level. Squeezing the spider's body provokes her to deliver almost twice the dose of venom than if her leg is pinched. Even so, more than half of her defensive bites contain no detectable venom at all. The defensive bites of some snakes and the defensive stings of scorpions

can likewise be less costly. The South African thick-tailed scorpion initially stings predators with a transparent pre venom that is poor in proteins but rich in pain-inducing potassium ions. The scorpion releases its milky venom, which is more protein-rich and toxic - but also more expensive to produce - only when its pre venom fails to do the job.

The cost of venom does not only affect how venoms evolve: it can also cause a loss of toxins. Some specialist venomous predators that have evolved to feed on only a limited number of species have streamlined their venoms by losing toxin types that are no longer useful. This removes the cost of producing obsolete toxins. Sea snakes, for example, have relatively simple venoms compared to their terrestrial relatives, with the venom consisting of only a few neurotoxins that effectively immobilize their fish prey. In a fascinating case of convergent evolution (in which species not closely related independently evolve similar traits), the venoms of sea kraits, which are related to sea snakes but independently adopted a marine fish-eating lifestyle, have undergone the same streamlining. In fact, the venom of sea kraits is so similar to that of sea snakes that it can be effectively neutralized by sea snake antivenom. Venom is such an expensive commodity that it is discarded if its ecological importance disappears. When the marbled sea snake switched to a diet consisting exclusively of fish eggs, it did not simply lose a few toxin types. It lost its venom, fangs, and even its venom gland.

75-2: The Prokaryotic Cell

Of all the types of cells revealed by the microscope, bacteria have the simplest structure and come closest to showing us life stripped down to its essentials. Indeed, bacteria contain essentially no organelles (recognizable substructures) - not even a nucleus to hold their DNA. This property - the presence or absence of a nucleus - is used as the basis for a simple but fundamental classification of all living things. Organisms whose cells have a nucleus are called eukaryotes (from the Greek words eu, meaning "well" or "truly", and karyon, a "kernel" or "nucleus"). Organisms whose cells do not have a nucleus are called prokaryotes (from pro, meaning "before"). The terms "bacterium" and "prokaryote" are often used interchangeably although we shall see that the category of prokaryotes also includes another class of cells so remotely related to ordinary bacteria that they are given a separate name.

Bacteria are typically spherical, rod-like, or corkscrew-shaped, and small - just a few micrometers long. They often have a tough protective coat, called a cell wall. In the electron microscope, the cell interior typically appears as a matrix of varying texture without any obvious organized internal structure. The cells reproduce quickly by dividing in two. Under optimum conditions, when food is plentiful, a prokaryotic cell can duplicate itself in as little as 20 minutes. In less than 11 hours, by repeated divisions, a single prokaryote can give rise to 5 billion progeny. Thanks to their large numbers, rapid growth rates, and ability to exchange bits of genetic material by a process akin to sex, populations of prokaryotic cells can evolve fast, rapidly acquiring an ability to use a new food source or to resist being killed by a new antibiotic.

Most prokaryotes live as single-celled organisms, although some join together to form chains, clusters, or other organized multicellular structures. In shape and structure, prokaryotes may seem simple and limited, but in terms of chemistry they are the most diverse and inventive class of cells. These creatures exploit an enormous range of habitats, from hot puddles of volcanic mud to the interiors of other living cells, and they vastly outnumber other living organisms on Earth. Some are aerobic, using oxygen to oxidize food molecules; some are strictly anaerobic and are killed by the slightest exposure to oxygen.

Virtually any organic material, from wood to petroleum, can be used as food by one sort of bacterium or another. Still more remarkable, some prokaryotes can live entirely on inorganic substances: they get their carbon from CO₂ in the atmosphere, their nitrogen from atmospheric N₂, and their oxygen, hydrogen, sulfur, and phosphorus from air, water, and inorganic minerals. Some of these prokaryotic cells, like plant cells, perform photosynthesis, getting the energy they need for biosynthesis from sunlight; others derive energy from the chemical reactivity of inorganic substances in the environment. In either case, such prokaryotes play a unique and fundamental part in the economy of life on Earth: other living things depend on the organic compounds that these cells generate from inorganic materials.

Traditionally, all prokaryotes have been classified together in one large group. But molecular studies reveal that there is a gulf within the class of prokaryotes, dividing it into two distinct domains, called the eubacteria (or simply bacteria) and the archaea. Remarkably, at a molecular level, the members of these two domains differ as much from one another as either does from the eukaryotes. Most of the prokaryotes familiar from everyday life - the species that live in the soil or make us ill - are eubacteria. Archaea are not only found in these habitats, but also in environments hostile to most other cells: there are species that live in concentrated brine, in hot acid volcanic springs, in the airless depths of marine sediments, in the sludge of sewage treatment plants, in pools beneath the frozen surface of Antarctica, and in the acidic, oxygen-free environment of a cow's stomach, where they break down cellulose and generate methane gas. Many of these environments resemble the harsh conditions that must have existed on the primitive Earth, where living things first evolved, before the atmosphere became rich in oxygen.



76-1: Strengths of the Byzantine Empire

In 330 A.D., Constantinople (modern-day Istanbul) became the new capital of the Roman Empire. Historians traditionally consider the destruction of the city of Rome by foreign invaders in 476 A.D. to mark the end of the Roman Empire. However, Constantinople continued to be the capital of another large empire, called the Byzantine Empire, for many centuries after the fall of Rome. By the seventh century, the Byzantine Empire faced enormous challenges. Not only did it have to fight off frequent attacks from neighboring peoples, there was great instability internally as well with many emperors coming and going. And yet the empire survived.

What gave it its extraordinary resilience was partly the strength of its deeply rooted administrative system and partly its geographical position. By the mid-seventh century the territory was divided into a number of administrative units or provinces known as "themes," each with its own military detachment under its own commander. The soldiers were allotted land and thus had a real incentive to protect their home district. The land grants were large enough to support the soldiers' families as well as tenants and hired laborers. In this way at least some of the financial burden of the army was met by its own labor. The military domination of the themes also led to a shrinking of the civil service (the class of professional government employees), which saved further costs. While the system had many advantages, not least in providing a degree of stability spread across the empire and making it difficult for any one commander to assume control of the whole army, there was understandable reluctance among the sedentary forces to respond to the emperor's demands if this meant fighting far from home. The problem was to some extent countered by the creation of six army units stationed in and around the capital. Although the soldiers in these units, too, were landholders, they were trained to be a mobile force capable of being deployed quickly either as a field army or in defense of the emperor and the city.

The other great strength of the Byzantine Empire was its unique geographical position sitting at the crossroads linking Asia with Europe and the Mediterranean with the Black Sea. While there can be little doubt that the volume of trade overall diminished and the number of coins in circulation fell, the large and moderately wealthy urban populations still required a basic range of commodities as well as luxuries when they could afford them. One marketing innovation of this period was the provision of provincial warehouses. Their function seems to have been to supply arms and equipment for soldiers to buy using the produce of their farms. The goods brought in from the countryside could then be further sold into the regional market system.

At the elite level two of the most sought-after products were silk and purple dye made from murex, a mollusk found in the eastern Mediterranean. Silken garments, particularly those dyed purple, were the privilege of the elite and were used extensively in the royal court and by religious leaders in the Church throughout the Roman empire, both east and west. As early as the sixth century, the emperor Justinian had made silk production the sole right of the Byzantine Empire. There were huge profits to be made, not least in meeting the demands of the western Roman Church. Silks and the rare purple garment could also be used as diplomatic gifts. In 705 Justinian II rewarded the Bulgar ruler with gifts of silk cloth and purple leather for helping him to regain the throne. That he also licensed him to trade in controlled Byzantine goods is a fair indication that trade was still essential to the fragile Byzantine state.

That said, there can be no doubt that the empire suffered a dramatic economic contraction during the seventh and eighth centuries, and the population declined significantly, partly as the result of a declining birth rate in times of stress and partly from repeated outbreaks of disease, which had begun in the sixth century and returned many times until the middle of the eighth century. The declining population and economic downturn were responsible for many changes. Once-great towns like Athens shrank to a half or a quarter of their original size, but nonetheless urban life continued.

76-2: The Story of Film Editing

Many filmmakers regard editing as the single most important creative step in determining the look and shape of the finished film. A good editor can save a film that has been directed in a mediocre fashion, and poor editing can damage the work of even the finest director.

Editing is the work of joining shots (scenes recorded by the camera, usually on film) to assemble the finished film. The editor selects the best shots from the large amount of film footage (raw, unedited film) provided by the director and the cinematographer, assembles these in order, and connects them using a variety of optical transitions. In theory, the process of editing begins with the completion of filming, or cinematography. In practice, however, the editor may begin consultations with the producer and director and may even begin cutting the film, taking scenes and putting them together, while principal filming is being completed. Most editors, however, will not watch the process of filming or view the locations where the film is shot. This allows them to view the footage unhampered by knowledge about the actual conditions that existed in front of the camera and to visualize with greater freedom various ways of combining the shots.

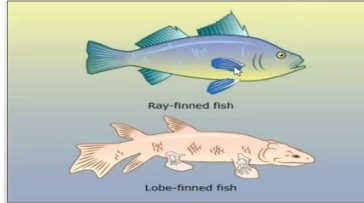
Over the years, the basics of editing have remained relatively constant. The first task is to assemble a first version or rough cut, which is done by eliminating all of the unusable footage containing technical or performance errors. These may include out-of-focus shots or shots containing unstable camera movement, lines where an actor has made mistakes in delivery, inaudible sound recordings, or lighting problems. Once all of this footage has been removed, the editor then assembles the remaining footage in a narrative sequence. This rough assembly will be pruned, refined, and polished to yield the final cut. The final cut is the completed product of an editor's work. It includes the complete assembly and timings of all shots in the film's finished form. It is in going from the rough cut to the final cut that the real art and magic in editing lies.

With the introduction of computers, editors began to use a nonlinear, computer-based editing system. Film footage was converted to videotape, which was then digitized and stored on computer disk, giving an editor instantaneous access to any frame or single image, shot, or edited sequence distributed anywhere in the existing footage. The editor would then decide which footage to work on by using notes that described the characteristics, strengths, and flaws of particular shots. Prior to the 1990s, when the film

industry adopted digital editing systems, editors worked directly with the actual film itself and had to search manually through all of the footage to find a desired shot or segment. This older approach was a linear system: the editor could only search for one shot at a time and had to do so by viewing footage sequentially, from beginning to end.

Digital systems made editing a much faster process, and the complex control they gave editors over the digitized footage helps explain why so many films had many more cuts and shot transitions than in earlier decades. Many shots were only a few frames long, less than a second of screen time.

While digital systems gave editors greater control over their footage and increased their ability to manipulate it in ever more elaborate ways, these systems had disadvantages. Unlike editors who worked in linear systems, those using digital systems did not view a film image but rather an electronic image on a small monitor screen, which was degraded in quality, with poor resolution. This could bias editors toward close-ups because they would look better on the monitor than long shots. Furthermore, because the monitor's image was a poor guide to the visual qualities of the actual film images, editors were forced to rely more heavily on their notes about the footage. It has been argued that editors using linear systems got to know their footage better because they had to manually search through all of it to find what they needed. The editor working on a digital system would not access footage that the notes had excluded. With recent advances in digital technology, even these disadvantages have been addressed.



77-1: From Water to Land

Fish were the earliest vertebrates (animals with backbones), able to live only in water. During the Devonian period (416-358 million years ago), a group of lobe-finned fish started a transition from water to land that ultimately led to the evolution of tetrapods—four-limbed animals including amphibians, reptiles, birds, and mammals (some tetrapods, like snakes, have later evolved to be limbless). Unlike ray-finned fish—the most common types of fish found today—lobe-finned fish have bones supporting some of their fins. This arrangement made it possible for fins to evolve into primitive limbs. It is likely that the main anatomical transition from fins to limbs was completed in an aquatic environment, where the limbs did not have to support the entire weight of the body due to the water's buoyancy (the upward force of water on a submerged object). The primitive fin-limbs probably first developed flexible joints, helping them to bend and push the fish forward as it crawled along in the shallows or negotiated thick waterweeds. The fin-limbs grew stronger as they developed simple knee, ankle, elbow, and wrist joints. Eventually, the primitive fish-tetrapod would have been able to crawl out of the water, where the lack of buoyancy would drive the evolution of limbs to support the weight of the animal's body. In turn, this would favor a stronger internal skeleton.

One such skeletal adaptation was the development of a rigid backbone, with sliding and overlapping processes to link adjacent vertebrae, while providing the necessary overall flexibility. In addition to securing support for the body, the hind limbs became the actual driving force for tetrapod locomotion. This required strong hip bones to anchor these rear limbs to the backbone. The fore limbs became the steering force, and so shoulder bones—which attach to the skull in fish—became detached in tetrapods, providing them with a flexible neck. Muscular changes accompanied these skeletal adaptations, promoting strong limb function and linking the hip and shoulder bones to the backbone.

Another requirement for the evolution of tetrapods, together with legs that work on land, is the ability to breathe air on land. Primitive air sacs existed in ray-finned and lobe-finned fish long before the first tetrapods appeared. In ray-finned fish, these air sacs were used

as swim bladders to maintain buoyancy. In lobe-finned fish of the Devonian period, however, the air sacs had already evolved into air-breathing lungs. Lobe-finned fish, such as living lungfish, breathe by gulping air into their mouths at the surface of the water. When they subsequently dive, the increasing water pressure bearing onto them forces the air in the mouth down into the lungs. The process reverses as the fish rises to the surface again, forcing stale air out of the lungs and mouth as the fish gulps in a fresh mouthful. Since this basic mechanism for respiration already existed in Devonian lobe-fins, the earliest tetrapods were already well on their way to the evolutionary stage of breathing on land.

The transition from fins to limbs was only one part of the evolutionary journey for terrestrial tetrapods. Many other anatomical changes must have taken place to help early tetrapods cope with life on the land. For example, the evolutionary change that resulted in a flexible neck increased the mobility of the head, helping them pick up food instead of grabbing items that passed by in the water. No one knows exactly which evolutionary forces prompted the transition from fins to limbs. The first tetrapods were probably still fully aquatic animals that lived in shallow waters. Some experts think they may have used their primitive limbs to stalk prey through the dense vegetation that lined the bottom of rivers and swamps. Others believe that environmental pressures, such as an increasingly dry climate, prompted the transition from water to land. As the heat dried up the shallow waters, primitive limbs may have helped the earliest tetrapods crawl between pools of water. Yet another hypothesis is that they were chased out of the water by bigger predatory fish. Any tetrapod that could survive out of water would have found itself in a new paradise, with an abundance of plant and insect foods and a notable absence of dangerous predators.

77-2: Economic Growth in Europe's Smaller Countries

Much has been written about the economic growth of England, France, and Germany in the nineteenth century, but the economies of a number of smaller European countries also performed impressively in this period, especially those of Belgium, Switzerland, the Scandinavian countries (Sweden, Norway, and Denmark), and, to a lesser degree, the Netherlands. Still, these countries suffered from various handicaps, the first being their geographic smallness: their home markets were narrow, and they were relatively poor in mineral resources. Belgium alone had coal (and in plenty), so it was a leader in the application of electricity to industry from the 1870s onward, but Switzerland, Sweden, and Norway had a rich waterpower potential. In the latter three countries, much of the territory was mountainous or arctic in climate and therefore scarcely populated. In the early nineteenth century, some of these countries were quite poor. The Netherlands, meanwhile, retained high per capita incomes but suffered from a difficult transition from its seventeenth-century golden age of commerce to industrial capitalism.

On the other hand, these small countries developed their educational systems quite early, and their workforces were literate and skilled, while wages (except in the Netherlands) were low. Thanks to the accumulation of valuable knowledge and skills among the workforce, the abundance of either local or foreign capital (money and other assets), and other advantages, such as a quiet political life—with only one short war for each country, except Sweden, which had none—they succeeded in overcoming the obstacles to industrialization. Most of them had an industrial tradition, but any growth of industry had to be driven by exports, and they became heavily involved in international trade. This was particularly true in the case of Switzerland, even though it alone was landlocked and surrounded by neighbors that put up barriers to trade; up to 1914, it was the world leader in exports of manufactured goods per capita. These countries benefited from the demand by the large advanced countries of Europe—especially Britain—for their manufactures and primary products, but they also exported to distant markets like Latin America. The adoption of free trade (allowing trading across boundaries without government interference) by Britain and the general liberalization of trade after 1860 were beneficial, but in order to export, these countries needed to be competitive, and they achieved efficiency by a strategy of specialization adapted to their resources.

Belgium experienced the most complete industrial revolution on the European continent, with an emphasis on heavy industry and large exports of coal and semifinished and finished metal products. In Switzerland, on the other hand, there were mainly light industries making high-quality goods: luxury cottons, watches, machinery (at first for the textile industries, with the manufacture of electrical equipment becoming important later), fine synthetic chemicals such as dyes and drugs (for which the Swiss held second rank, far behind Germany), and processed foods.

Sweden was rather a latecomer. For a long time, its exports were mainly of primary and semifinished products—bar iron, cut timber, paper pulp, and iron ore. From the 1880s onward, however, it developed a high-tech engineering industry, often on the base of Swedish inventions. This allowed it to become a large-scale exporter of various specialized products such as ball bearings, telephones, cream separators, and capital goods (machines for making products). Norway specialized in exports of fish, lumber, pulp, and paper, and in shipping services. Its industry was small, but electrochemistry was introduced in the 1900s with the help of French and Swedish capital.

Denmark's main export was initially grain, but in the face of competition its farmers reconverted to animal products: bacon, butter, cheese, and eggs—mainly for British breakfast tables. Danish agricultural productivity became the highest in Europe, making the country proof that industrialization is not the only road to wealth. The Netherlands also benefited from high-productivity agriculture and from its exports. Industry there, which had suffered in the wars of the early nineteenth century, had a slow but steady recovery, with large-scale industry emerging in the late nineteenth century.

Each country thus had its own specialized products and services. Their growth rates were fast. From 1870 to 1913, the Scandinavian countries outperformed the rest of Europe in the growth rates of individuals and achieved a spectacular convergence with the richest countries, especially in standards of living. Belgium and Switzerland had been "rich" for some time.

78-1: Monochrome Landscape Painting in Medieval Japan

During Japan's Muromachi period (A.D. 1336-1573), an important art form was monochrome painting (painting using one color), done in the manner evolved several centuries earlier by artists of the Song dynasty of China. Like their Japanese counterparts of this later age, the Song monochrome artists painted a variety of subjects, including Zen Buddhist religious officials, folk deities, and flowers and birds. But their primary interest lay in landscapes (known in Japanese as *sansui*, or pictures of mountains and water). And indeed, Song monochrome landscapes are among the more striking works of Chinese art. They are, moreover, perhaps the most supremely moving tributes of any people to the grandeur and vastness of nature.

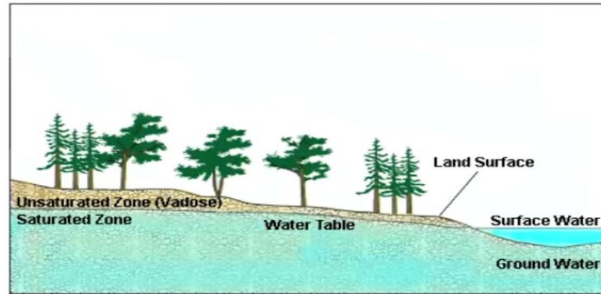
The Song masters did not attempt to reproduce nature as it really was; rather, they employed bold and even daring brushwork to capture in stylized outline misty scenes of forests, jagged cliffs, waterfalls, and awesome mountains (the most distant of which seem to be on the point of vanishing into space). Human figures sketched into these landscapes are usually ant-like in size. We see them, insignificant figures engulfed by the universe, as lone travelers moving slowly along mountain trails or as solitary beings seated in pavilion-like huts nestled on the sides of towering peaks. Song monochrome painting appealed particularly to the medieval Japanese because its medium of black ink was so compatible with the tastes of the age. In the first phase of painting in the Song manner during the fourteenth century, Japanese artists devoted themselves primarily to portrait and figure work, but in the fifteenth century they turned increasingly to landscapes.

Among the greatest masters of monochromatic ink work of the fifteenth century was Shubun. Although Shubun, who was active during the second quarter of the century, is reputed to have painted many different subjects in a variety of media, the only extant works attributed to him are landscapes, mostly on folding screens and sliding doors. A typical Shubun landscape is "visionary" in that it is a depiction, derived wholly from imagination, of a scene set in China. Like that of other Japanese artists of his time, Shubun's work is also impressionistic, since space is not clearly differentiated (that is, it is difficult to judge the relative depths of the various sections of a painting) and mountains, cliffs, and other pictorial elements often appear to be suspended or not properly integrated with the rest of the landscape. By contrast, Song-style landscapes by Chinese artists are notable for the care with which they are constructed: Foregrounds, middle

distances, and backgrounds are clearly distinguishable, and all parts of the picture "fit together" into a coherent reproduction, even if stylized, of a view of nature.

Thus, there appears to have been a fundamental difference in the approach to landscape between the Song-style Chinese artist and such Japanese painters as Shubun, a difference that seems to consist in the fact that the Chinese artist was as much concerned with presenting a philosophical idea as with representing visual qualities. The Chinese artist sought to portray in nature the kind of harmony and overall agreement of parts that ideally ought to prevail in human society. In other words, the Chinese artist tried to make a social statement; and the greater sense of structure and depth he could incorporate into his landscapes, the greater the philosophy of his work.

The Japanese, on the other hand, have never dealt with nature in their art in the universalistic sense of trying to discern any grand order or structure; much less have they tried to associate the ideal of order in human society with the harmonies of nature. Rather, they have most characteristically depicted nature - in their poetry, painting, and other arts - in specific distinct glimpses. The Chinese Song-style master may have admired a mountain, for example, for its enduring, fixed quality, but the typical Japanese artist (of the fifteenth century or any other age) has been more interested in a mountain for its changing aspects: for example, how it looks when covered with snow or when partly obscured by mists or clouds.



78-2: The vadose Zone and its Structure

When rain falls on dry land, it doesn't become part of the groundwater immediately. First, it must penetrate the unsaturated ground (the ground that is not soaked with water and still has space to absorb more water). The water table, or water level, marks the end of the unsaturated zone and the beginning of the saturated (water-soaked) zone. Reaching the water table may take considerable time, and not all of the incoming water necessarily gets as far as the water table; usually, after a temporary stay in the unsaturated zone, some of the water returns to the atmosphere.

Water in the unsaturated zone typically is moving, up and down and horizontally; it is seldom at rest. The unsaturated zone has two other names: sometimes it is called the zone of aeration, because its pores usually contain some air, and sometimes the vadose zone (from the Latin *vadosus*, meaning shallow). From this point on, we will use the term vadose zone; its brevity makes up for its being a jargon term, and it is more accurate, too; the vadose zone is not unsaturated always and everywhere, as we will see. Nor is it a single, homogeneous zone: in most places it has three distinct layers. The topmost layer is the soil; below it is the intermediate zone, consisting of the subsoil together with any solid bedrock that happens to lie between the subsoil and the water table; and at the bottom, in contact with the water table, is the capillary fringe, a layer into which groundwater soaks upward, like water into a sponge.

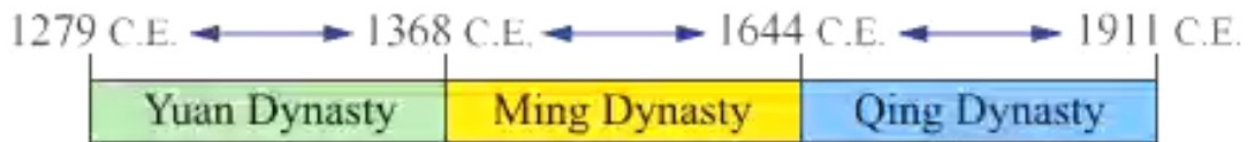
The soil is the most familiar, and certainly the most complex, of these layers. The distinction between soil and subsoil is that the former contains living organisms larger than bacteria, while the latter does not. The soil is the layer occupied by the roots of plants; it is also the home of myriads of mostly unnoticed soil organisms. This abundance of living material affects the soil's structure, its color, its chemistry, and the way water flows through it. Indeed, when water descends from soil into subsoil, it encounters entirely different conditions.

The thickness of the soil varies enormously: in some places it is no more than a thin layer of dust on bare rock; in others, it may consist of peat beds, areas made up of partly decomposed plant materials, 4 or 5 meters thick. Soil is continually subject to two opposing forces: dead organic material, accumulating ceaselessly, continually builds it up; at the same time never-ending erosion, as water flows through and over it, continually washes it away. If the soil thickness remains unchanged for long periods, as it does in many places, the reason is that these two processes balance each other.

The intermediate zone is the zone of loose mineral sediments, and sometimes bedrock, sandwiched between the soil and the capillary fringe. In places where the soil is deep or the water table shallow, there may be no intermediate zone; then all that lies between the surface and the capillary fringe is true soil.

The two upper layers of the vadose zone together form a temporary holding ground for water on its way to some other destination: some flows down to join the groundwater, some flows laterally to adjacent streams and lakes, and some is transported back into the atmosphere by the action of plants.

The lowermost of the three layers of the vadose zone is the capillary fringe. It is not necessarily different in texture from the material above and below it, but it is permanently wet, saturated with water sucked from the groundwater beneath. The height to which water will rise above the water table because of this spongelike suction depends on the texture of the material holding the water. The water moves up to fill the pores and interstices (spaces) of this material and the tiny channels that link them. It is constantly being pulled downward by gravity at the same time as it is being pulled upward by the suction force of surface tension; the level at which these opposing forces balance determines the thickness of the capillary fringe.



79-1: Chinese Salt Production and Government Power

By the year 2000 B.C.E., techniques had been developed in China for the production of salt. Archaeological evidence from northwestern China suggests that local inhabitants were by then using the technique of boiling water from saltwater lakes to extract its salt. In the region of Shansi and Shensi provinces in northern China, salt was sometimes even collected from evaporated lakebeds in the dry season. Inhabitants also learned to divert water from salt lakes into shallow holding pools, allowing the water to evaporate over a period of months. The natural salts would then be collected. Deposits of several million round-bottomed ceramic pots found in Sichuan Province in southwestern China suggest an organized system of salt processing by boiling during the same period. Scholars believe that the salt was then traded to the salt-poor regions of Hubei and Hunan. Similarly, evidence suggests that sometime prior to 2000 B.C.E., salt also began to be extracted directly from seawater. Over time, the coastal regions became the center of Chinese salt production, eventually providing roughly 80 percent of China's annual salt output. Given the increasing scope and value of salt in China, the control of its trade became a perpetual goal of the Chinese government and administration. The first formal attempt to establish the state's monopoly (exclusive control) over the production and distribution of salt goes back to the Eastern Zhou Dynasty (770-221 B.C.E.). The creation of this monopoly appealed to those in power because it provided the funds necessary to create a formidable standing army.

By the time of the Ming (C.E. 1368-1644) and Qing (C.E. 1644-1911) Dynasties, the government had developed intricate systems for the control of salt production and distribution. Beginning in 1370, the Ming government administered the salt monopoly through the "certificate system." Under this arrangement, each salt-producing household was required to provide an annual quota of salt to the government. In return, the household was provided with a fixed payment to cover wages and expenses. The salt was then transported to one of over two hundred official salt depots (storage centers), which were located within fifteen salt administration regions. Salt merchants, for their part, received certificates giving them the right to transport and market salt from the

government depots, but only after they had delivered grain to troops (military) serving on the northern frontier. This replaced the previous system, used by the Yuan Dynasty (C.E. 1279-1368), wherein merchants bid for the right to distribute salt. Thus, the Ming government guaranteed a supply of food for troops through the monopoly on salt. Salt was then delivered to local retailers and sold to the public. In the late Ming period, beginning in 1631, a new process known as the "syndicate system," which eventually created a hereditary class of salt merchants, placed salt distribution into the hands of a small group of families. This system lasted until 1831.

While both the certificate and the syndicate systems directed considerable amounts of money into the government treasury, they had significant drawbacks. First, the mandatory system of wholesale depots run by the government meant that salt had to be transported long distances even when there was a local market. Thus, salt might be produced in a region, transported to a depot far away, and then returned in the hands of licensed merchants to be sold to the public by local retailers. Such a process greatly elevated the cost of salt to the common buyer and created a substantial demand for smuggling and even the raiding of vehicles transporting salt. It is estimated that anywhere from one-quarter to one-half of the salt consumed during the Ming and Qing periods was manufactured and sold on the illegal black market outside of state control, despite the continued efforts of the state to restrict such activities. The state derived considerable revenue from the control of salt, but it did so at the expense of the workers, who not only had to provide labor to the government but also paid elevated prices for a daily necessity of life. In turn, the people resisted the government's control over salt by creating an underground (unofficial) economy in smuggled and stolen salt. In effect, the state encouraged the populace to undertake an illegal trade in salt.

79-2: Temporary Pools

Temporary pools are freshwater habitats that retain water for only three to four months of the year or even shorter periods: typically for some eight or nine months they are dry basins without surface water. At the latitude of Canada's southern Ontario, for example, the basin of a typical temporary pool fills with water when snow melts in March and April. Because temporary pools (by definition) have neither a surface inlet nor outlet, the water steadily recedes through evaporation and seepage into the surrounding soil, disappearing altogether from the surface in June or July. Most of the animal residents one sees in temporary pools are aquatic insects such as caddis flies, whose larvae (wingless, immature insects) are sometimes the most conspicuous animals in these communities.

At least two advantages to living in this type of habitat can be recognized. One involves predation. Fish are major predators of aquatic insects and other small freshwater creatures; however, fish, at least in temperate regions over much of North America, are unable to survive the dry phase, and hence are eliminated from temporary pools. Predatory insects such as aquatic beetles and bugs and dragonfly nymphs do live in temporary pools, but there are far fewer species, and therefore a less efficient predator population than in permanent waters. Moreover, populations of the invertebrate predators in temporary pools have to develop anew every year from colonizing adults or from drought-resistant eggs. All of this means that the impact of predation is reduced for aquatic animals that have adaptations to survive the dry phase of temporary pools.

A second advantage is concerned with nutrition and growth. Temporary pools, especially in wooded sites, are fuelled primarily by the energy and nutrients stored in fallen leaves and other decomposing plant detritus (nonliving matter that originates from plants). Furthermore, the dry phase of temporary pools usually supports a community of specialized herbaceous plants that can tolerate springtime flooding but grow rapidly in the moist, organically enriched soil after surface water disappears. These plants contribute to the detrital base every year. Although the food web is based mainly on detritus, the processes of decomposition in temporary pools are somewhat different from those in the detritus-based communities of streams and the shore zones of lakes. Detritus decomposes faster when exposed to air during dry periods and has a higher protein content upon flooding in spring than when submerged continuously in permanent water. The reason for this difference is that the terrestrial fungi causing the decomposition of plant detritus in the absence of surface water flourish where oxygen is not limited. In permanent waters, where detrital materials are covered by water, oxygen is not plentiful

and decomposition is brought about by aquatic fungi and bacteria, which do not grow as abundantly as terrestrial fungi. Colonizing terrestrial fungi contribute much of the protein of decomposing plant materials, and in feeding experiments limnephilid caddis fly larvae of temporary pools showed a preference for decaying leaves with higher protein levels. Consequently, rapid development, which is so important to animals in transient waters, would be enhanced by the protein-rich detritus as well as by the higher summer temperatures of small bodies of water.

These principles have led to some practical applications. An increase in productivity following a dry phase underlies the practice in wetland management of alternately draining and flooding wetlands. Aquatic birds such as ducks feed on larval caddis flies and other aquatic invertebrates and also on plants; the increased nutrient levels resulting from exposure of more of the bottom detritus to air and terrestrial fungi enhance productivity throughout the food web. By the seventeenth century, fish farmers in Europe had learned that periodically leaving ponds dry and empty of fish yielded higher productivity in controlled ponds of carp fish than did constant water levels. Through manipulation of water levels, the ponds were wintered as dry basins and periodically over a full year as well-practices that led to increased production of fish. In retrospect then, it appears that after centuries of experimentation to increase productivity in marshes and fish ponds, humans succeeded in duplicating the natural cycle of temporary pools.



80-1: The Medieval European Recovery

For centuries after the Roman Empire's fall in the fifth century C.E., economic activity in western Europe was negligible compared to that of civilizations to the east. Agricultural production suffered from repeated invasions, which seriously disrupted the economy and society. Moreover, the decay of urban centers resulted in decreased industrial production and trade. By the tenth century, however, political stability served as a foundation for economic recovery, and western Europeans began to participate indirectly in the larger trading world of the Eastern Hemisphere.

With the establishment of the Frankish kingdom and later the Carolingian Empire between the sixth and the ninth centuries in what is now part of France and Germany, the European center of gravity shifted from the Mediterranean to more northern lands, particularly France. But the agricultural tools and techniques inherited from the classical Mediterranean world did not transfer very well. In light, well-drained Mediterranean soils, cultivators used small wooden plows that basically broke the surface of the soil and disrupted weeds. This type of plow made little headway in the heavy, moist soils of the north. About the sixth century a more serviceable plow became available: a heavy tool equipped with iron tips that dug into the earth and turned the soil so as to aerate it more thoroughly and break up the root networks of weeds. The northern plow was more expensive than the light Mediterranean plow, and it required cultivators to harness much more energy to pull it through heavy soils. As a result, it was slow to come into wide use. Beginning about the eighth century, however, the heavy northern plow contributed to increased agricultural production.

As the heavy plow spread throughout western Europe during the ninth and tenth centuries, cultivators took several additional steps that increased agricultural production. Peasants cleared new lands for cultivation, and they also constructed watermills, which

enabled them to take advantage of a ready and renewable source of inanimate energy, thus freeing human and animal energy for other work. Moreover, they experimented with new systems of rotating crops (systems for alternating the particular crop grown in a particular field). As a result, peasants could cultivate land more intensively than before, as key soil nutrients were less likely to be depleted.

The agricultural surplus of medieval Europe was sufficient to sustain feudal lords and their subjects, but not substantial enough to support cities with large populations of artisans, craft workers, merchants, and professionals. Whereas cities had thrived and trade had linked all regions of the Roman Empire, early medieval Europe was almost an entirely rural society that engaged in little commerce. Manors and local communities produced most of the manufactured goods that they needed, including textiles and heavy tools, and they provided both the materials and the labor for construction and other large-scale projects. Towns were few and sparsely populated, and they served as economic centers for the areas immediately surrounding them rather than integrating the economic activities of distant regions.

By the tenth century, however, political stability and increased agricultural production had begun to stimulate trade and the development of urban settlements in western Europe. This revived trade took place in the Mediterranean, the North Sea, and the Baltic Sea. Merchants from the coastal cities of Italy regularly traded with Muslims in Sicily and Tunisia, who linked Europe indirectly with the larger Islamic world of communication and exchange. Even more prominent was trade in the North Sea and Baltic Sea, where Scandinavian seafarers played the role of trader as well as raider.

In C.E. 200, before the Roman Empire began to experience serious difficulties, the European population had stood at about thirty-six million. It fell sharply over the next four centuries, to thirty-one million in C.E. 400 and twenty-six million in C.E. 600 - a decline that reflected both the ravages of epidemic diseases and the unsettled conditions of the early Middle Ages. Then, gradually, the population recovered, edging up to twenty-nine million in C.E. 800 and thirty-two million in C.E. 900. By C.E. 1000 the European population once again amounted to thirty-six million - the level it had reached eight centuries earlier. By this time, western Europe was poised for extraordinary economic and demographic expansion that occurred in the late Middle Ages.

80-2: Abandoning Hunting and Gathering

For much of human history, people survived by hunting and gathering, but in the Neolithic period, starting around 10,000 B.C.E., this was gradually given up in favor of growing grains and other plants or domesticating animals (pastoralism). Why did early societies in so many parts of the world gradually abandon a way of life based on food gathering? Some theories assume that people were drawn to food production by its obvious advantages. For example, it has recently been suggested that people settled in what is now the Middle East so they could grow enough grains to ensure themselves a ready supply of beer. Beer drinking is frequently depicted in ancient Middle Eastern art and can be dated to as early as 3500 B.C.E.

However, most researchers today believe that climate change drove people to abandon hunting and gathering in favor of pastoralism and agriculture. So great was the global warming that ended the great Ice Age that geologists give the next era which began around 9000 B.C.E. a new name: the Holocene. Scientists have also found evidence that temperate lands were exceptionally warm between 6000 and 2000 B.C.E., the era when people in so many parts of the world adopted agriculture. The precise nature of the crisis probably varied. In the Middle East taking up food production may have been a response to shortages of wild food caused by a dry spell or population growth. Elsewhere, a warmer, wetter climate could have promoted rapid forest growth in former grasslands, reducing the supplies of game and wild grains.

Additional support for an ecological explanation comes from the fact that in many drier parts of the world, where wild food remained abundant, agriculture was not adopted. The inhabitants of Australia continued to rely exclusively on foraging until recent centuries, as did some peoples in all the other continents. Many Amerindians in the arid grasslands from Alaska to the Gulf of Mexico hunted bison, while in the Pacific Northwest others took up salmon fishing. Abundant supplies of fish, shellfish, and aquatic animals permitted food gatherers east of the Mississippi River in North America to become increasingly sedentary. In the equatorial rain forest and in the southern part of Africa, conditions favored retention of the older ways. The reindeer-based societies of northern Eurasia were also unaffected by the spread of farming.

Whatever the causes, the effects of the gradual adoption of food production in most parts of the world were momentous. A hundred thousand years ago, there were fewer than 2 million people, and their range was largely confined to the temperate and tropical regions of Africa and Eurasia. During the last glacial epoch, between 32,000 and 13,000 years ago, human populations may have fallen even lower. Then, as the glaciers retreated, people moved into new land and adopted agriculture. Their numbers gradually rose to 10 million by 5000 B.C.E. and mushroomed to between 50 million and 100 million by 1000 B.C.E. This increase brought important changes to social and cultural life.

The evidence that an ecological crisis may have driven people to food production has led researchers to reexamine the assumption that people in agricultural societies were better off than foragers. Modern studies demonstrate that food producers have to work much harder and for much longer periods than do food gatherers. In return for modest harvests, early farmers needed to put in long days of arduous labor clearing and cultivating the land. Pastoralists had to guard their herds from wild predators, guide them to fresh pastures, and tend to their many needs. There is also evidence that even though the food supply of early farmers was more secure than that of food-gathering peoples and pastoralists, the farmers' diet was less varied and nutritious. Skeletal remains show that on average Neolithic farmers were shorter than earlier food-gathering peoples and more likely to die at an earlier age from contagious diseases. People in permanent settlements were more exposed to diseases from water contaminated by human waste, to disease-bearing vermin and insects that infested their bodies and homes, and to new diseases that migrated from their domesticated animals (especially pigs and cattle).



81-1: Climate Change in Early Urban Centers

During the third millennium B.C.E., the first urban centers in Egypt, Mesopotamia, Iran, Central Asia, and South Asia flourished and grew in complexity and wealth in a wet and cool climate. This smooth development was sharply, if not universally, interrupted beginning around 2200 B.C.E. Both archaeological and written records agree that across Afro-Eurasia, most of the urban, rural, and pastoral societies underwent radical change. Those watered by major rivers were destabilized, while the settled communities on the highland plateaus virtually disappeared.

After a brief hiatus, some recovered, completely reorganized, and used new technologies to manage agriculture and water. The causes of this radical change have been the focus of much interest. After four decades of research by climate specialists working together with archaeologists, a consensus has emerged that climate change toward a warmer and drier environment contributed to this disruption. Whether this was caused solely by human activity, in particular agriculture on a large scale, or was also related to cosmic causes such as the rotation of Earth's axis away from the Sun, is still a hotly debated topic. It was likely a combination of factors.

The urban centers dependent on the three major river systems in Egypt, Mesopotamia, and the Indus Valley all experienced disruption. In Egypt, the hieroglyphic inscriptions tell us that the Nile no longer flooded over its banks to replenish the fields with fresh soil and with water for crops. Social and political chaos followed for more than a century. In southern Mesopotamia, rivers changed course, disrupting settlement patterns and taking fields out of cultivation. Other fields were poisoned by salts brought on through overcultivation and irrigation without fallow periods (periods when nothing is planted). Fierce competition for water and land put pressure on the central authority. To the east

and west, pastoralists faced with shrinking pasture for their flocks, pressed in on the river valleys, disrupting the already challenged social and political structure of the densely urban centers.

In northern Mesopotamia, the responses to the challenges of dryness were more varied. Some centers were able to weather the crisis by changing strategies of food production and distribution. Some fell victim to internal warfare, while others, on the rainfall margin, were abandoned. When the region was settled again, society was differently organized. The population did not drastically decrease, but rather it distributed across the landscape more evenly in smaller settlements that required less water and food. It appears that a similar solution was found by communities to the east on the Iranian plateau, where the inhabitants of the huge urban center of Shahr-i Sokhta abruptly left the city and settled in small communities across the oasis landscape.

The solutions found by people living in the cities of the Indus Valley also varied. Some cities, like Harappa, saw their population decrease rapidly. It seems that the bed of the river shifted, threatening the settlement and its hinterland. Another city, Mohenjo Daro, on the other hand, continued to be occupied for another several centuries, although the large civic structures fell out of use, replaced by more modest structures. And to the south, on the Gujarat Peninsula, the population and the number of settlements increased. They abandoned wheat as a crop, instead cultivating a kind of drought-enduring millet that originated in West Africa. Apparently, conditions there became even more hospitable, allowing farming and fishing communities to flourish well into the second millennium B.C.E.

The evidence for this widespread phenomenon of climate change at the end of the third millennium B.C.E. is complex and contradictory. This is not surprising because every culture and each community naturally had an individual response to environmental and other challenges. Those with perennial sources of freshwater were less threatened than those in marginal zones, where only a slight decrease in rainfall can mean failed crops and herds. Likewise, certain types of social and political institutions were resilient and introduced innovations that allowed them to adapt, while others were too rigid or shortsighted to find local solutions. A feature of human culture is its remarkable ability to adapt rapidly. When faced with challenges, resilience, creativity, and ingenuity lead to cultural innovation and change.



81-2: Commedia dell'Arte

In contrast to opera, which is chiefly a musical form, another equally popular type of entertainment in Renaissance Italy was pure theater. This is *commedia dell'arte*, which is Italian for "comedy of art" and means "play of professional artists" (as opposed to the kind of scholarly play performed by nonprofessionals).

Commedia companies usually consisted of ten performers-seven men and three women, though the number sometimes varied. They were traveling troupes, possibly the successors of Greek and Roman mimes, or clowns. Although there were instances when *commedia* performers staged serious forms of drama, they usually staged comedies, and through the years the term "*commedia dell'arte*" has come to be associated primarily with comedy.

Commedia thrived in Italy over a considerable period of time, from 1550 to 1750. It was not a written literary form but, rather, consisted of improvised presentations. Scenarios-short scripts without dialogue-were written by members of a company, and these scripts provided plot outlines; in other words, the performers had no set text but invented the words and actions as they went along. Over a thousand such scenarios survive from the Italian Renaissance. Using these outlines, actors would create the dialogue and would be expected to move the action along through improvisation.

The conventions of *commedia dell'arte* made the actors' task simpler than its improvisatory nature would suggest. For one thing, *commedia* actors played the same stock characters throughout most of their careers. Among the popular comic figures were a lecherous, miserly old Venetian, Pantalone; a foolish pedant who was always involved in his neighbor's affairs, Dottore; a cowardly, braggart soldier, Capitano; and sometimes foolish, sometimes sly servants known as zanni. Arlecchino, or Harlequin was the most popular of the comic servants. *Commedia* scenarios also included serious young lovers whose romances were often blocked by Pantalone and Dottore. Since the performers fused their own personalities with those of the characters, improvisation

was easier. And since the performers worked together, playing the same characters for extended periods of time, they became adept at creating comic interaction on the spur of the moment.

Improvisation was also made easier because all the commedia characters employed standard lazzi- repeated bits of physical comic activity or "business." Capitano, for example, would get entangled with his sword. (Twentieth-century film and television comics had their own lazzi; the great American film clowns-Laurel and Hardy and the Marx Brothers, for example-had pieces of physical business that they repeated in all their performances.)

In addition, commedia actors used conventional entrance and exit speeches as well as prepared musical duets. Surviving from the Renaissance are manuscripts put together by commedia actors that contain jokes, comic business, and repeated scenes and speeches. (These books were referred to by many different names, but the most common term for them was zibaldoni.)

Costuming also facilitated improvisation. Commedia characters all wore traditional costumes, such as Harlequin's patchwork jacket and Dottore's academic robe, so that audiences could recognize them immediately. Usually, each character would wear the same outfit, and its exaggerated details reflected his or her comic personality. A significant addition to Harlequin's costume was the slapstick, a wooden sword used in comic fight scenes. Sometimes the slapstick consisted of two thin slats of wood, one on top of the other; when a performer was struck with it, the effect was greatly exaggerated by the sound of the two pieces of wood smacking together. Today we use the term "slapstick" for comedies emphasizing physical horseplay or roughness. Masks, covering either the whole face or part of the face, were an essential element of commedia costumes. Pantalone's mask, for example, always had a huge hooked nose. The young lovers, however, did not wear masks.

Commedia dell'arte was enormously popular with audiences. One measure of its success was its popularity outside Italy, particularly in France.

The most successful commedia companies were often organized by families and chose names that were meant to characterize them: I Gelosi (The Zealous), I Fideli (The Faithful), I Confidenti (The Confident), and I Accesi (The Inspired). Most companies were based on a profit-sharing plan: a company's members shared in its profits as well as its

expenses and losses. Commedia performers were flexible; they could perform in town squares, in unused theater spaces, in the homes of wealthy merchants, or at royal courts.

82-1: Seasonal Bird Populations

Humans have hunted birds for their meat and harvested their eggs since earliest times, and therefore humans have an interest in understanding the seasonal movements of bird populations. Birds are regarded as resident when they spend all year in the same area, and migratory if they move seasonally between the breeding and nonbreeding areas. Most birds of the tropics and some temperate species (species that live in areas with moderate temperatures) are resident, but many mid-latitude species, and most which breed in high latitudes, are migratory. The most common reason for an annual migration is to allow the bird to occupy a part of the world that provides food at only one time of year. In continental North America and Eurasia, insects are available only in summer in northern forested regions, so insect-eating birds migrate south in winter. Other reasons for migration include the weather, which becomes too cold or too wet at certain times of year, and competition from other species for food or nest sites. Nearly all migratory species are from the northern hemisphere, partly because seasons there are more distinct than in the southern hemisphere. Current migration patterns of northern hemisphere birds have evolved only in the past 10,000 to 5,000 years, since the end of the last Ice Age. They are thought to have been preceded in the Pleistocene (2,500,000 to 11,700 years ago) by shorter migrations because the temperate climatic zone at that time was more compressed geographically.

Many of the waterfowl today breed in the tundra of northern Eurasia, northern Canada and Alaska, Iceland, and Greenland, where the vegetation and insect life can support the birds during what is quite a brief breeding season. Geese, which breed up to the edge of the Arctic, migrate south in winter to escape frozen water; they remain in the northern hemisphere, but in milder regions. For example, the brant (brent goose) in North America migrates from Alaska and northern Canada to the Pacific and the Atlantic coast, and in Eurasia it migrates from Siberia to the coasts around western Europe. Many of the plovers follow a similar strategy, breeding inland and north, but returning to more southerly latitudes and to coastal locations in winter. Birds of these families are most easily caught on migration or in winter. Migration in seabirds takes a different form. Many come to land to breed but spend the rest of their life cycle at sea.

Even birds that do not migrate change their social behavior in winter. Many of the species that congregate and breed colonially (in large groups) become solitary in winter and occupy a single territory. Others congregate in flocks at that time. They also become more mobile, and both are adaptations to less predictable food supplies: a tree that may be full of berries one winter may have few or none the next. Hunters often target flocks rather than solitary birds, so hunting of those species that congregate is often a winter activity. Some species may be both migratory and sedentary. The woodcock, for instance, has a sedentary British population, which is enhanced in winter by migrants from Scandinavia.

From the point of view of humans who live in one place, certain birds are present in one season only, either in summer to breed or in winter; or they are passage migrants, appearing for a short time in spring and autumn during temporary stops to feed and rest while en route between summer and winter territories. Migratory birds can be a reliable seasonal source of food for people because they return each year at a predictable time but then leave, thus preventing numbers from being reduced too much by predation. In the main, migration routes are fairly fixed. For instance, many ducks follow the shoreline, and other species follow major rivers. Some avoid crossing wide expanses of water, so they concentrate at narrow crossing points such as Gibraltar at the entrance to the Mediterranean and Falsterbo at the mouth of the Baltic. Migrating birds store fat reserves under the skin, an adaptation to the journey to be taken, so many are at their most desirable from the point of view of human hunters as they start the migration.

82-2: Hunting the American Bison

When nonnative settlers arrived on the flat, treeless North American plains in the nineteenth century, American bison—a large herd animal—thrived despite hunting by Native American peoples. Scholars once assumed that Native American cultural values and hunting practices were designed to limit demands on natural resources. Native Americans harvested only around half a million bison every year for domestic consumption (a sustainable figure, given that it was such a successful species). Almost all parts of the animal had value. Most organs could be used for food, skins for shelter and clothing, and bones for tools. The Blackfeet (a Native American tribe) obtained more than 100 specific items from the bison that sustained daily life. But in emphasizing Native American ingenuity in turning a variety of bison parts into useful products, scholars overlooked considerable evidence of waste and excessive killing.

In recent years, the role played by Native Americans in the near extinction of the bison has been more closely studied. Archaeological evidence has revealed that traditional methods of hunting, such as forcing bison off cliffs, could be remarkably wasteful. For instance, one such site in southern Colorado contained the bones of some 200 bison (males and females, adults and juveniles), but over 40 animals at the bottom of the pile had either not been touched or been only partly butchered. But by the late eighteenth century, the adoption of the horse and the rifle had made Plains Indians (Native Americans who lived in the plains) more effective and selective hunters. They preferred to kill female bison because their meat was juicier and their skins were easier to work. After 1700, almost all Plains Indians had become involved in commercialized hunting, exchanging bison tongues and skins for European trade goods and horses. Native Americans began to slaughter the bison for their tongues and skins alone, leaving the rest of the body to rot. Between the 1830s and the 1860s, as demand increased, they supplied firms such as John Jacob Astor's American Fur Company with over 100,000 bison robes annually. In combination with environmental factors, especially lack of rain, the Plains Indians' preference for killing females (thus limiting the herds' reproductive capacity), the competition for grazing on prairie grasses from their horses, and the pressures of hunting both for their own use and for the market began to speed the decline in bison numbers.

Contrary to the old belief, there were elements in Native American belief systems that were also destructive to the bison. The Plains Indians believed that when the great bison

herds dispersed during the winter months, they went to grasslands underground, reemerging every spring in inexhaustible swarms "like bees from a hive." If bison herds did not reappear in expected numbers, it was because many had not yet left their underground prairies. Such a notion undoubtedly discouraged the conservation of what was a diminishing resource, as Plains people could still hunt in enormous quantity, secure in the knowledge that there were always more animals grazing down below. Today, few historians would accept uncritically the idealized image of the environmentally friendly Native American who consciously pursued a lifestyle that left few traces on the land. But the pressure that Plains Indians exerted on the bison were slight in comparison with that exerted by Euro-American market hunters.

The bison population collapsed rapidly from the mid- nineteenth century. The demand for bison skins for the tanning industry, which turned animal skin into leather, was the primary cause. In 1850, when tanners still mostly used the skins of cows, tanning was already the fifth largest industry in the United States, and it grew in value by supplying markets at home and overseas with a variety of leather products--most importantly, the heavy belting that powered machinery in mills and factories. Demand outstripped the supply of cowskins in the United States, forcing tanners to import them from Argentina and elsewhere in Latin America. But in 1870 a process for transforming bison skin into tough, elastic leather was discovered by tanners in Philadelphia, making bison hide an ideal material for use as industrial belting. Thousands of Euro-American commercial hunters flooded onto the plains, killing around 2 million bison for their skins annually between 1870 and the early 1880s. At the same time, the extension of the railroads to the plains made transporting bison skins to tanneries both quick and inexpensive.

83-1: In the Darkest Depths

Although scientists in the early 1800s were unable to explore the deepest regions of the ocean, they knew that sunlight could not penetrate depths greater than 275 meters. Without sunlight, there can be no photosynthesis (the process by which plants convert sunlight into energy), and therefore, no plants or algae to serve as food for other organisms. Since ocean depths can exceed 10,000 meters, scientists hypothesized that the deepest areas of the ocean could not support life.

Throughout the nineteenth century, scientists exploring the ocean collected organisms from ever greater ocean depths. In an 1873 expedition, researchers aboard the ship HMS Challenger dragged an open-sided box suspended from the ship across the floor of the Atlantic Ocean. This box--known as a dredge--sample the sea floor in different parts of the ocean at depths of up to 4,57 meters. The scientists were astonished to discover nearly 5,000 previously unknown species. When it became clear that life flourished at depths beyond the penetration of light, scientists were forced to reject their earlier hypothesis that no life existed in the deep ocean waters. After discovering this rich abundance of deep-sea life, scientists were faced with the need to understand how it could exist. The lack of light suggested that deep-sea organisms were somehow sustained by energy that did not come from photosynthesis on the ocean floor. Scientists had observed that the surface waters of the ocean produced a steady descent of tiny particles that were produced by the death and decomposition of organisms living there. These particles are known as "marine snow". Large organisms such as whales also occasionally died and fell to the ocean floor. Scientists hypothesized that marine snow and the remains of large organisms provided the energy needed to sustain organisms in the depths of the ocean.

In the 1970s, scientists were finally able to send submersibles--small manned submarines--to take a first-hand look at the deepest ocean areas. Their discoveries were striking. They not only confirmed that much of the ocean floor supported living organisms but that areas near openings in the floor of the ocean, which later came to be known as hydrothermal vents, contained a great diversity of deep-sea species. Hydrothermal vents release plumes of hot water with high concentrations of sulfur compounds and other mineral nutrients. A tremendous diversity of species surrounded these hydrothermal vents. Indeed, the total amount of life at these depths rivaled that seen in some of the most diverse places on Earth. It became clear that the amount of energy contained in organic matter resting on the sea floor was not sufficient to support such a diverse and abundant set of life forms. That earlier hypothesis now had to be rejected.

How could so much life exist at the bottom of the ocean? That this life existed near the hydrothermal vents suggested that the vents were somehow responsible. Scientists had known for a long time that some species of bacteria could obtain their energy from chemicals rather than from the Sun. The bacteria use the energy in chemical bonds, combined with carbon dioxide (CO₂), to produce organic compounds--a process known as chemosynthesis--similar to the way that plants and algae use the energy of the Sun and CO₂ to produce organic compounds through photosynthesis. Based on this knowledge, scientists hypothesized that the hot vents, which release water with dissolved hydrogen sulfide gas and other chemicals, provided a source of energy for bacteria and that these bacteria could be consumed by the other organisms living around the vents.

After several years of investigations, scientists found that the immediate area around the hot vents contained a group of organisms known as tube worms. These animals have no digestive system, but possess specialized organs that house vast numbers of bacteria that live in a symbiotic (mutually beneficial) relationship with the tube worms. The tube worms capture the sulfide gases and CO₂ from the surrounding water and pass these compounds to the bacteria, which then use the sulfide gases and CO₂ to produce organic compounds. Some of these organic compounds are passed to the tube worms, which use them as food. These bacteria also represent a food source for many of the other animals that live near the vents. In turn, these bacteria-consuming animals can be consumed by larger animals.

83-2: Early Native American Cultures

In the seventeenth century, the gulf separating European colonists in North America from those who were native to America was defined not only by their material cultures, but also by how they viewed their relationship to the environment and how they defined social relations in their communities.

Regarding the soil as a resource to be exploited for human benefit, Europeans believed that land should be privately possessed. Individual ownership of property became a fundamental concept, and an extensive institutional apparatus grew up to support it. Fences symbolized private property, inheritance became the mechanism for transmitting land from one generation to another within the same family, and courts gained the power to settle property disputes. Property was the basis not only of sustenance, but also of independence, material wealth, political status, and personal identity. The social structure in Europe directly mirrored patterns of land ownership, with a land-wealthy elite at the apex of the social pyramid and a mass of individuals lacking property forming the broad base.

Native Americans also had concepts of property, and tribes recognized territorial boundaries. But they believed that land was invested with sacred qualities and should be held in common. Communal ownership sharply limited social layers in most Native American communities. Accustomed to wide disparities of wealth, Europeans often found this remarkable. Not all Europeans were wealthy, competitive individuals. The majority were peasants scratching a subsistence living from the soil, living in kin-centered villages with little contact with the outside world, and exchanging goods and labor through barter. But in Europe's urban centers, a wealth-conscious, ambitious individual who valued and sought wider choices and greater opportunities to enhance personal status was coming to the fore. In contrast, Native American traditions stressed the group rather than the individual. Holding land and other resources in common, Native American societies were usually more egalitarian, and their members were more concerned with personal courage than personal wealth.

European colonizers in North America also found the matrilineal (female-based) organization of many tribal societies odd and unfamiliar. Contrary to European practice, family membership among the Iroquois, for example, was determined through the female

line. A typical family consisted of an old woman, her daughters with their husbands and children, and her unmarried granddaughters and grandsons. When a son or grandson married, he moved from this female-headed household to one headed by the matriarch of his wife's family. Divorce was also the woman's prerogative. If she desired it, she merely set her husband's possessions outside the door to their dwelling. European women, with rare exceptions, were entirely excluded from political affairs. By contrast, in Native American villages, again to take the Iroquois example, designated men sat in a circle to deliberate and make decisions, but the senior women of the village stood behind them, lobbying and instructing. The village chiefs were male, but they were named to their positions by the elder women of their clans. If they moved too far from the will of the women who appointed them, these chiefs were removed.

The role of women in the tribal economy reinforced the sharing of power between male and female. Men were responsible for hunting, fishing, and clearing land, but women controlled the cultivation, harvest, and distribution of food. They were responsible for probably three-quarters of their family's nutritional needs. When the men were away on hunting expeditions, women directed village life.

In economic relations, Europeans and Native Americans differed in ways that sometimes led to misunderstanding and conflict. Over vast stretches of the continent, Native Americans had been involved in trading networks for centuries before Europeans arrived. This experience made it easy for them to engage in trade with arriving Europeans and incorporate new metal and glass trade items into their culture. But trade for Native American peoples was not only a way of acquiring desirable goods. It was also a way of preserving interdependence and equilibrium between individuals and communities. This purpose displayed itself in elaborate ceremonies of gift-giving that preceded the exchange of goods. For Europeans, trade was an economic matter with the additional benefit of building goodwill between two parties, but the social and spiritual functions of trade were sharply limited in comparison with Native Americans.

84-1: Habitability of Planets

Magnetic fields deflect (turn away) charged particles. Earth has a magnetic field generated by the movement of liquid in its core. One reason that Earth's magnetic field is considered important for Earth's habitability (ability to support life) is that it helps protect the planet from bombardment by cosmic rays. Cosmic rays are energetic charged particles that originate from space, either from the Sun or from farther away in the galaxy. Solar cosmic rays are relatively low-energy particles that are released in a stream of charged particles (the "solar wind") from the upper atmosphere of the Sun. Galactic cosmic rays are high-energy particles and gamma rays that originate from outside the solar system. Prolonged exposure to either of these types of radiation can create health risks, including cancer—a serious consideration for astronauts out in space.

Although Earth's magnetic field helps protect the planet from cosmic rays, that protection is only partial: the magnetic field barely deflects the most energetic cosmic rays that have energies of many gigaelectron volts. What then shields Earth's surface from high-energy cosmic rays? The answer is the atmosphere itself. Most cosmic rays, including those falling at the poles, are absorbed by the atmosphere at altitudes of 80 km or more. The high-energy ones create cascades of secondary particles, and some of these do indeed make it down to the surface. By the time they get there, however, they have lost most of their force, and so they account for less than 10 percent of the radiation exposure that an average person receives during the course of a year.

Planetary magnetic fields have another effect that may actually be more important in keeping a planet habitable. Earth's magnetic field prevents the solar wind from interacting directly with its atmosphere. (This statement is equivalent to saying that it protects Earth from solar cosmic rays.) Mars, which lacks an intrinsic magnetic field, does not have such protection, and consequently its atmosphere may have been sputtered away by the solar wind. "Sputtering" is the term used to describe the process whereby collisions between solar wind particles and electrically charged atoms in a planet's upper atmosphere can lead to loss of atmospheric gases. Based partly on this observation, scientists Peter Ward and Donald Brownlee suggest that planets that lack magnetic fields are unlikely to be habitable.

This argument, though, overlooks something obvious: Venus, which also has no intrinsic magnetic field, has a very dense atmosphere-100 times thicker than Earth's. Furthermore, Venus is closer to the Sun than is either Earth or Mars, and hence it is exposed to a denser, more energetic solar wind. If solar wind sputtering is so effective, then why didn't Venus lose its atmosphere? The answer probably has to do with planetary size. Venus is roughly the same mass as Earth, and so its upper atmosphere is less extended than that of Mars. That makes it harder for the solar wind to strip away atmospheric gases. Both Venus and Mars do have induced magnetic fields (fields generated in their upper atmospheres by their interaction with solar wind), and in Venus's case this induced field is evidently sufficient to protect its atmosphere. Mars's problem is the combination of its lack of an intrinsic magnetic field and its small size, which is clearly not conducive to planetary habitability.

This brings us to the issue of planetary size itself. Mars's small mass ($1/9$ of Earth's mass) also allowed the planet to cool more quickly. Without enough internal heat to sustain volcanism, it was unable to recycle carbonates (consisting of one carbon and three oxygen atoms) back into CO_2 (carbon dioxide, a gas that traps heat at the planet's surface). Mars appears to have already cooled off by about 3.8 billion years ago. No one knows exactly how big a planet needs to be to stay volcanically active for 4.5 billion years, as Earth has done, but one can guess that it needs to be $1/3$ of Earth's mass or more. A higher mass should also give a planet a better chance of sustaining a magnetic field because its core should take longer to solidify.

84-2: Early Metallurgy

European smiths (metalworkers) were working copper in the Balkans as early as 3500 B.C. In the Americas, copper working achieved a high degree of refinement in coastal Peru among the Moche and Chimu cultures, and also in western Mesoamerica. In North America, the Archaic peoples of Lake Superior exploited the native deposits of copper ore (rock containing copper) on the southern shores of the lake, and the metal was widely traded across the eastern continent and cold hammered into artifacts from Archaic (650-490 B.C.) to Woodlands (1000 B.C.-A.D. 1000) times.

The real explosion in copper metallurgy took place midway through the fourth millennium B.C., when southwest Asian smiths discovered that they could improve the properties of copper by alloying (combining) it with a second metal such as arsenic, lead, or tin. Perhaps the first alloys came about when smiths tried to produce different colors and textures in ornaments. But they soon realized the advantages of alloys that led to stronger, harder, and more easily worked artifacts. Most early bronze—an alloy of copper and tin—contains about 5 to 10 percent alloy material; 10 percent is optimum for hardness. An explosion in metallurgical technology occurred during the third millennium BC, perhaps in part resulting from the spread of writing. The use of tin alloying may have stimulated much trading activity, for the metal is relatively rare, especially in the eastern Mediterranean region. Bronze working was developed to a high pitch in northern China after 2000 B.C.

Smiths during the Bronze Age (3000-1200 B.C.) certainly knew about iron, but it was considered of little apparent use. They knew where to find iron ore (rock containing iron) and how to fashion iron objects by hammering and heating. But the crucial process in iron production is carburization, in which iron is converted into steel. The result is a much harder object, far tougher than bronze tools. When an iron object is carburized, it is heated in close contact with charcoal (a form of carbon) for a considerable period of time. The solubility of carbon in iron is very low at room temperature but increases dramatically at temperatures above 910°C, which could easily be achieved with charcoal and a good Bronze Age bellows—a device used to supply air. It was this technological development that led to the widespread adoption of iron technologies in the eastern Mediterranean area at least by 1000 B.C.

Iron tools are found occasionally in some sites as early as 3000 B.C., but such artifacts were rare until around 1200 B.C., when the first iron weapons in eastern Mediterranean tombs appeared. The new metal was slow to catch on, partly because of the difficulty of extracting it from its ore. Its widespread adoption may coincide with a period of disruption in eastern Mediterranean trade routes as a result of the collapse of several major kingdoms, including that of the Hittites, after 1200 B c Deprived of tin needed to make bronze, the metalworkers turned to a much more readily available substitute-iron. It was soon in use even for utilitarian tools and was first established on a large scale in continental Europe in the seventh century B c by people of the western and central European Hallstatt culture.

Iron ore is much more abundant in nature than copper ore is, being readily obtainable from surface rock formations and wetland deposits. Once its potential was realized, it became much more widely used, and stone and bronze were relegated to subsidiary, often ornamental, uses. It made available abundant supplies of tools with tough cutting edges for agriculture. Moreover, iron tools could be lighter than bronze ones, and they could hold a sharper edge. With iron tools, clearing forests became easier, and people achieved even greater mastery over their environment.

85-1: Essentials of Chinese Art

One important effect of recent research is that scholars now question ideas of essential, permanent characteristics of Chinese art. Many long-held conceptions turn out to be based on the tastes and values of ruling and literate (mostly male) elite patrons and audiences of certain periods. Often left out of such accounts are features of popular, religious, or craft arts and the cultural activities and values of women. There is also now an increasing awareness of the interactions and cross-borrowings between China and other cultures. But even while the ideas of essentialism are questioned, we can still ask about distinctiveness. In what ways were Chinese arts special and noteworthy within the broad history of world art? A few long-term characteristics stand out, although, as important as these culturally distinctive elements were in the history of Chinese art, they still leave out much of significance and interest.

China is unique among the great cultures of the world in the degree to which its civilization was identified with its craft products. For the ancient Greeks and Romans, China was Seres, "the land of silk." In early modern Europe, the word "China" was as likely to refer to porcelain tableware as to the tableware's country of origin. Indeed, Chinese productivity and technical excellence in many crafts were unrivaled for centuries or even millennia. Ceramic production and the carving of the hard stones known collectively as jade are part of the earliest horizons of Chinese cultures in the Neolithic period (ending 2000 B.C.E.), and the products of these activities have been made continuously in large numbers—sometimes approaching an industrial scale—down to the present. Chinese silks were prized luxury items for both domestic and international consumption for nearly two millennia. Chinese lacquerware factories employed an early version of mass production more than 2,000 years ago. Bronze vessel casting, gold and silver metalworking, wood-block printing of books and pictures, and hardwood furniture making are among many other media with long histories of large-scale, technically distinguished production. Far from being mere cultural embellishments, such craft industries were major economic and social forces.

Thus, the use of such terms as "craft arts," "craft skill," and "craft production" in the context of Chinese production refers to major traditions of highly accomplished technique and organization and should not convey the negative evaluations often implied by

distinctions between fine arts and crafts in the modern West. Critical traditions in China do indicate distinctions between literate arts—especially those such as calligraphy, painting, and architecture that acquired an accompanying theoretical, historical, and critical literature—and art forms that depended more on specialized skills and sometimes on laborious or cooperative production. Such distinctions were fluid over time, however, as antique ceramics were prized and inscribed by imperial collectors, ancient ritual bronze vessels attracted the interest of antiquarian scholars, and individual potters, carvers, and metalworkers achieved independent renown.

Ritual performances of respect for ancestors are a Chinese cultural feature with roots that extend at least as far back as the Shang dynasty (1766–1122 B.C.E.) and continue down to the present. Among the artistic products directly associated with ancestor veneration are monuments of tomb architecture and memorial sculpture, as well as a wide array of ritual and grave goods: bronze offering vessels, along with tomb furnishings of stone and ceramic sculpture, metalwork, lacquer, and jade. More generally, the importance of ancestral cult practices contributed to a strong historical consciousness in Chinese culture and to such artistic by-products as antiquarian tastes, retrospective calligraphy and painting styles, and an art-historical literature.

The Chinese written language has been a unifying cultural force since the Shang period, facilitating communication across temporal and geographical divides. At the time, writing was in some ways exclusionary, since it was limited to literate elites who were always a distinct minority of the population until the recent past. Nevertheless, writing has been a powerful vehicle for forging cultural identity, and the high status accorded the written text, at times verging on the divine or magical, lent prestige to the arts of calligraphy, inscribed monumental vertical stone slabs called stelae, printed and illustrated books, and paintings inscribed with written texts.

85-2: Atwater Food Energy Value System

The Atwater system of ascertaining energy values in foods was developed by Wilbur Olin Atwater, a college chemistry professor, around the end of the nineteenth century. Atwater knew that food contained three main substances the body uses for energy: protein, fat, and carbohydrate. Using a simple laboratory device called a bomb calorimeter, he recorded how much heat was released when typical proteins, fats, and carbohydrates were completely burned. He found that there was not a lot of variation among the different types of each item. For example, all proteins tended to produce a little more than four kilocalories of heat per gram (kcal/gram).

After that, Atwater needed to know two more things. First was how much of the major macronutrients—protein, fat, and carbohydrate—a food contains. Fat was easy, because unlike protein and carbohydrates, fat dissolves in ether (a liquid used to dissolve other substances). So Atwater chopped foods finely, shook them up with ether, and weighed how much material was dissolved in the ether. That gave him a food's fat content (or, more strictly, lipid content: lipids include both fats, which are solid at room temperature, and oils which are liquid). The same method is used today. Protein was harder to index because no test identifies proteins in general. However, Atwater knew that about 16 percent of the weight of an average protein was nitrogen. So, he found a way to measure the amount of nitrogen, which gave him the concentration of protein.

Carbohydrates were the hardest. There was no test then, nor is there now, for identifying the concentration of carbohydrates in general. But Atwater knew the main organic matter in foods was the three big items: protein, fat, and carbohydrate. He also knew how to calculate the total amount of organic matter. He simply burned the food completely, leaving only the mineral ash that did not burn and was therefore the inorganic part. Knowing how much organic matter the food contained and how much fat and protein it held, he obtained the amount of carbohydrate by subtraction: the weight of the carbohydrate was what was left when the weights of the fat, protein, and mineral ash had been subtracted from the total weight of the original food item.

Atwater was thus able to estimate the amount of protein, lipid, and carbohydrate in his food. The second piece of information he needed was how much of the food a person eats is digested, as opposed to being passed through the body unused. This required him

to analyze the feces (waste matter) of people who were eating precisely measured diets, which he duly did. He was then able to estimate, for each of the three nutrients, how much of what was eaten was also digested. Once again, he found that there was little variation within the categories of protein, fat, and carbohydrate, so he assumed the variation could be ignored.

The chemist now had what he wanted. He knew how much energy each of the three big types of macronutrient contained, how much of each macronutrient was present in the food, and how much of it was used in the body. Ignoring variation within each type of macronutrient, he proposed the convention that still dominates the food industry and government standards. By taking into account the proportion of the food that he found was not digested, which was rarely more than 10 percent, he claimed that on average proteins and carbohydrates each yield four kcal/gram, while lipids yield nine kcal/gram. These are known as Atwater's general factors.

This simple and convenient system forms the basis of the Atwater convention and is essentially what the United States Department of Agriculture's National Nutrient Database and McCance and Widdowson's *The Composition of Foods* use to produce their tables of nutrient composition. But nutritionists have long recognized important limitations in the Atwater convention, so it has been modified in various ways, such as by making the general factors more specific in 1955 when the Atwater specific-factor system was introduced to take advantage of a half century of nutritional biochemistry research. For example, the energy value of different types of protein is known to vary: egg protein produces 4.36 kcal/gram, whereas brown rice protein produces 3.41 kcal/gram. An exhaustive list of such variants has been compiled.

86-1: Egyptian Hieroglyphs

Sumer and Egypt were two of the world's earliest civilizations and the sites of two of the oldest known writing systems. Egyptian hieroglyphs came into existence early in the third millennium BCE, a little after Sumerian cuneiform, and were probably invented under the influence of the latter. There are many instances of early Sumer-Egyptian relations, but there is no direct evidence for the transfer of writing, and a credible argument can also be made for the independent development of writing in Egypt. Like cuneiform, hieroglyphs consisted of pictograms-pictorial symbols that initially stood for objects-some of which over time became phonetic signs standing for sounds. Unlike the symbols in Sumerian writing, which eventually lost their resemblance to the original pictograms, Egyptian hieroglyphs retained their recognizable form for a very long time. Even after the hieroglyphs became phonetic signs, they were recognizable as things: a man with hands raised, a crown, a hawk. Eventually the Egyptians evolved determinants, signs placed next to a hieroglyph to show whether it served as a phonetic symbol or as a pictogram. But neither hieroglyphic script nor cuneiform evolved into a fully phonetic form-an alphabet.

Indeed, Sumerian never had the chance to develop an alphabet. It was replaced by Akkadian, the language of Sumer's conquerors, before its development was complete. Hieroglyphs, on the other hand, existed for thousands of years without losing their character as pictures. Probably this can be explained by the Egyptian attitude towards writing. For the Egyptians, writing brought immortality. It was a magic form in which the lines themselves carried life and power. Some hieroglyphs were too powerful to be carved in a place considered to have magic power; they could only be written in a less powerful area, so they wouldn't bring unwanted forces into existence. The name of a king, carved in hieroglyph on a monument or statue, gave him a presence that went on past his death. To deface the carved name of a king was to kill him eternally.

This attitude towards writing preserved the pictorial form of hieroglyphs, since the pictures themselves were thought to have such power. In fact, far from being phonetic, hieroglyphs were designed to be indecipherable unless you possessed the key to their meaning. The Egyptian priests guarded this knowledge in order to keep this tool in their own hands. Ever since, the mastery of writing and reading has been an act of power. As a matter of fact, hieroglyphs were so far from intuitive that the ability to read them began to fade even as Egypt still existed as a nation.

Hieroglyphs could preserve their magical and mysterious nature only because the Egyptians invented a new and easier script for day-to-day use. Hieratic script was a simplified version of hieroglyphic writing, with the careful pictorial signs reduced to a few quickly dashed lines. Hieratic script became the preferred handwriting for business matters, bureaucrats, and administrators. Its existence depended on another Egyptian invention: paper. No matter how simple the lines were, they could not be written quickly on clay.

Clay had been the traditional writing material of both the Sumerians and the Egyptians for centuries. It was plentiful and reusable. The writing on a smooth-surfaced clay tablet that had been dried in the sun would last for years; but simply dampen the surface of the tablet, and the writing could be smoothed and altered. to correct or change a record. Records that had to be protected from tampering could be baked instead, fixing the marks into a permanent, unalterable archive. But clay tablets were heavy, awkward to store, and difficult to carry from place to place, severely limiting the amount of writing in any message. Sometime around 3000 BCE, an Egyptian scribe realized that the papyrus (a thick paper made from the papyrus plant) used as a building material in Egyptian houses could also serve as a writing surface. With a brush and ink, hieratic script could be laid down very rapidly on papyrus. Of course. papyrus was also much simpler to carry. And while hieroglyphs continued to be carved on the stone walls of tombs and on monuments and statues, letters and other important documents were written on papyrus.

86-2: The Industrial Revolution in Britain

In Britain, the eighteenth century was marked with a series of inventions that brought new uses to known energy sources (coal) and new machines to improve efficiencies (steam engines) and enable other new inventions (water pumps and railroads). Funding the inventions and financially supporting inventors and inventions through several trials required money. The eighteenth century was marked with a flow of capital (money or wealth) from Britain's colonies and from global trade into Western Europe. The flow of capital into Western Europe enabled investors to fund inventors and to perfect inventions. For example, James Watt is credited with improving the steam engine by creating a separate chamber to house the steam and by improving the engine's moving parts, getting them to perform correctly. The invention did not happen overnight: a series of attempts over a few decades finally worked when Watt partnered with toymaker and metalworker Matthew Boulton. Boulton, who had inherited money, financed the final trials and errors that made Watt's steam engine functional and reliable.

Innovations in iron manufacturing enabled the production of the steam engine and other products made of iron. In Coalbrookdale, England, in 1709, ironworker Abraham Darby found a way to smelt iron (remove the oxygen from rock containing iron). By burning coal in a vacuum-like environment, the English already knew they could cook off the impurities, leaving behind coke, the high-carbon portion of coal. Darby put iron ore and coke in a blast furnace, and then pushed air into the furnace, a combination that allowed the furnace to burn at a much higher temperature than wood charcoal or coal allowed. Mixing the iron ore with limestone (to attract impurities) and water and smelting it with coke enabled ironworkers to pour melted iron ore into molds (instead of shaping it with heat and hammers), making cast iron. The use of molds allowed more consistency in iron parts and increased production of iron components.

The steam engine alone had dramatic effects on production. It was used to pump water out of coal mines, enabling coal workers to reach deeper coal seams; to power spinning wheels that spun 100-plus spools of thread at a time; to power dozens of looms in a factory all at once; and to create a new mode of transportation, the railroad. The first railroad in England was opened in 1825. In 1830, Manchester (a center of textile manufacturing) was connected by rail to the nearby port of Liverpool, and in the next several decades thousands of miles of iron tracks were laid.

Before the railroad connected places and reduced the transportation costs of coal, manufacturing needed to be located close to coalfields. Manufacturing plants also needed to be connected to ports where raw materials could arrive and finished products could depart. In the late 1700s, plants were usually connected to ports by broad canal or river systems. In Britain, densely populated and heavily urbanized industrial regions developed near the coalfields. In the early 1800s, as the innovations of Britain's Industrial Revolution diffused into mainland Europe, the same set of locational criteria for industrialized zones applied: nearby coalfields and connection via water to a port.

Once the railroad was well established, some manufacturing moved to or grew in existing urban areas with large markets, such as London and Paris. London was an attractive site for industry because of its port location on the Thames River and more importantly because of its major role in the flow of regional and global capital. By locating in London, an industry was at the pulse of Britain's global influence. Paris was already continental Europe's greatest city, but like London, it did not have coal or iron deposits in its immediate vicinity. When a railroad system was added to the existing network of road and waterway connections in Paris, it strengthened the city's position as the largest local market for manufactured products for hundreds of miles. Paris attracted major industries, and the city, long a center for the manufacture of luxury items, experienced substantial growth in such industries as metallurgy and chemical manufacturing. These urban centers became, and remain, important industrial complexes not because of coalfields' proximity but because of the centers' commercial and political connectivity to the rest of the world.

87-1: The Chytrid Fungus Epidemic

The scientific community first suspected that frogs were dying out in the 1980s. Several cases of sudden disappearances of frogs from places where they were formerly common were noticed, and worryingly, these were places where the habitat was protected and in good health in other respects. Furthermore, these extinctions occurred in widely separated places, among them: litter frogs in Puerto Rico, stream-breeding frogs in Australia, western toads in Colorado's Rocky Mountains, and frogs and toads from the mountainous rain forests of Costa Rica and Panama.

When scientists working on amphibians around the world began to exchange notes on their experiences, it became clear that the amphibian declines in pure and widespread habitats were occurring much more quickly than they could explain in terms of habitat destruction, pollution, or climate change, even though these factors have undoubtedly had some effect on frog populations in some places, and may even have caused local extinctions. A fungal disease was discovered in amphibians in the early 1990s. The fungus belongs to a group called the chytrid fungi and, although there are over 1,000 different forms of chytrid fungi, this was the only species known to become parasites on a vertebrate host. In 1999, it was named *Batrachochytrium dendrobatidis*, or Bd for short. It is now known that Bd only lives on amphibian skin, and often causes the death of frogs. Importantly, Bd infects all species of amphibians, and so it remains a threat even after one or more species has disappeared from a region. The researchers reached the conclusion that Bd was the main cause of amphibian declines in pure habitats and set out to investigate it.

There were several factors at work. Frogs at higher elevations are more likely to be affected than those at lower ones, indicating that the fungus is stronger at the cooler temperatures experienced in mountains than the warmer conditions lower down. It seems that the optimal temperature for the fungus is 17°-25°C (62.5°-77°F), which is also, unfortunately, the temperature at which frog diversity tends to be greatest. Frogs restricted to cool mountains, especially stream-breeders, are the most affected. Higher temperatures reduced the activity of the fungus and can even be used to cure infected frogs under laboratory conditions. Species that tend to aggregate, under rocks, for example, are more likely to catch Bd, and periods of stress, perhaps as a result of unusually warm or dry weather, may also trigger outbreaks (and may also increase the tendency of the frogs to aggregate). Researchers discovered that although all amphibians

have bacteria and other microbes on their skin that help them to fight off diseases, including fungal infections, the effectiveness of this defense appears to vary from one species to another. Partially immune species can act as reservoirs of infection, or carriers, allowing the fungus to spread throughout a community of frogs. They also found that Bd can survive away from its hosts, at least for limited periods of time, and can be collected from wet rocks on which frogs were previously sitting, for instance.

So far, nobody has definitely established where Bd came from. One theory that is gaining favor among experts is that it has always existed in parts of sub-Saharan Africa, as the fungus was discovered on a museum specimen of an African clawed frog, *Xenopus fraseri*, collected in West Africa as far back as 1933. It was also found in *Xenopus laevis* collected in Uganda in 1934 and *Xenopus gilli* from the Cape regions of South Africa in 1982. Furthermore, the fungus appears not to be fatal to frogs from the southern half of Africa, suggesting that they had acquired immunity to it over a long period of time. Certainly, Bd was present on all clawed frogs collected in the region in 1973 with no apparent ill-effects. Clawed frogs were transported around the world in the middle years of the twentieth century for testing and, just as some of the frogs were accidentally introduced to new regions, this too could have been an opportunity for the fungus to spread. Regardless of its origins, Bd has spread to much of the world since the 1980s.

87-2: The Effects of the Tambora Volcano Eruption on the Atmosphere

In 1815, the Tambora volcano in Indonesia erupted and sent millions of tons of ash and sulfuric acid into the atmosphere. Tiny drops of liquid ash and sulfuric acid formed an aerosol cloud that spread throughout the winter of 1815-1816, carried by winds until it covered the Earth, cooling global temperatures by reflecting and scattering sunlight. Although the cloud reflected only 0.5 to 1 percent of the incoming energy, it reduced the Northern Hemisphere average temperature in 1816 by about 1 degree Celsius, a seemingly small amount of cooling that had a considerable impact on global weather patterns, with devastating consequences for agriculture on both sides of the Atlantic Ocean.

Ironically, however, the effects of Tambora's aerosol cloud could have been far worse if the eruption had been slightly weaker. While immense in size and scope, Tambora's aerosol cloud was not particularly efficient at reflecting sunlight. Stronger volcanic eruptions tend to eject more sulfur dioxide into the stratosphere than do weaker eruptions, and they lead to more sulfuric acid droplets within the same volume of atmospheric gases. A greater number of droplets increases the chance that droplets will meet and collide, forming larger droplets that will be removed more quickly from the stratosphere by gravity. A single, larger droplet also has less total surface area than two smaller droplets, and so is less effective at scattering sunlight. There is therefore a balance to be struck between eruptions that are too weak to penetrate the stratosphere—and so produce a small, short-lived amount of cooling—and eruptions that produce large, less effective sulfuric acid droplets. By measuring the remnants of Tambora's aerosol cloud in material taken from ice sheets and lake sediments, modern scientists have determined that the climatic consequences—while undoubtedly devastating—could have been far worse if the particles had been roughly half their size.

Unlike the sudden drop in temperatures in Indonesia that occurred immediately after the eruption of Mount Tambora, the planetwide cooling was a gradual process that took up to a year to be fully realized. While air temperatures can, and frequently do, change rapidly in response to variations in solar energy, soil and ocean temperatures adjust much more slowly. The land and sea possess considerable capacity to store heat, while the atmosphere has practically no storage capacity. When the atmosphere is cooler than the land and sea, heat will flow from these reservoirs back into the air; but since the air cannot store heat for long, much of this is soon lost to space. If, on the other hand, the

atmosphere is warmer, some of that excess heat will be stored in soil and water until a balance is reached.

As Tambora's stratospheric aerosol cloud began to cool air temperatures by subtly reducing the amount of solar energy reaching Earth, the land and oceans would have resisted this cooling by transferring stored heat into the atmosphere and cooling themselves as a result. By early 1816, the land, ocean, and atmosphere were shifting toward a new balance of energies, largely as a result of the solar-dimming effect of the aerosol cloud. The adjustment cooled first air, then land, and finally ocean temperatures across the globe. Using information from tree rings—the width of each ring is related to the growing conditions (mostly temperature and rain or snowfall) that year—climatologists have determined that 1816 was the second-coldest year in the Northern Hemisphere since 1400, surpassed only by 1601 following the eruption of the Huaynaputina volcano in Peru.

In the meantime, the aerosol cloud had produced other noticeable optical phenomena, most notably a series of spectacular red, purple, and orange sunsets in London in the summer and autumn of 1815. Sunsets typically appear yellow, orange, or red because atmospheric gases scatter blue light more effectively than they do light of other colors, skewing the visible-light spectrum toward red. The effect is even more pronounced when the Sun is low on the horizon, since its light must pass through a thicker layer of the atmosphere to reach the ground, resulting in less blue light and more red light. Stratospheric ash, dust, and soot particles from volcanic eruptions enhance this atmospheric scattering effect, leading to brilliant red sunsets.

88-1: Iron, Steam, and Factories in Eighteenth-Century Europe

Industrial production in eighteenth-century Europe was facilitated by the use of a new energy source, coal. Traditionally, charcoal had fueled the smelting of iron—the melting process used to extract iron from iron ore. Britain, however, ran low on wood—the source of charcoal—before other European countries did and needed an alternative fuel. There was plenty of coal, but it contained impurities, particularly sulfur, that contaminated the materials with which it came into contact. In 1708, the British ironmaster Abraham Darby discovered that coal in a blast furnace could smelt iron without these attending complications. His discovery triggered the iron industry's use of coal. In 1777, the introduction of a steam engine to operate the blast furnace considerably increased efficiency. In 1783, a steam engine was used to drive a forge hammer to shape the iron; three years later, steam-driven rollers flattened the iron into sheets. With these innovations, the output of the British iron industry doubled between 1788 and 1796 and again in the following eight years.

The greater supply of iron stimulated other changes. Relatively cheap and durable iron machines replaced wooden machines, which wore out rapidly. The new machines opened the door to further advances. Improvements in manufacturing methods and techniques led to the production of ever-larger amounts of goods, usually at lower prices. Industrial change started with cotton, but breakthroughs in the use of iron and coal continued and sustained these changes.

Before the age of industry, the basic sources of power were humans, animals, wind, and water. Humans and animals were limited in their capacities to drive the large mills needed to grind grain or cut wood. Wind was unreliable because it was not constant. Water-driven mills depended on the seasons—streams dried up in the summer and froze in the winter. And water mills could be placed only where a downward flow of water was strong enough to drive a mill. Clearly, the infant industries needed a power source that was constant and not confined to riverbanks. The steam engine, invented and improved on in Britain, met that need and stoked Britain's industrial growth. As late as the 1860s, people, animals, and wind- and water-operated machines still supplied more than half of the energy needs of manufacturing in Great Britain and the United States. But the steam engine was clearly the power source of the future.

The steam engine was first used to pump water out of coal mines. As mining shafts were dug ever deeper through groundwater, drainage became a critical factor. In 1712, Thomas Newcomen invented a steam-operated water pump. Its use spread rapidly. The first steam engine in the Americas was a Newcomen engine installed in New Jersey in 1753. James Watt improved on the Newcomen engine considerably, making it twice as efficient in energy output. Eventually, by developing a separate condenser, Watt devised an engine that enabled the steam engine to power a variety of machines. Thus, steam engines could operate mills that had previously been powered by water or wind. The high-pressure steam engine was even more powerful and energy efficient. The use of steam engines spread in Britain and then to the rest of Europe and to the United States.

The steam engine centralized the workplace. With the machine as a central power source, it became practical and commonplace to organize work in a factory. Locating a manufacturing plant where it was most convenient eliminated the expense of transporting raw materials to be worked on at a natural but fixed power source such as a waterfall. The central factory also reinforced work discipline. These factories were large buildings without decoration, sometimes inspired by military architecture and therefore resembling soldiers' housing. With the introduction of blast furnaces and other heat-producing manufacturing methods, the tall factory chimney became a common sight on the industrial landscape.

The steam engine powered a dramatic growth in production. It increased the force of blast furnaces and the mechanical power of machinery used to forge iron and to produce equipment for spinning and weaving. Assisted by machines, workers were enormously more productive than when they depended solely on hand-operated tools.

88-2: Formation of Earth's Atmosphere and Oceans

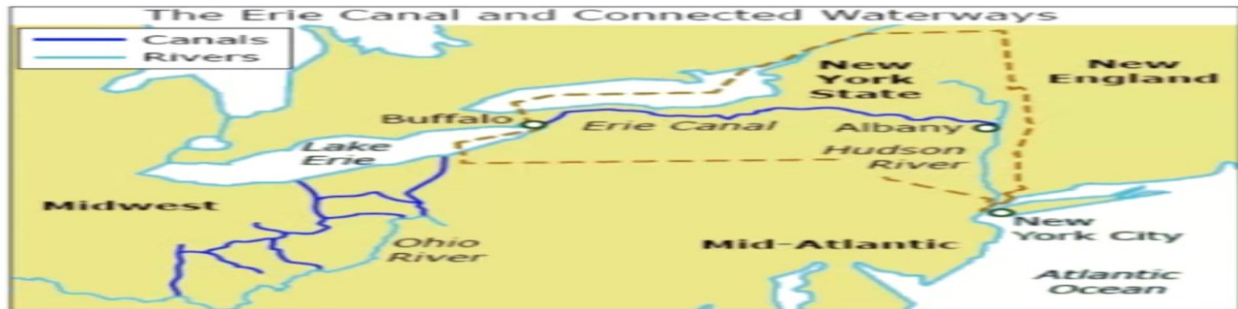
Early in its history, Earth was molten and had neither atmosphere nor oceans. However, as the gradually cooling surface began to solidify into rock, intense heat trapped below Earth's outer crust had to get out somehow. The result was surely volcanoes, geysers, earthquakes, and a variety of other geological events that literally blew off steam and pent-up heat through cracks in the surface. "Outgassing" of this sort happens even today, though at only a few locations on Earth and rather infrequently at that. But several billion years ago, this type of geological activity was surely more widespread and frequent. Scrutiny of modern volcanoes shows that lots of steamy water vapor, carbon dioxide, and nitrogen would have then undoubtedly emerged, along with vast quantities of ash and dust. Smaller amounts of hydrogen, oxygen, and other gases doubtless accompanied these early planetary eruptions. Calculations imply that over the course of Earth's history, enough gas was exhaled through fissures from Earth's interior to create much of our current atmosphere. The rest presumably came from comets and meteorites that could have scattered large quantities of matter on young Earth. Even today, some 40 thousand tons of extraterrestrial matter falls to Earth each year, almost all of it burning in the air or splashing in the ocean.

The origin of our present atmosphere is therefore a combination of terrestrial outgassing and interplanetary assault (further changed by later biological events). In truth, Earth's atmosphere is perhaps still adjusting, as present-day volcanoes occasionally sputter gas and heat amid incoming debris arriving from space. Today's atmosphere was not, however, derived directly from a mixture of interstellar gases. The composition of Earth's atmosphere thus differs considerably from the average cosmic abundance of the elements.

Since the atmosphere and oceans of planet Earth are so closely linked, they almost certainly originated, at least partly, from the same sources—our planet's interior and materials from space. As regards the ocean, geologists argue that as the surface cooled sufficiently, the first store of liquid water pooled as water vapor condensed. After all, steam is the main component of volcanically vented matter, and the hydrated rocks (mostly silicates, with water trapped inside) that today make up Earth's mantle (the layer immediately below the crust) store several times more water than in all the seas combined. But outgassing from within may not be the whole story, since our planet's water has chemical subtleties implying that some of it may have come from beyond.

Debate swirls among geologists concerning the rate and timing of ocean formation. We are unsure whether Earth's mantle outgassed the global seas all at once early in our planet's history—known among geologists as the "big burp" theory. Or perhaps the seas took some time to form, having secreted from Earth's interior in a series of volcanic events that occurred more gradually. Researchers argue that some of the waters of Earth have resulted from a rain of water-rich comets and meteorites that collided with our planet in great numbers during its first billion years. A minority of researchers go so far as to argue that all of Earth's water originated in this way, but this extreme view does not fit with the fact that the makeup of water on Earth does not match that trapped in interplanetary bodies. Three comets that recently bypassed Earth—Halley's in 1986, Hyakutake in 1996, and Hale-Bopp in 1997—all emitted radiation that revealed a higher concentration of water with heavy hydrogen atoms than is found in Earth's oceans.

Most likely—as with many other aspects of the cosmic evolutionary story that are neither black nor white, neither solely this nor clearly that—Earth's large bodies of seawater emerged from both within and without, and then, as the rate of outgassing and bombardment declined, a global recycling system began to operate. Water locked in rocks was expelled back into the ocean whenever the rocks were heated, such as those near volcanoes. To be sure, most of today's seawater is thought to have been recycled many times, perhaps as frequently as every ten million years. Recently, water has been directly observed emanating from certain underwater vents. Whether this is "juvenile" water still originating directly from Earth's mantle and incrementally adding to the world's supply or merely existing seawater cycling through the vents is not yet known.



89-1: A Transportation Transformation

In the northeastern United States, from Independence (1776) until the Civil War (1861-1865), state and local governments, rather than the federal (national) government, were the sources of most initiatives to improve transportation, although private entrepreneurs were the principal builders of toll roads in the early nineteenth century. All toll roads required payment for use, but few toll roads were profitable, and improvements in water transportation made greater contributions to commerce and economy. The use of steamships on rivers from the second and third decades of the nineteenth century reduced the cost and increased the speed of transporting goods up such major rivers as the Hudson, Ohio Mississippi, and Missouri and contributed to the growth of cities such as Albany, St. Louis, Cincinnati, Louisville, and Pittsburgh. Since there were over twenty thousand miles of navigable rivers in the pre-Civil War United States, the fuller use of natural inland waterways was a major step in integrating the new nation. Steamships soon lowered transport rates on large lakes, along the sea coasts, and in international trade.

Failing to obtain federal aid, New York State, with help from British investors, financed the 363-mile Erie Canal. The canal opened in 1825 and linked Buffalo on Lake Erie with Troy and Albany on the Hudson River and thus with the Atlantic Ocean. The Erie Canal lowered transportation costs from Buffalo to New York City by 90 percent and established New York City as the premier United States port. The canal that linked the Ohio river with Lake Erie in 1833 was another example of connecting major waterways and large regions with each other. Nearly four thousand miles of canals were operating by the 1850s, with three-quarters of the financing coming from state and local governments, borrowed mostly from British investors.

The principal canal networks connected the Northeastern Seaboard with the Upper Midwest. The low cost of canal and steamboat shipping made it feasible to convey even

bulky, low-priced goods long distances. This put northeastern and Mid-Atlantic grain farmers at a competitive disadvantage in respect to grain from the rich soils of the Ohio River valley and beyond. The beleaguered farmers had to switch to dairying or vegetable and fruit crops to remain viable, but instead they often left farming or migrated west. New England farmers were most likely to migrate, and recurrent sightings of stone walls in deep New England forests are modern-day reminders of those abandoned farms.

The canals were successful, but then came the railroads. The first railroad in the United States—the Baltimore and Ohio—opened in 1830 with a locomotive imported from Great Britain. Thirty years later, the United States boasted over thirty thousand miles of tracks with a total expenditure five times greater than that invested in canal construction. Like the canals, most railroad lines were located in the northern regions of the nation and facilitated east-west transport, and Chicago, with four thousand miles of track converging there by 1860, became one of the major transportation hubs. Rail competition led to the closure of many transportation canals and the cancellation of new canal projects.

Railroads were more flexible than canals because they could move goods and people over a variety of terrains with fewer problems posed by mountains, water levels, freezing weather, and trans shipments. They reduced overland travel time by about thirty times, and it was the greater speed that made rail travel attractive to people. But railroad freight (shipped goods) revenue exceeded passenger revenue by the end of the 1850s, also the point when the volume of freight sent by rail surpassed that of canals. River transport continued to be widely used

The telegraph was a spectacular change in communication. It complemented the expansion of the railroads by enabling railroads to coordinate and monitor schedules. The railroad companies often allowed a telegraph company to string lines along railroads' rights of way in exchange for discounted service fees.

There were over fifty telegraph companies in the early 1850s when a group of Rochester, New York businessmen began consolidating small companies into what became Western Union (a famous telegraph company). This consolidation led to better efficiencies and improved service, facilitating the increased use of telegraph service for money orders, business news, newspaper reporting, personal messages, and conduct of war. Western Union completed the first coast-to-coast line across the nation in 1861, leading the United

States Post Office to cancel the Pony Express (mail delivery by riders on horseback) when the Civil War began.

89-2: Communication: Bumblebees Versus Honeybees

The waggle dance of the honeybee has been frequently described and much studied, and it is one of the most complex systems in insect communication. In contrast, it has long been assumed that for bumblebees the search for food is essentially a solitary endeavor, that workers do not communicate with each other about good sources of forage, so that each individual has to learn for itself which flowers provide reward. Indeed, it has been known for many years that bumblebees (of a range of species) are unable to recruit nest mates to particular places. However, it has become apparent that bumblebee foragers (bees that look for food) do communicate, indicating that they have found an available food source, but they do not indicate a specific location. In an elegantly simple experiment, Dornhaus and Chittka demonstrated that on their return to the nest, successful foragers of *B. terrestris* (a species of bumblebee) stimulate other workers to forage and communicate to them the scent of the food source that they have located. The returning forager runs around on the surface of the nest in an excited manner, frequently bumping into nest mates and buzzing her wings (very similar behavior occurs in some stingless bees). This stimulates workers to leave the nest and search for the source of the scent. This communication system is less complex than that of honeybees, for the new recruits do not appear to be given any positional information about the location of the food source. It would appear that there is some sort of pheromone signal (a chemical factor that triggers a social response in members of the same species) released by the returning forager, for activity in adjacent nests is stimulated unless air flow between the nests is prevented.

Why do bumblebees not communicate positional information? Dornhaus and Chittka argue that conveying the location of food sources may be less important to bumblebees than to honeybees; honeybees evolved in tropical ecosystems where they rely mainly on flowering trees, which grow close to one another and may be several kilometers from the nest and therefore difficult to locate. In the temperate habitats in which bumblebees probably evolved, the plants on which they feed are generally more scattered. There is nothing to be gained in recruiting workers to a specific small patch that one bee can adequately exploit single-handedly. However, communication about the types of flowers

that are providing rewards will allow the colony to rapidly encourage feeding on a rewarding plant species when it comes into flower, and so keep track of the changing seasonal availability of different species.

A second possibility is that bumblebees may forage over shorter distances than honeybees, perhaps as a result of their smaller colony size, thus rendering communication about the precise location of forage less important.

There is also a cost to conveying location. Honeybee recruits can take over an hour to decide where to forage when presented with just two returned foragers advertising different locations. They also take a long time to find the food source that is being advertised, for the locational information is not precise. Thus, in bumblebees, it may be that the costs of conveying this information outweigh any gains.

Communication also occurs between bees while foraging. Both bumblebees and honeybees can distinguish between rewarding and nonrewarding flowers of the same species without sampling the reward available. When doing so, they are often observed to hover by and "inspect" each flower, accepting some and rejecting others. In some circumstances, the bees may be directly assessing the reward levels, or perhaps examining correlates of reward such as flower size and symmetry. However, there is now clear evidence that perhaps the most important cue used by bees to decide whether to probe or reject a flower is scent marks left by bees on previous visits. Such marks may increase foraging efficiency by reducing the time spent handling flowers that have recently been emptied by another bee. Bumblebees, honeybees, and carpenter bees leave short-lived repellent marks on flowers that they visit, and members of the same species use these to discriminate between visited and unvisited flowers.

90-1: Natufians and the Domestication of Grain

It is known that the Natufian people used sickles to cut and harvest wild grains in the Mediterranean region as early as 12,500 B.C. But did this practice lead to domesticated grains? Wild grains are brittle (easily broken) and the seeds burst forth when ripe, whereas nonbrittle "domesticated" grains remain on the plant until beaten off. In wild grains, there are one or two nonbrittle mutants to every 2 to 4 million brittle individuals. If ripe grain is cut with a sickle and taken to a village to dry and be beaten out, a higher percentage of the nonbrittle mutants is harvested since many of the brittle seeds are lost. If this grain is then planted, this ratio will increase each harvest.

Gordon Hillman, an archaeologist, and Stuart Davies, a biologist from the University of Wales, have used their knowledge of plant genetics and ancient gathering techniques—much of it acquired by experimentation—to estimate how long the change from wild to domestic strains would take. By using computer simulation, they showed that in ideal circumstances as little as twenty cycles of harvesting and replanting in new patches could have transformed a wild, brittle type of wheat into the domesticated nonbrittle variant. Under more realistic conditions, 200 to 250 years is the most likely period of transition.

The archaeological evidence makes it clear that this transition did not happen during the Natufian Period (about 12,500 to 10,200 years ago). There are microscopic differences between the shape of the grain from domestic cereals and wild varieties, and although cereal grains are rare in the Natufian archaeological record, all of those known are clearly from wild cereals. Not for another millennium at least do we encounter the first domesticated grains—from the settlements of Abu Hureyra and Tell Aswad in Syria and Jericho in Palestine. So, the Natufian people appear to have cut the wild cereal stands with their sickles for as much as 3,000 years without the evolutionary leap from brittle to nonbrittle plants.

There seems to be a very simple explanation for this, identified in some remarkable research by Romana Unger-Hamilton during the 1980s while she worked at the Institute of Archaeology in London. Under the guidance of Gordon Hillman, she spent many months replicating the Natufian style of harvesting wild cereals. Using identical sickles made with bone handles and flint blades, she cut stands of wild wheat and barley on the slopes of Mount Carmel, around the Sea of Galilee and in southern Turkey in a series of controlled experiments. The blades were then microscopically examined for signs of

"sickle-gloss" (traces of wear which cause a blade to shine)—the texture, location, and intensity of gloss will vary with different types of cereals and at different stages of ripeness.

Unger-Hamilton found that the sickle-gloss on the true Natufian blades was most similar to that on the blades she used to harvest cereals that were not yet ripe. In that state, the brittle plants would have shed only a little of their grain, so that it would be collected from the nonbrittle variants in virtually the same minute proportion as from those plants within the stand. So even if the Natufian people were planting seed to generate new stands of wild cereals, nonbrittle variants were unable to become dominant. Harvesting the unripe ears was perfectly sensible as it avoided the loss of most of the grain from the brittle plants, which would have already been shed to the ground.

Another factor probably prevented the emergence of domesticated cereals among the Natufians: their sedentary lifestyle. Patricia Anderson, of the Jales Research Institute in Paris, undertook a similar program of research to that of Romana Unger-Hamilton and confirmed many of her results. She also found that when wild stands are cut with sickles, even when still in a green stage, the grain that falls to the ground is quite sufficient to provide for the next year's crop. So, the Natufians would have only needed to plant if they were beginning a brand-new plot of cereals—otherwise they could have relied upon growback at the existing stands. Even if the grain collected by the Natufian people did have a higher proportion of the nonbrittle variants, unless they were creating new plots of cereals in new places, these variants would never have had the opportunity to become the dominant form.



90-2: Egypt in Early World History

The origins of Ancient Egypt represent some of the first instances of complex human societies. Indeed, the Nile Valley is the perfect place to start a state-level society. The annual flood brings fertile silts from the Ethiopian highlands to the plains along the Nile River Valley, allowing agriculture to continue indefinitely without overtaxing the soil. Evidence suggests that the inhabitants of the upper Nile region (that is, the "upstream" or southern regions of Egypt) were harvesting wild barley as early as 10,000 B.C.E. Perhaps as early as 5000 B.C.E., permanent agricultural settlements had been established both in the upper and lower Nile regions. These settlements were similar in some ways to other sites found in what is now known as Egypt's Western Desert. Pottery found at the early Nile sites is of a style also found across the West African Sahel and North African region. This evidence suggests that the early Nile settlers may have come to the river valley to escape a savanna (grassland) that was beginning to make the transition to desert. Still, during the period up to the third millennium B.C.E., the region around the Nile Valley was well enough watered by rain to support some degree of agriculture, so it is probably more accurate to think of the drying of the Sahara as a series of gentle pushes rather than as a single dramatic shove out of the region.

During the period from 3600 to 3300 B.C.E., the settlements along the upper Nile appear to have begun the process of creating what anthropologists call complex societies by developing a specialization of labor and moving toward more hierarchical political and economic systems. Two of these early Nile states have been excavated and examined in some detail: Nagada and Hierakonpolis. Nagada was located just north of where the city of Thebes would later be established, and Hierakonpolis lay just to the south. The inhabitants of both proto-states undertook a process of environmental modification. They cleared trees and heavy growth to expand the area of cultivated land and also built dykes and canals to control the flow of water. These efforts were successful enough that by around 3500 B.C.E., they had expanded the amount of arable land fourfold and led to a population explosion of up to 1,000 percent—perhaps reaching a population density of

1,000 persons per square kilometer. Such an increase is a testimony both to their ingenuity and to the bounty of the Nile; the major towns of Nagada and Hierakonpolis were walled, suggesting that the inhabitants were, sometimes, forced to defend their newly created wealth from their neighbors.

Both towns show increasing economic activity by the middle of the fourth millennium B.C.E. Hierakonpolis seems to have developed an economic specialization in the manufacture of a fine ware pottery known to archaeologists as "Plum Red", and examples of this pottery were exported throughout the Nile Valley. The town was also the home of one of the world's first large-scale breweries producing an estimated 1,200 liters of beer a day. Such economic specialization led to social differentiation. Rather than a relatively egalitarian society with very few differences in wealth from individual to individual or family to family, now some people were rich and others were not. Within the towns, a wide variation in the size of homes suggests growing social distance between the rich and poor. Cemeteries found in the Nagada region during this period show an increasing social stratification. Some tombs include luxury items such as gold jewelry, gems, and finely decorated pottery. The wealthy tombs tended to be clustered together in a certain part of the cemetery, maybe as family groups or maybe just to be separate from the poor, even in death.

Even in this period before the unification of the Nile, there was considerable contact with other regions. Although the archaeological record for the lower Nile is less complete, it is clear that there was some degree of exchange along the Nile Valley. This was no doubt aided by the fact that whereas the current runs north, the prevailing winds blow south, allowing for relatively easy round-trips.

91-1: Physiological Adaptations of Tuna

Tuna fish not only swim fast, but they can also cover great distances over short periods of time or during long migrations. At night, tuna are known to cover some 15 kilometers during offshore excursions (possibly for feeding). For swimming at such great speeds and over long distances, the tuna has evolved a muscle structure and circulatory system that athletes can only dream of.

In general, fish move by a series of muscle contractions that begin at the head and progress through the body toward the thruster, the tail. Two types of muscle, white and red, are used in swimming and are arranged along the flanks (sides) of the fish. White muscle contracts quickly and is used for bursts of high-speed swimming. It operates anaerobically (without oxygen), generating lactic acid as a by-product (similar to the lactic acid that can build up in humans after prolonged vigorous exercise and produce fatigue or muscle cramps). Red muscle contracts more slowly, is used for slow, continuous swimming, and operates through an aerobic (with oxygen) metabolic pathway. Because red muscle does not produce lactic acid, it can continue to contract almost indefinitely with the necessary fuel (oxygen and glucose). In most fish, the bulk of the muscle is white, with only a small lateral strip of red. The tuna, however, is equipped with a much larger proportion of red muscle than the average fish. This allows the tuna to cruise at high speeds aerobically, without lactic acid buildup. White muscle is used to generate additional thrust to go from cruising to extremely high speed. To sustain such athletic skill, the tuna requires an efficient means of supplying blood and oxygen to its hard-working muscles and a way of releasing the heat generated during muscle contractions.

Fish, like other animals, need oxygen to respire and to fuel their working muscles. Dissolved oxygen is some 30 times as dilute in the sea as in the air. Fish use their gills to extract oxygen from the water by pumping water in through their mouths, then over and out through the gills. The pumping is done by repetitively opening and closing the mouth, thus sucking or gulping the water in and pushing it past the gills. The tuna's skull and jaw are rigid to enhance swimming speed, so it cannot physically pump water in. Instead, it breathes by means of ram ventilation. As a tuna swims, its mouth stays open and water is literally rammed in and over its extraordinarily large gills. Like some of the other fast-swimming fish, the tuna must constantly move or else it will suffocate.

Tuna and some sharks have evolved a specialized circulatory system that uses excess heat production to stay warm in their often-chilly medium. Fish are typically cold-blooded; their body temperature is regulated by the surrounding water. But the tuna has a special internal heat exchanger made up of intermingling arteries and veins. Cool, oxygen-rich blood flows from the gills to the tissues through arteries and a network of small blood vessels. Running parallel to and intermingling with the arterial network (the vessels carrying blood from the gills to the organs) is the venous system. In the venous system, warm, oxygen-depleted blood is carried from the organs and tissues back to the gills. Because the veins and arteries are intermingling and in close contact, heat is readily transferred from the warm blood in the veins to the cold blood in the arteries. With elegant physiological simplicity, this countercurrent heat exchange uses the tuna's own internal heat from muscle contractions to warm incoming blood. In most other fish, heat is lost directly from the gills; but in the tuna, there is little heat loss, and by the time the blood reaches the tuna's interior, it is nearly as warm as the internal body tissues. This exchange is so effective that a tuna's core temperature has been measured at 10 to 20 degrees Celsius warmer than the surrounding water. Scientists suspect that elevated body temperatures help keep the tuna's metabolism high for food processing, allow its red muscle to contract more quickly, and may enhance lactic-acid breakdown. Tuna may also be able to partially shut down the heat exchanger if they become too warm, or they may make excursions into deeper, colder water to avoid overheating.

91-2: The Carboniferous Atmosphere

An explosion of new animal forms took place about 530 million years ago, and from that era onward, oxygen built up rapidly in the atmosphere. As oxygen levels climbed, the new ecosystems grew in stature and complexity until lush, swampy forests had developed. Thus began the Carboniferous period (roughly 360 to 300 million years ago), named for the deep beds of coal—a form of carbon—that formed from the remains of trees and other organic matter during this time. We know a lot about life in the Carboniferous world because of the many fossils found in coal seams (layers). In 1979, miners working in the town of Bolsover, England, stumbled upon the fossil of an enormous dragonfly, with a wingspan of more than half a meter. Carboniferous scorpions were a meter long, and mayflies had 40-centimeter wingspans. Plants were huge too. The lycopods, which we know today as tiny club mosses, reached heights of 50 meters. All these life forms were giants by today's standards, so what was going on? A hint at the answer comes from the way the animals breathed. Insects do not have lungs. Instead, their bodies are riddled with tiny tubes called trachea[e] that let air diffuse in and out passively. This puts a size constraint on insects because the passive breathing system cannot deliver oxygen quickly to deep tissues. In today's atmosphere, a dragonfly the size of the Bolsover specimen would not be able to beat its wings fast enough to fly because its flight muscles would be deprived of oxygen. To get airborne, it must have flown through air much richer in oxygen than ours.

Calculations suggest the Carboniferous air was 30 to 35 percent oxygen, as compared to 21 percent today. The explanation for this high level of oxygen lies in the coal itself. Coal forms when dead plant material doesn't fully decompose, instead becoming buried and then compressed over vast spans of time. By 375 million years ago, the swampy forests of the Carboniferous were already laying down the first coal seams at a prodigious rate, and they continued doing so for millions of years. In fact, about 90 percent of the world's coal stems from this time. The Carboniferous plants must have been piling up in the ground when they died, perhaps because the standing water of the swamps protected them from decay—still (unmoving) water lacks oxygen that is needed in order for decomposition to occur. Or perhaps bacteria had not yet evolved the ability to digest the tough fibrous material (lignin) in wood. And because the dead plants weren't being fully recycled, there was an imbalance in the carbon and oxygen cycles. The trees were taking in carbon dioxide and giving off oxygen through the process of photosynthesis, but the reverse process wasn't happening—the carbon was staying trapped in the plants, and the

oxygen wasn't being used up by the process of decay. So as forests flourished, producing more and more oxygen, the atmosphere's oxygen level rose. Life had created an atmosphere in which it could make a leap of evolution.

The Carboniferous was a green world of mosses, vines, horsetails, and ferns, and vast areas of equatorial land were given over to forest swamps. But the oxygen-rich atmosphere brought something else. This is the time when charcoal first appears in the fossil record, and charcoal results from organic matter exposed to high temperatures, which tells us that the atmosphere now had sufficient oxygen to support fire. From around 365 million years ago, ferocious wildfires swept across the land, and they were fires like no other, for in air that was 30 percent oxygen, even the damp swamps would have burned. Today, studies of Carboniferous charcoal reveal not only the great intensity of the fires but the nature of the plants that were consumed by them. This, along with fossil evidence, reveals that the plants had adapted to become resistant to fire, with thick bark, high crowns, and long tubers that were protected deep in the soil. Destructive as the wildfires that swept the planet must have been, they were also part of the mechanism that kept the atmosphere in balance, because they would have consumed oxygen and prevented levels from rising too high while also, by burning the forests, injecting carbon dioxide back into the air.



92-1: Escape Mechanisms of Ypsolopha Dentella

Ypsolopha dentella (*Y. dentella*) is a moth that lives in the northeastern United States. It appears as a larva (the immature form that differs entirely from the adult moth) early in the spring, feeding on the young leaves of honeysuckle plants. *Y. dentella* larvae are well camouflaged. Predominantly green, they match the color of the leaves they eat. As additional defenses, the larvae also have the ability to squirm and to jump off a leaf and hang in the air using a silken thread. Squirming, or wiggling, is an escape mechanism they share with many small caterpillars, but *Y. dentella* manifests this behavior in the extreme. Release a *Y. dentella* larva on a smooth surface and poke it gently with a toothpick, and it will propel itself over the surface by twisting and turning with such speed that it will literally vanish from view. The response may last only seconds, but that is long enough to allow the larva to distance itself from the scene.

When a larva has wiggled its way to the edge of a leaf and can go no farther, it jumps, plunging into space, suspended by a slender silken thread that it squeezes from a small spigot that projects from just beneath the mouth. The spigot bears the openings of two large sac-like glands that take up a considerable portion of the larva's body cavity and are bulging with the viscous liquid from which the silk is made. Jumping does not require that the larva take time to anchor the silken thread to the edge of the leaf before jumping. The silken thread is fastened to the larva's cocoon, and the larva needs only to let out a piece of silk long enough to cover the distance to the leaf edge. It is therefore connected to home base at all times and free to squirm away and take a plunge at a moment's notice.

When they plunge, the larvae drop some 10 to 20 centimeters or more, a considerable distance relative to a body length of 15 millimeters or less. In essence their plunging is another vanishing act, a trick that in combination with squirming enables the larvae to avoid trouble by leaving the scene. When the larvae plunge they drop virtually

instantaneously. The rate at which they are able to convert liquid glandular precursor into silken strand is therefore quite remarkable.

The suspended larva, as a rule, does not remain hanging from its thread for long. Within minutes, it begins the slow process of climbing up the thread to return to its cocoon. To ascend, the larva pulls itself up along the cord, a short distance at a time, using its mouthparts to exert the pull and to tuck away the loops of slack reeled in as it proceeds upward. The process is elegant. The entire thread is stuffed into a holding space between the third pair of legs. When the larva completes its climb, it relinquishes the package, leaving it stuck to the leaf margin the moment it is back at home base. In so doing, the larva is essentially squandering a resource. Other silk producers are more conservation oriented. Spiders, for instance, when they take down their webs, eat the silk.

Jumping is doubtless effective against many predators, including ants and jumping spiders, but it comes at a price. Aside from the cost of the silk itself, which the larva needs to conserve for eventual cocoon production, jumping takes time. Although the larvae usually ascend within minutes, they sometimes remain suspended for over an hour. Potentially this wait could cut significantly into feeding time and therefore have a retarding effect on development. Whether it does depends on how often the larvae are attacked under normal circumstances, which has not been determined. But there is no question that predation pressure could be high at times upon *Y. dentella* larvae, particularly from ants.

One intriguing possibility is that jumping protects *Y. dentella* larvae against predation by wasps. No data on *Y. dentella* itself are available, but other caterpillars have been shown to avoid wasp attacks by resorting to squirming behavior.

92-2: The Medieval Agricultural Revolution

After around 900 A.D., the North Atlantic region began a 300-year spell of elevated temperature known as the medieval warm period, which affected agriculture both in North America and in Europe- including a burst of productivity that has sometimes been called a revolution. Although some have questioned whether the term "agricultural revolution" is appropriate for western Europe during this era, it is clear that dramatic changes in food production coincided with the medieval warm period. The central innovation was the three-field system of crop rotation, which began to take hold in much of northern Europe and Britain during the tenth and eleventh centuries. Although historians are not as certain as they once were about either the extent of its use or its novelty, the three-field system replaced in many areas an earlier agricultural model that Germanic peoples of the northern parts of western Europe had inherited from the Roman Empire. Many farmers had formerly used two sets of fields; one was cultivated and the other left unplanted, and the cultivation would alternate from year to year. The main crop had been wheat, sometimes joined by plots of rye or other grains, planted in the autumn and harvested in the spring.

The new three-field pattern took advantage of climatic changes during the warm period to add summer crops to the ancient mix. The two-field system had always been better adapted to the wet, mild winters and dry summers of the ancient Mediterranean region than to the very different environments north of the Alps and the Pyrenees. The three-field system was not only better adapted to northern conditions but substantially more productive than the Mediterranean model. For any given place, the three-field revolution increased the amount of land under annual cultivation by a third; a community that formerly cultivated, say, 300 of its 600 acres annually now farmed 400. The transition may not have been quite as earthshaking as the replacement of the crops first grown by Native Americans in eastern North America with a regime of corn, beans, and squash, but it was dramatic nonetheless. On both continents, new agricultural patterns transformed life in temperate zones.

In northwestern Europe, more was involved than new crops and a new organization of the landscape. A novel kind of yoke allowed horses to be employed to pull heavy loads, augmenting the slower and less powerful oxen that still did much of the work. Horsepower was particularly useful in pulling improved heavy, wheeled plows, which were now frequently fitted with moldboards (curved iron plates) that turned as well as

dug the soil. As these innovations spread, simple wooden tools called harrows also began to replace dragged thornbushes in the chore of breaking up and smoothing freshly plowed ground. An even simpler device—two pieces of wood attached together to make a flail—improved the process of separating the seed from the plant.

None of these changes occurred overnight, and all built upon precedents that were hundreds of years old. Nonetheless, the cumulative impact was striking. Vast areas either abandoned since the decline of Rome or never before cultivated were put into production. More important, the three-field system diversified the diet of ordinary people. Among the new summer crops were oats, barley, and a variety of legumes, including vetches, peas, lentils, and beans. The expanded availability of legumes (apparently grown previously, if at all, mostly as garden vegetables) was a major nutritional improvement to a diet formerly composed largely of daily bread. Western Europeans and North Americans discovered the virtues of legumes at almost exactly the same time.

Most European households kept gardens, which, like the gathered vegetation of North America, contributed essential nutrients for at least part of the year. The extent to which this phenomenon was novel is unclear, and, even for the elite, the yield was what the most careful student of early medieval nutrition calls "a somewhat monotonous diet heavily oriented to members of the cabbage and onion families." Still, the nutritional gains from western Europe's agricultural revolution were substantial. As the system took full hold, in the eleventh through fourteenth centuries, the region's population seemed to have roughly tripled.

93-1: Animals and Forests

Many species in the deer family, particularly white-tailed deer and moose in North America and red deer in Europe, browse (feed on) immature trees such as seedlings and saplings and in so doing curtail the regeneration of trees. In Scotland, high numbers of red deer pose a particular problem for the regeneration of Scots pine forests, some of which have not seen substantial regeneration for 300 years. Many smaller mammals, including mice, squirrels, voles (small, mouse-like animals), and rabbits, also regularly eat tree seeds and seedlings, sufficient to alter the course of woodland development.

These animals have a preference for particular tree species and can therefore potentially change woodland composition and future development. Moreover, some tree species can tolerate repeated browsing, but others cannot. Many broad-leaved tree saplings, for example, will readily resprout if the leading stem is browsed, but conifers (trees with needlelike leaves) generally lack this ability. This is significant in northern zones, where voles more than make up for the comparative absence of large animals. In northern Europe, vole outbreaks occur on a three- to four-year cycle (and somewhat more irregularly in North America), during which populations multiply several hundredfold. As voracious feeders, they rapidly consume their preferred plant roots and bulbs and then switch to seedlings of pines, spruces, and other conifers. This can have substantial economic impact for commercial forestry, and in Finland alone the cost of seedling losses to voles has been estimated at €20 million during such outbreaks.

Browsing mammals feed on trees, leaves, and shoots and consequently suppress tree growth. Conversely, grazers feed on grasses and can enhance tree establishment. This is most obviously played out in the African plains. By suppressing trees in African dry forests, elephants, giraffes, and other large browsers facilitate the spread and growth of grasses. This creates a substantial fuel load (amount of burnable material), and the resulting intense fires reduce woody vegetation further and prevent tree regeneration. Grazers such as zebras and wildebeests reverse this process by cropping the grasses, which reduces fire intensity. Trees reestablish, and the resulting forest shade suppresses the grasses. The change of African savannas from woodland to grassland and back again is therefore maintained by different kinds of herbivores (plant eaters).

Yet it is predators that have the key role in triggering the change from woodlands to grasslands and back. Trees provide predators with cover within which they can stalk and ambush herbivores. Herbivores therefore exist within a "landscape of fear," in which they

avoid wooded areas in favor of more open grasslands, where they can more easily spot predators. The concentration of grazing herbivores in grasslands is therefore brought about by predators, and this serves as the main driver of conversion to woodland. Elephants, by contrast, are more or less immune to predators on account of their bulk and have no aversion to wooded areas. As large browsers they feed on leaves, breaking branches and even knocking over trees to get to them, thereby initiating a return to grassland.

No matter the destruction done by mammalian herbivores, it pales into insignificance compared to that caused by insects. Since 1990, some 30 billion conifers from Alaska to Mexico have been killed by tiny bark beetles. Individually, bark beetles are smaller than a grain of rice, but when conditions are right, their populations explode in epidemics that can last a decade or more. In recent decades, a few bark beetle species, such as the mountain pine beetle in North America and the European spruce bark beetle, have been particularly destructive of coniferous forests. The increased incidence of summer droughts in these regions has rendered many trees vulnerable to attack, while milder winters have facilitated the growth of bark beetle populations. Healthy trees defend themselves by drowning the tiny pine beetles in resin, a sticky substance produced by some trees. Female beetles therefore target vulnerable drought-stressed trees, and once located, they release a pheromone (chemical signal) to attract other beetles. The tree responds with sticky resin and poisonous gases, but sheer force of numbers eventually overcomes a tree's defenses. The beetles, which might now number several thousand, lay their eggs under the bark, and the larvae (young) feed on the living tissue, which ultimately kills the trees.



93-2: Painting in the Dutch Golden Age

The seventeenth century in the Netherlands is often called its Golden Age, not least because of its painting, featuring such masters as Johannes Vermeer, Jacob van Ruysdael, and Frans Hals. Many factors allowed the Dutch (Netherlandish) artistic scene to flourish, but two of the most important were innovations in the stylistic treatment of paintings and in the process of producing them. In the early 1600s, Esaias van de Velde began to paint landscapes with a simplified color scheme, fewer people, and a lower horizon, creating a sense of space that helped make the image more attractive. Painter Jan Porcellis and others modified their seascapes in a similar way, using simpler compositions, as well as a more monochromatic approach with less varied and less vivid colors--now known as the "tonal" approach. This helped create an interest in pictures of everyday life, moving away from historical or religious subjects that had long been dominant.

The art historian Michael Montias presented some of the inventions by van de Velde, Porcellis, and others as innovations in process. By combining a swifter painting technique with simpler compositions (fewer figures and objects) and more restricted colors, Porcellis and van de Velde are considered to have set in motion a trend for producing cheaper paintings that could penetrate a broader market and thus attract new groups of potential purchasers. By using more sky, more shade, and less crowding in their pictures, painters effectively reduced the amount of labor they needed to invest in the painting. Such specialized and "painterly" works took much less time to complete than their meticulously executed counterparts, and since labor costs were the prime determinant of production costs, this had a dramatic impact on the price of paintings. Montias asserted that the works of the realistic "tonal" school of landscape painting, for instance, brought

substantially lower prices than those of their predecessors (typically 15 to 30 florins, versus 70 to 100 florins for the older works).

At roughly the same time, Dutch publishers cut back on production costs by reducing the size of books. It has been suggested that painters also applied this strategy, but as of yet, there is no quantitative evidence to convincingly corroborate this. Fortunately, the few quantitative studies that exist on the size of Dutch paintings provide some clues. Historian Ad van der Woude's analysis of the average size of paintings in the Dutch Rijksmuseum produced by Dutch painters indicates a gradual decline in the size of paintings over time. On average, painters born between 1550 and 1599 produced larger paintings than the groups born in 1600-1649 and 1650-1699. His data, however, are not extensive enough to show exactly when this trend set in. Another clue can be found in the distribution of paintings by subject matter. The sizes of traditional subjects of religion, mythology, and history paintings produced by the 1550-1649 group were on average significantly larger than landscapes, figure painting, or still lifes--paintings of inanimate objects, such as a bowl of fruit. We can combine this finding, which is based on a limited sample of paintings, with the relative distribution of subjects in probate inventories (court documents created for inheritance purposes) in seventeenth-century Amsterdam (a Dutch city) and collected by Montias. He compared inventories from the periods 1620-1649 and 1650-1679 and found that the share of landscapes, figure paintings, and still lifes increased, while the share of history paintings dropped significantly. This suggests that an increasingly large share of paintings in Amsterdam inventories were of smaller than traditional size.

By offering quality paintings for reasonable prices, the new generation of painters unlocked the demand for paintings. Whether the primary motives of the trendsetters were artistic or economic, the consequences were unambiguous as productivity increased and paintings could be offered for lower prices without losing out on quality or necessarily threatening painters' profits. The middle-income groups that previously could only afford copies or prints were now able to own new and original paintings by living masters. Because the artistic novelties did not replace or exclude other subjects, styles, and techniques, the range of paintings on offer expanded dramatically.



94-1: The Waterfall Environment

Waterfalls can act as barriers that pose a number of challenges to plants and animals. There is evidence that waterfalls can so isolate the upper parts of rivers that distinct varieties or even unique species of fish may evolve in the river on each side of the waterfall. Waterfalls commonly form obstacles to migration but, as the well-known example of salmon testifies, some creatures are able to overcome low waterfalls and rapids, particularly where the descent is broken into separate waterfalls, forming a natural staircase. While no fish can swim against falling water, many jump effectively. Atlantic salmon can clear over three meters. To enable it to jump higher, the salmon leaps from the peak of the standing wave that commonly forms near the foot of a waterfall. Here, there is an upward component in the turbulence which assists the fish in its leap. To facilitate the movement of fish upstream, fish passes, which are artificial stepped channels bypassing waterfalls, have been built on many rivers, while in some places, the waterfalls themselves have been deliberately modified or destroyed for this purpose.

When descending rivers, fish may have no difficulty dealing with waterfalls and rapids, particularly where the drop is small enough for them or their parents to have ascended. Here, water acts as a cushion and lessens the effect of the fall, helping prevent damage. Waterfalls can be destructive of aquatic life, however. Some researchers have found that many species of plankton--minute water animals and plants usually living in colonies--are eliminated by waterfalls and rapids. Others have shown that some species can survive the descent of steep rapids and even major waterfalls.

Swimming and jumping are not the only means by which fish can overcome the barriers imposed by rapids and waterfalls. Species with suckers (organs that help an animal cling to a surface) can often ascend wet vertical rock faces beside or beneath falling water in stream courses. Some fish belonging to the large family of gobies have the ability to climb waterfalls in this way. Other fish that have the ability to overcome waterfall barriers include young eels, which make use of wet areas beside the stream to bypass falling water.

While waterfalls often act as barriers to movement and form obstacles in the stream that separate parts of the stream where life can be readily sustained, they provide a habitat to which some species are well adapted. There are plants and other organisms that survive and even flourish in the fast-flowing, turbulent water of waterfalls and rapids. Its waters well-supplied with oxygen by numerous falls, the tumbling stream provides an ideal environment for the production of chemical energy by living organisms in the form of plants such as algae, so long as there is adequate sunlight for photosynthesis. Particularly important are the small plants which grow closely attached to other plants, rocks, stones, and dead branches in the stream. These form a dense, slimy surface of algae or mosses, which are the most important photosynthetic organisms in running streams. The highly productive level at which living material is generated is due largely to the rapid current. This prevents the development of a layer of nutrient- and carbon dioxide-depleted water that would otherwise build up around the plants, as they absorb nutrients and use carbon dioxide for photosynthesis. Such a layer would reduce photosynthesis. Within the mass of algae and other plants lives a very rich collection of small animals, such as water mites and insect larvae. Clinging to solid holds in the tumbling flow and in the splash zones of waterfalls, these communities of plants and tiny animals flourish, providing food for the rest of the inhabitants of the stream. Many of the latter, however, are unable to live in the fast-moving, turbulent waters of rapids and waterfalls, except for some species that pass through them for short periods on their way to more favorable environments. Indeed, the animals that are commonly found in and around fast-moving streams and waterfalls normally take advantage of any available protection from the powerful currents, finding refuge behind or beneath rocks, or in gentle currents and quiet pools outside the main flow.



94-2: Easter Island's Statues

Easter Island, a small, isolated island in the South Pacific Ocean, is ringed by monumental statues known as moai. Researchers have long wondered how the islanders were able to construct and transport the statues using only the resources available to them. The statues range from 6 feet to almost 33 feet high, but there is a standard style and shape: a long human head and torso with prominent chin and stretched earlobes, arms held tightly at the sides and hands resting on the stomach. To some of the statues were added eyes of red and white stone and coral and red stone pukao (topknots) on top of the head, which could represent hair or the red feather headdresses seen by European visitors in the early eighteenth century. Some 230 of the statues were once set upright on platforms, some with multiple statues in a row.

There are questions about how the statues were carved, how they were moved, and how they were put in place. Locating the source of the stone used to make nearly all the statues is no great achievement, for it forms an impressive monument in itself. The stone quarry inside the old volcano at Rano Raraku is an extraordinary sight, with hundreds of carved-out niches left behind after the finished statues were removed and nearly 400 mostly uncompleted examples.

As for carving the stone, the surface of the volcanic rock is quite hard when it is weathered. Yet, once this surface crust has been broken through, the rock beneath is only a little harder than chalk and can easily be shaped, helped along by softening it with

water. This difference between the surface and the inside of the rock led some early explorers to conclude mistakenly that the statues had a hard pebble coating and a soft interior of clay. The tools used to carve and free the statues from the ground were undoubtedly the pointed picks of hard stone discarded in vast numbers at the quarry. In a well-known experiment, Thor Heyerdahl, leader of the 1955 Norwegian archaeological expedition to Easter Island, arranged to carve the outline of a statue at Rano Raraku. Six men hammered away with stone picks for three days, wetting the rock as they worked, at the end of which they had produced the outline of a statue some 16 feet long. From this, Heyerdahl estimated that six men could have carved the whole statue in about a year.

Once the giant statues had been freed from the rock, some of them were transported to their eventual resting places on platforms up to six miles away, along the tracks that radiate out from Rano Raraku, although the larger the statue, the shorter the distance it was moved. This was not necessarily because of their weight but more likely because of the fragility of the carved statues. The largest statue to be transported is that known as Paro, a giant 32 feet tall and over 80 tons in weight, which was shifted some four miles across rough country.

The early explorers who assumed that the island had always been treeless were completely baffled as to how the statues could have been transported without the help of wooden levers and rollers. However, archaeologists have been able to show that the landscape of Easter Island was once very different. By analyzing the pollen deposited by vegetation in the three lake beds on the island, they have drawn up a picture of the changing environment, confirming the idea that the island was once forested, with palm pollen being the dominant type. Therefore, trees or ropes could have been used to move the statues.

By attaching ropes to the head and base of a 13-foot statue, Thor Heyerdahl used a fifteen-man crew to move the statue by turning it on its base when it was tilted forward. They moved it only a few yards but damaged the base in the process. Geologist Charles Love placed a statue on a small wooden platform and ran it across wooden rollers. Twenty-five men managed to move the statue 150 feet in only two minutes, but misplacing the rollers caused the statue to come crashing down.



95-1: Silver in North America in the Late Eighteenth Century

Silver and silver objects had multiple meanings in late-eighteenth-century North America. Insights about social attitudes and the buying habits of consumers come partly from the records kept by silversmiths such as Paul Revere, the most politically active, artistically innovative, and successful silversmith of the period. His customers came from a wide range of socioeconomic groups-artisans, middle-income merchants, the business elite, and government appointees. Revere produced more low-end goods-- eating utensils, shoe or knee buckles, buttons, and harness fittings -- than high-end. The elite preferred to buy more elaborate silver goods from England or other European countries. Much of his success after the American Revolution (1776-1783), however, lay in his ability to respond to demands for specific kinds of more elaborate silver objects, in particular, sugar bowls, creamers, and teapots. One expert notes that after the Revolutionary War, Revere dominated silver teapot production, taking advantage of a return to popularity of a beverage that earlier symbolized a resistance to English taxes and the consumption of which had thus diminished immediately prior to and during the war.

Before the war, inhabitants of the American colonies had preferred to purchase silver as single pieces, thus slowly acquiring complete tea sets; after the war, the purchase of larger tea services as a complete set became popular. Revere's production of such en suite tea services was facilitated by the availability of rolled sheet silver, which was a more economical method of producing silver vessels, in particular, fluted teapots-- which have grooves or ridges -- like the one at the center of the set he made for a wealthy Boston merchant and his wife, John and Mehitable Templeman, in 1792. The rolled sheet metal also made it easier to achieve the fluting and straight edges that mark these objects. Their clean lines and finely engraved designs of draped fabric and tassels recall neoclassical forms (inspired by ancient Greek and Roman art) that were then popular in much of

Europe and associated with newly emerging republics on both sides of the Atlantic. These neoclassical attributes, found on all manner of goods and buildings in the immediate post-Revolution period, were the hallmarks of what became known as the Federalist style. Revere also engraved Templeman's initials--JMT--on the spoon handles, stands, and sugar urn, reinforcing the association between these objects and their owners, and serving as a kind of insurance, making it easier to track down the owner if the item was stolen.

Much of the silver in the homes of the well-to-do in the eighteenth century took the form of coins that were often melted down and transformed into domestic objects, and sometimes, remelted to pay off debts or to fund revolutionary armies. This transference from craft to currency and back suggests the particular place that silver held in colonial and post-Revolutionary America. According to historian Richard Lyman Bushman, silver was "a powerful material for establishing identity and configuring hierarchical relationships... Everywhere, silver was used to command assent, to assert authority, or to claim respect." Silver's power, notes Bushman, lay in the fact that it could signify three things at once: money, beauty, which was associated with divinity, and rank. In an era of dramatic fluctuations in the value of paper currency, silver maintained its value.

Yet the sheen of silver was tarnished by both its origins and its association with thievery and deceit. Much American silver originated in mines in Mexico and Peru and was extracted through a system of forced indigenous labor marked by long hours and appalling conditions, conditions that were remarked upon in political treatises and literary tracts available to American colonists. Many merchants, through their participation in the slave trade, then acquired the coins created from this silver. Once acquired, owners of such silver had to worry about thieves who recognized the wealth and status associated with this valuable metal and who were not averse to counterfeiting it as well as stealing it. Thus, as Bushman writes, "silver figured in a complex and ambiguous narrative of power." Whether as coinage or as teapots, it represented both elite status and a more democratic consumer revolution, as well as the ugly underside of the economic system that made both possible.



95-2: Measuring Climate Change on Mountains

Climate change has been linked to greenhouse gases in Earth's atmosphere that allow sunlight to reach the planet's surface while trapping the infrared energy that it emits. While the most abundant greenhouse gas is water vapor, human activities have only a small direct influence on its overall concentration, and its residence time (the time it stays in the atmosphere) is only nine days. The other major greenhouse gases are carbon dioxide and methane, and their concentrations in Earth's atmosphere have been increasing since the mid-eighteenth century, due particularly to the burning of fossil fuels, the clearing of forests, and other changes in land use. Molecules of carbon dioxide and methane stay in the atmosphere for years after they are emitted--their residence times are between 5 and 200 years (carbon dioxide) and 12 years (methane)--so human activities have long-term effects.

Evidence of these effects can often be found on mountains. For instance, the clearest direct evidence that concentrations of carbon dioxide are increasing around the world comes from continuous measurements taken since March 1958 near the summit of Mauna Loa, Hawaii, at an altitude of 3,397 meters. The site was chosen by the United States Weather Bureau after it was unable to find anywhere in the continental United States where the air was clean enough. The atmospheric scientist Charles Keeling's original reason for monitoring carbon dioxide was to better understand the daily cycle of its concentration in the atmosphere. However, within a few years, he had identified a seasonal cycle--because vegetation in the Northern Hemisphere absorbs carbon dioxide in summer and releases it in winter--and also discovered that, each year, concentrations were higher. Since 1958, the concentration of carbon dioxide at Mauna Loa has increased from 317 to over 400 parts per million. There is now a global consensus that Earth's

climate is changing because concentrations of carbon dioxide and other greenhouse gases are increasing as a result of human activities.

Mountains are not only where the primary cause of climate change has been measured, but they are where crucial evidence of its effects can be seen. Two of the most compelling types of evidence are the melting of glaciers and the fact that plants are growing at ever higher elevations on mountain summits. An international program of glacier monitoring began in 1894, as it was hoped that long-term observations would provide insights into processes of climate change, such as the occurrence of ice ages. Initially, the program focused on measuring the lengths of glaciers. Subsequently, the monitoring program was expanded to include the measurement of the mass balance of glaciers--the difference between the inputs of snowfall and the losses, mainly by melting--and the use of images from satellites. The great benefit of these images is that they provide complete coverage of areas that are often not easily accessible, such as many glaciated areas. Since 1986, the resulting standardized data on changes in glaciers, provided by scientists in over thirty countries, has been compiled by the World Glacier Monitoring Service. These data show that the world's glaciers are shrinking--and at an accelerating rate. While the primary reason is a warming atmosphere, an additional cause, also of human origin, is the deposition of fine particles of black carbon, or soot, on the surface of glaciers. This comes mainly from engines and the burning of coal and other biofuels. Because soot absorbs more radiation than ice, its presence increases rates of glacier melt. This phenomenon occurred in the European Alps in the nineteenth century, so this was the region where the rate of glacier melting was greatest at the time, and it is currently happening in the Himalaya.

While long-term measurements of carbon dioxide and glaciers were planned, finding evidence of alpine plants growing at higher elevations was largely a lucky accident. Early in the twentieth century, the Swiss botanist Josias Braun-Blanquet climbed many peaks in the Swiss Alps. While on their summits, he recorded the highest specimens of each plant species. In the early 1990s, two young Austrian botanists, Michael Gottfried and Harald Pauli, returned to the peaks that Braun-Blanquet had climbed and repeated his measurements. They found a statistically significant increase in the number of species on the summits. These changes could then be correlated to changes in regional climates.

96-1: Cool Early Earth Hypothesis

The oldest Earth rocks known to date, found in Quebec, Canada in 2008, were determined to be about 4.3 billion years old. However, the oldest known materials of the solar system—Moon rocks and meteorites—are around 4.55 billion years old. Why the difference? Earth's surface is constantly being recycled and remodeled due to the motion of tectonic plates. The Moon, by contrast, has a dead surface with no such movement. Meteorites that formed with the early solar system, like carbonaceous chondrites, have been unaltered since they cooled, so they give the oldest dates of all.

These are the dates for the oldest rocks on Earth, but they are not the oldest Earth materials known. That distinction goes to a handful of zircon sand grains from a much younger sandstone found in the Jack Hills of Western Australia. Each individual grain can be dated by uranium-lead methods, so they give a range of ages. But the oldest grains of all give an age of 4.404 billion years, at least 100 million years older than the 4.3-billion-year-old materials from Quebec. Thus, the current record holder for the oldest material from Earth (that is, not a Moon rock or meteorite) is 4.4 billion years. These sand grain dates put us closer and closer to the age of Moon rocks and meteorites, but we still have a gap of about 200 million years. This is about the same time span as the beginning of the Age of Dinosaurs (Late Triassic) until today, so it is not a trivial amount of time.

But those same tiny zircon sand grains held even more surprises. Not only did they give the oldest known dates, but when scientists analyzed the tiny bubbles of gases trapped inside them, they found evidence of the early atmosphere from over 4 billion years ago. These bubbles had oxygen isotopes (variants) in them that suggested Earth had liquid water on its surface as early as 4.4 billion years ago! This is surprising because when Earth formed 4.55 billion years ago, it was in a molten state (liquefied by heat). Prior to this discovery, geologists had always assumed that Earth took a long time to cool from its molten state. Most thought that Earth took about 700 million years to cool down below the boiling point of water (100°C), because that was the age of the oldest sedimentary rocks that would have been produced by running water (the Isua Supracrustals from Greenland, which are 3.8 billion years old). But the Jack Hills zircons turn that assumption inside out. If the zircons truly indicate the presence of liquid water on Earth 4.4 billion years ago, then it took only 200 million years for Earth to cool from its molten state to a

condition that was below the boiling point of water. This also suggests that there were not as many meteorite impacts during this time interval, or the oceans would have been vaporized over and over again. Taken together, these data suggest what is now called the "cool early Earth hypothesis."

So where did this early Earth water come from? Traditionally, geologists thought that it was water trapped inside Earth's mantle (middle layer) when it cooled, gradually escaping through volcanoes in a process called degassing. But lately, chemical analyses of extraterrestrial objects match the chemistry of Earth's oceans (especially carbonaceous chondrite meteorites). This suggests that there was a lot of water trapped in the debris of the early solar system (of which the chondrites are remnants), which represent the material from which Earth formed. The same is true of Moon rocks, which do not have much water in them today, but apparently, they were pretty wet when the solar system formed. If this is so, then Earth was born with its water already present as it cooled and condensed. It only required its surface temperature to drop below 100°C for that water to form the first oceans. One thing we can rule out is comets. Although comets are often called "dirty snowballs" because they are made mostly of dust and water ice, chemical analyses of four comets now show that their geochemistry is very different from Earth's water. Thus, the popular idea that comets impacted early Earth and melted to form its oceans can be dismissed.

96-2: Text in Dutch Painting

The seventeenth century was a golden age for Dutch painting, with artists producing genre paintings (depictions of scenes from everyday life), still lifes of fruit or flowers, and portraits of prominent citizens or community groups. Such paintings were typically intended not only to create visual pleasure but also to serve other purposes: to convey a moral lesson, to serve as legal documentation, or to commemorate an event or person. One important element in achieving these functions was the use of text--information expressed in words.

Text figured in paintings in various ways. Like their counterparts elsewhere, many seventeenth-century Dutch artists signed and dated their works. A few sculptors and manuscript painters from earlier periods had signed their works, but the signing of pictures as standard practice was a recent development in the Netherlands. Although it may now seem a self-evident procedure, the introduction of signatures and dates on paintings had significant effects on viewing pictures. The inscription of the artist's name on a picture calls attention to the making of the image and thus disturbs the illusion that the painting is a simple, faithful record of reality. Perhaps aware of this illusion-breaking effect, and to avoid it, the architectural painter Pieter Saenredam often countered it by inscribing his name on a column in his painting of a church as if it were graffiti, and by recording the precise location and date as a guarantee that he personally recorded the scene: "This is the Cathedral great church of Haerlem in Hollandt. Pieter Saenredam finished painting this, the 27th of February 1648." As in legal documents, a signature also had the function of claiming the initial ownership of the work by the master of a workshop. Even if he or she had not painted all of the picture, the signature was a guarantee to the customer that the work had come from a particular workshop.

Painters and printmakers also inscribed texts to convey information about the themes of their images. Portraits often give the sitters' ages and occasionally their names, thus verifying that the work presents an image of a particular person at a specific time. Like signatures on pictures, the making of portraits in the sense of records of specific people had become widespread Netherlandish practice only in the fifteenth century. Both developments reflect a new European understanding of personal identity.

Many other types of images included texts, especially still lifes and genre paintings. In his painting *Room Corner with Curiosities* of 1712, Jan van der Heyden (1637-1712) depicted books, globes, and fine objects from foreign cultures as a miniature collection, apparently in use by its owner, who has left open a book of maps and a Bible. The Bible is opened to the beginning of Ecclesiastes, one of the gravest Old Testament sections, with the opening words, "Vanity of vanities, all is vanity." Dutch preachers, writers, and still-life painters frequently cited this text to underscore the vanity, in the sense of futility (uselessness), of earthly goods and pursuits in the face of death and Christian concepts of an eternal afterlife. The text was so well-known that it gave its name to a subtype of "vanitas still lifes," paintings filled with objects to mark the shortness of life. Still-life paintings could include images of books by famous Dutch poets or praise for naval heroes, thereby celebrating their contributions to Dutch literature and history.

Some writers and artists even blurred the border between picture making and writing. Early in the century, schoolmasters and engravers turned handwriting into a pictorial art form of its own, in calligraphic exercises (exercises in stylized handwriting) that indicate both learning and extreme manual control. A handwriting manual by Jan van de Velde (1569-1623) offers intricate designs that attest to the skills of the author as well as those of the engraver Simon Frisius (1580-1629), who created the illustrations for the book. Perhaps not accidentally, these spectacular celebrations of writing and its reproduction in print were developed in a period of European self-consciousness about literacy and printing as feats that distinguished European civilization from cultures elsewhere in the world.

97-1: The Chemical Defenses of Trees

Woody plants produce a great variety of chemical compounds to provide protection against other plants, diseases, and herbivores (plant-eating animals) big and small. These chemicals can be found in just about any part of the plant (for example, the waxy surface of apple leaves contains toxins to repel certain insects). They can also be emitted into the air in considerable quantities, which explains why pinewoods have a strong fragrance and the eucalyptus-dominated bushlands of Australia have a blue haze. These chemicals seep out through the bark or the holes in a leaf (the stomata) or ooze out of special glands, often on teeth around the leaf edge (as found, for example, in pines, alder, and willows) usually when the leaf is young but sometimes (as in certain willows) even late in life.

Some chemical defenses are highly toxic and will kill attackers in small doses. Known as "qualitative" defenses, they work in low concentrations. These include alkaloids present in plants such as the tansy ragwort, which is notorious for killing cattle and horses when ingested. Other chemicals work by building up in the herbivore. Referred to as "quantitative" defenses, they become more effective the more they accumulate in the animal's body. Classic examples are phenolic resins in creosote bushes and tannins found in a number of plants. Tannin is a protein precipitant which gradually makes food less digestible, so herbivores end up starving and having stunted growth. These often have the effect of causing the herbivore to move elsewhere. It is possible, for example, that the endangered British red squirrel is declining not through competition with the introduced North American gray squirrel, as commonly thought, but because of acorns. Acorns (nuts from oak trees) contain digestion inhibitors that gray squirrels can disarm but red squirrels cannot. Red squirrels do well in conifer tree plantations, feeding on the more nutritious pine seeds, and where there are no oaks to give the gray squirrels a competitive edge.

Plants are not immune to the effects of chemical defenses. Spanish moss, which hangs from trees in the southern United States and in South America, never grows on pines, presumably because of the resins pine trees secrete. Perhaps the best-known example of antiplant chemical defenses was first reported in the first century AD by Pliny the Elder, who wrote that "the shadow of walnut trees is poison to all plants within its compass." Walnuts and related trees contain the chemical juglone, which, seeping from the roots, reduces the germination of competitors, stunts their growth, and even kills nearby plants,

resulting in open-canopied walnut groves having very little growing under them. Tomatoes, apples, rhododendrons, and roses are very susceptible, but many grasses, vegetables, and Virginia creeper will happily live under walnuts.

Defensive chemicals are costly for the plant to produce and store; many are toxic to the plant producing them. Oaks may put up to 15 percent of their energy production into chemical defenses (which explains why stressed trees are most prone to attack: they have less energy available to produce defenses). Thus, although there may be large pools of defenses available where attacks are common and ferocious, it makes evolutionary sense in lesser situations to produce the chemicals in earnest only when they are needed. Oaks (and a number of other trees) manage with low concentrations of tannins in their leaves, but once part of the canopy is attacked, the tree will produce tannins in large quantity. More than that, the tannins released into the air are detected by surrounding oaks and they in turn will produce more tannins in preparation for the onslaught. In a similar way, willow trees in Alaska that had their leaves eaten by snowshoe hares produce shoots with lower nutritional concentration and higher levels of lignin and deterrent phenols, thus rendering them less palatable. These "induced" defenses may have an effect for some time: birch trees can remain unpalatable for up to three years after being eaten. Expensive defenses are saved not only when they are not needed but also where. The North American aspen has been shown to produce fewer tannins in trees growing on fertile soils: it is cheaper to replace lost material than to defend it. Similarly, more is invested in defenses where herbivores and diseases are more common.

97-2: The Porcelain Industry at Jingdezhen

Porcelain is a ceramic material that is prized for its translucence, hardness, and smoothness. Porcelain was first produced in China, perhaps as early as C.E. 100–200. By the Tang period (C.E. 618–960), porcelain objects were being exported from China to the Islamic world. Over the centuries, manufacture of porcelain for both domestic consumption and export grew. The large scale of production for export in the Yuan period (1279–1368) is indicated by a shipwrecked transport vessel bound for Japan that was found off the Korean coast in 1976. It was loaded with some seventeen thousand pieces of export wares (goods to be sold). Included among the porcelains were many Longquan celadon (a particularly beautiful type of green porcelain from the period) and white or bluish-white porcelains from Jingdezhen.

The Yuan period was pivotal in the history of porcelain for several reasons. There was a definitive concentration of production at the Jingdezhen kilns (ovens) in the southeastern province of Jiangxi. This region was favored by abundant concentrations of the necessary special clays, as well as forests for wood to fire the kilns and a relatively central location well connected for water transport and distribution. The Yuan rulers established an official agency to supervise ceramic production at Jingdezhen in the thirteenth century, and it continued as the official governmental porcelain center under the succeeding Ming (1368–1644) and Qing (1644–1911) dynasties. In addition to official kilns that supplied wares for palace use and for diplomatic gifts or awards to officials, the Jingdezhen region had many private kilns that produced wares for broad domestic consumption and for export.

The concentration of resources, artisan expertise, and imperial prestige at Jingdezhen led to the development of a porcelain industry of national and international scale. Specialization of labor and rationalization of manufacture, the process of improving the means and methods of production to improve efficiency in mass production, characterized porcelain production as it did later industries (and some earlier productions, such as lacquerware in the Han period, from 206 B.C.E. to C.E. 220). By the mid-sixteenth century, porcelain production involved more than ten thousand craftsmen in the region, and a single piece might undergo seventy-two processes before completion. The result was a centralized and sophisticated craft industry of vast scale that supplied markets ranging from the imperial households to export markets in Southeast Asia and, eventually, Europe and America.

In terms of technique and style, the Yuan period was important for introducing painted decoration in which copper-red and, most famously, cobalt-blue pigments were used to paint the porcelain and then covered over with a transparent protective glaze, known as underglazing. Although the previous Song period (C.E. 960–1279) had favored wares decorated in a single color with pictorial designs limited to carved or molded patterns, earlier precedents existed for lively pictorial designs brushed on ceramics. Ink-like iron oxide or colored enamel pigments were used on Cizhou types, and painted designs appeared on the still earlier Changsha wares of the Tang period. These were not porcelains but less refined stoneware, fired at lower temperatures, and Yuan-period Jingdezhen saw the first large-scale combination of porcelain technique with underglaze painting. The taste for underglaze blue-and-white porcelains may have been both stimulated and enabled by the international connections of the Mongol rulers of the Yuan regime. The best sources for the cobalt used for the blue ornamentation came from Western Asia and so were more easily supplied during the period of Mongol domination of Asia, with its open trade routes. Much early blue-and-white production seems to have been aimed at export markets in the Middle and Near East and Southeast Asia. Some of the design of these wares may have aimed at replacing cruder painted Near Eastern wares with more durable and refined porcelains decorated in similar style, and much of the demand that stimulated production of blue-and-white wares may have originated from Arab clients in West Asia.

One of the great advantages of underglaze-painted decoration was its adaptability to a wide range of tastes and markets. Skilled painting specialists produced lively designs of decorative flowers and plants, dragons, ducks, fish, or figural scenes drawn from popular drama, as the occasion demanded; in addition, designs in Islamic script or incorporating Tibetan Buddhist motifs were produced in large numbers for those markets and recipients.

98-1: The River Nile in Ancient Egypt

The special character of the Nile, which made it central to ancient Egyptian culture, was its annual inundation (flooding). During June, the river began to rise, and a quantity of green water appeared. The color is said to have resulted from the brief period of reproduction of myriad minute organisms. During August, the Nile rose rapidly and assumed a muddy red color created by the rich red Earth brought into its waters by its tributaries. The Nile continued to rise until mid-September, then remained at that level for two or three weeks. In October, it rose again slightly, then began to fall gradually until May, when it reached its lowest level.

The Nile has created a convex floodplain. In convex floodplains, sediments (clays and silts) are deposited by flood waters, making the land nearest the river have the highest elevation. The convex floodplain is marked by natural levees that form elevated barriers immediately adjacent to the river. These levees rise a few meters above the seasonally inundated lowlands. When the Nile floods, the water covers most of the low-lying land up to the edge of the desert. When the floods subside, the waters are trapped behind the levees and prevented from returning to the river. The benefit of such topography is obvious: the water can be used where it stands or can be channeled to other areas as dictated by agricultural needs.

Ancient records, those preserved both in texts and in the visible evidence on ancient devices for measuring water levels called nilometers, indicate that a flood of six meters was perilously low and that one of nine meters was high enough to cause damage to crops and villages. A flood of seven to eight meters was ideal in that low-lying areas and basins throughout the whole valley would be flooded up to the edge of the rising ground of the desert, but towns, villages, and dikes that served as paths and water barriers remained above the water level.

The ancient Egyptians fully understood the extent to which their lives and prosperity depended on the unfailing regularity of the inundation. The occasional low flood and consequent shortage of food were enough to cause much anxiety among the populace at the beginning of each flood season. Ancient Egyptians, therefore, never became completely confident about the annual inundation and its gifts, even though it usually brought a layer of fresh, rich silt and waters for irrigation that made agriculture in the Nile

Valley relatively easy. The generally predictable crops and resulting surplus freed a significant segment of the population from agricultural labor, allowing for the development of nonfarming occupations, such as full-time craftspeople, bureaucrats, and priests.

The importance of the Nile to Egyptian civilization is reflected in the role that it played in religion and the myths that revolved around the river. The Nile was known in antiquity by the Egyptian name Iteru, meaning "great river." The personification of the inundation was a god named Hapy, who was associated with fertility and regeneration. The ancient Egyptians had various conceptions of the origin of the inundation. Some texts relate that it began in a cavern at Philae, while others credit the site Gebel Silsila (about 100 kilometers to the north) as the source. It was believed that veneration of the gods associated with these sites in the Aswan area could ensure a sufficient inundation. The Famine Stela, a text carved on rocks at Sehel near Philae, records a famine that was averted by donations of land and goods to the Temple of Khnum at Aswan. This text was formerly thought to date from the reign of Djoser (2687-2667 B.C.), but in reality, it dates to the Ptolemaic period some 2,500 years later.

98-2: The Faint Young Sun Paradox

Billions of years ago, in the first part of what is called the Archean period, our Sun was much weaker than it is today, emitting only about seventy percent of its current light and heat. This means that Earth should have been a much colder place, covered in ice. Yet there is good evidence that it was not as cold as we would expect. This is called the faint young Sun paradox. Something must have provided Earth with an atmosphere that retained the Sun's heat. It was initially hypothesized that this atmosphere most likely included carbon dioxide, a greenhouse gas that induces this effect.

This question has been debated for the best part of half a century, as the puzzling non-iciness of early Archean times continued to emerge from the evidence of the rocks. The atmosphere, we know, had little or no oxygen in it, as layers of sediment of that age deposited by rivers include grains of minerals such as pyrite and uraninite that, in today's oxygen-rich climate, would quickly oxidize and rust away. Among the ingredients of the atmosphere, many have suggested high levels of carbon dioxide from sources such as volcanic eruptions. However, extremely high levels of this gas would be needed to compensate for the faint early Sun—perhaps at about a hundred times current atmospheric levels. How can we tell if an ancient atmosphere had a lot of carbon dioxide or a little?

One way is to look at layers thought to represent ancient land surfaces—soils. Some minerals, such as siderite (iron carbonate), need high levels of carbon dioxide in order to form. So, if we find siderite in fossil soil, can we infer that carbon dioxide levels were high at the time of its formation? Not necessarily, because it depends on what other minerals were there to influence the reactions, and on the microenvironment. Even at a small depth within a soil, the carbon dioxide levels may be very much higher than in the air above, because of the activity of soil bacteria. And even moderately preserved Archean soils are exceedingly rare. The tentative studies made so far mostly suggest levels twenty or so times higher than today—but not a hundred times higher, and so insufficient to overcome the faint young Sun effect on its own.

So, what other additional greenhouse gases might there have been? Methane (CH_4) is a good candidate. This simplest of the hydrocarbon gases is about twenty-five times as powerful a greenhouse gas as carbon dioxide. Virtually, all methane on Earth today is biologically produced, much of it by the action of microbes called methanogens. It can,

however, form without the involvement of life, such as by the reaction of water with certain types of iron- and magnesium-rich igneous rock. In today's atmosphere, methane is oxidized to carbon dioxide within a few years. But in a low-oxygen world such as in the Archean, it could have survived for many years in the atmosphere. So, that is one possibility. There are other candidates too—ammonia, for instance. This is also a greenhouse gas, but it is easily broken down to nitrogen when exposed to ultraviolet light. Now, the best way to keep out ammonia-destroying ultraviolet light is to have a global sunscreen made of ozone: to make ozone, a planet needs to have significant amounts of oxygen in its atmosphere, and that state was yet a billion years in the future.

Was the Archean world, perhaps, simply hazy? A haze (cloud) of organic-rich particles might take on the function of ozone and block out significant amounts of ultraviolet light. The trouble is, it can also block out the warmth-giving sunlight, as the early studies on this possibility indicated. Those early studies, though, simply treated the haze particles as simple spheres. On that mightiest of Saturn's moons, Titan, there is an organic haze—but the particles seem on closer examination to be anything but spheres, rather forming highly complex, fluffy collections of material. Shaped in this way, they are not much of a barrier to incoming light—but they can still absorb ultraviolet radiation. So, if early Archean Earth was in that respect, at least, Titan-like, it might have harbored significant amounts of ammonia.



99-1: How the First People Got to the Interior of North America

Most researchers believe that the first people to come to North America came from Asia to what is now Alaska across a land bridge called Beringia, which was exposed as a result of low sea levels during the Ice Age. Evidence in the heart of North America points to a settlement time of around 14,000 years B.P. (before present). But there is some disagreement about which route these settlers took to the interior of a still uninhabited North America from Alaska. Until recently, everyone agreed that the first settlers traveled southward by land through the interior between two great ice sheets, the Cordilleran and the Laurentide; but in recent years, a growing number of people have favored the idea of a coastal route, triggering a major debate.

In the 1950s, Canadian geologists reported that Cordilleran ice did not fuse with the western margins of the Laurentide, creating between them an ice-free corridor sometimes called the Mackenzie Corridor. The Mackenzie Corridor became, in the minds of some, a kind of superhighway to the south, where mammoths and people traveled. This appealing picture is a geological myth, for the boundaries of the ice sheets are slowly being mapped, revealing only a partial corridor, and one that was very inhospitable indeed. With rugged land, mazes of meltwater lakes, crumbling rocks, the scantiest of vegetation, and few animals hunted for food, there was little incentive to travel through this ice-free zone at the height of the last Ice Age. If human beings did travel south, the most likely times for them to do so would have been either before the last glacial maximum (the period when ice sheets were at their maximum extension) beginning 25,000 years ago, or soon after 15,000 years ago, as the ice sheets began to retreat. Recent research has suggested that the corridor may have been quite wide by 11,000 years ago.

If the ice-free corridor was an inhospitable and late migration route to the south, what about other paths? Recent evidence suggests that long stretches of northern coastline from the Kuril Islands off Japan to Beringia and south through the Aleutians (a chain of volcanic islands in the northern Pacific Ocean) were more extensive 13,000 years ago than was once suspected. Did the Pacific coast provide a route into the heart of North America? Advocates of coastal settlement believe that fishing and sea mammal hunting, as well as a sophisticated maritime tradition, were well established in northeastern Asia during the late Ice Age, and that small groups of coastal people traversed the ice-strewn shoreline of the Beringia land bridge, then hunted and fished their way southward through the fjords, bays, and more sheltered waterways of the Alaskan coast.

Such a theory is entirely plausible, but its advocates have to live with a reality that any sites likely to document such migrations are buried under 300 feet (90 meters) of rising seawater. Furthermore, there is absolutely no archaeological evidence for sophisticated sea mammal and maritime technology in the form of skin boats (umiaks) or skin kayaks in the far north before about 3000 years ago, when the Bering Strait region became a major center for sea mammal hunting. Maritime adaptations are considerably earlier along the Aleutian chain but certainly much later than a supposed, and perhaps conservative, crossing date of 14,000 years ago. No one has examined the difficulties that sailors would have encountered crossing from Siberia to Alaska in small crafts. Apart from ice flows and pack ice, both of which may have been thicker during a period of warming, the rowers would have had to develop the clothing and boats to deal with very cold water and hypothermia (low body temperature), as well as often savage weather conditions, even in midsummer.

The manual British Admiralty Ocean Passages for the World describes these waters in summer as wet, fog-ridden for days on end, and subject to frequent, severe westerly winds, conditions that challenge even today's well-equipped fishing boats and freighters. Even in more southerly areas like southeastern Alaska, conditions can deteriorate rapidly, with strong headwinds that are notoriously difficult to row against. Southeast Alaska has plenty of sheltered bays and inlets, but the same cannot be said of the exposed coasts of the Gulf of Alaska, or of the shoreline south of Cape Flattery on the Olympic Peninsula, the Pacific route to lands far to the south.

99-2: Martian Volcanoes

Earth is not the only planet to have volcanoes; in fact, Mars—though all of its volcanoes have long been inactive—has the two largest ones in the solar system: Olympus Mons and Alba Patera. Mars has spectacular lava fields, large craters formed by a volcanic explosion, and shields (broad, low volcanoes shaped like shallow domes).

The most striking characteristic of Martian volcanoes is in fact their great size and longevity. The giant shields of Tharsis and to a lesser extent the volcanoes of Elysium and Hellas pack in many times more lava than the largest volcanoes on Earth, and yet Mars is a smaller planet. The reason lies in the planets' internal makeup. Because it is larger and hotter, Earth has a nearly molten (hot liquid) mantle that is overlain by a very thin rock crust—less than 10 kilometers thick under the ocean basins. The hot mantle churns over in great loops, and rips the overlying rigid crust into great slabs, called tectonic plates, that shuffle around the globe. In those places where deep-seated "hot spots" send plumes of molten material to the surface, the magma (the trapped lava) starts piling up on the crust to build a volcano, but because the crust is drifting, the volcano eventually breaks away from its source and goes extinct. In its place, a new volcano starts to grow, so that a hot spot under a mobile plate creates a string of medium-sized volcanoes rather than one giant shield.

On the smaller, faster-cooling Mars, the crust has solidified and thickened to the extent that it remains stationary with respect to the underlying hot spots. When a plume of magma erupts at the surface, it will build a single volcano in one place for as long as the supply lasts—typically, hundreds of millions of years. Not surprisingly, then, the volume of one Martian shield like Olympus Mons or Alba Patera is comparable to that of an entire chain of volcanoes on Earth, such as the Hawaiian Emperor chain that stretches across the Pacific and spans nearly 100 million years of hot-spot activity.

Another characteristic of Martian volcanoes is the large size of their craters and the long run-out distances of their lava flows. This is principally due to the lower gravity on Mars: 38 percent, or about one third, of Earth's gravity. Gravity controls to some extent how solid rock behaves under stress. When the crust stretches and breaks in a low gravity field, as it does on Mars, the fissures that open up are wider than on Earth and can funnel larger amounts of magma toward the surface.

Magma chambers grow correspondingly larger and when they empty out and collapse, they yield larger craters. Eruption rates also tend to be higher on Mars, since greater quantities of lava can flow out of the wider fissures. These larger volumes guarantee a better retention of heat. Because it stays hot and molten for a longer period of time, lava on Mars travels greater distances than it does on Earth before cooling and slowing to a halt.

Explosive eruptions are also affected by the unique set of conditions that exist on Mars. The atmospheric pressure is so low that any gas bubbles trapped in the magma will undergo tremendous expansion upon reaching the surface. As a result, it will take lesser amounts of dissolved gases in Martian (versus terrestrial) magma for the mixture to foam and shoot out explosively from the vent. On Earth, the confining pressure of the atmosphere requires that magma contains close to 1 percent (by weight) of light gases for it to spray upward as a lava fountain. About 3 to 4 percent is needed for the bubbles to grow large enough to blow the magma to shreds and create an ash cloud. On Mars, thresholds for such disruptive behavior are much lower: respectively 0.03 and 0.2 percent gases by weight. Twenty to thirty times less gas is needed on Mars to achieve similar results.

100-1: Characteristics of Tropical Rain Forests

Almost all the action in a rain forest (not just photosynthesis but also flowering, fruiting, predation, and herbivory) happens high in the canopy, the leafy green roof of the forest. Apart from the trees, the vegetation is largely composed of plant forms that reach up into the canopy by climbing the trees (for example, vines) or root in the damp, upper branches (epiphytes). The epiphytes depend on the sparse resources of mineral nutrients that they extract from crevices and pockets of the organic matter of soil found on the tree branches. The plants and animals of the canopy are not easy to study; even to gain access to the flowers in order to identify the species of tree is difficult without the erection of tree walks. It is a measure of the problems of doing research in rain forests that botanists have trained monkeys to collect and throw down flowers and a research team has used hot air balloons to move over the canopy and work in it.

Most species of animals and plants in tropical rain forests are active throughout the year, though the plants may flower and ripen fruit in sequence. In Trinidad, for example, the forest contains at least 18 closely related trees in the genus *Miconia* whose combined fruiting seasons extend throughout the year; this contrasts with the situation in temperate latitudes.

Dramatically high species richness is the norm for tropical rain forests, and communities rarely, if ever, become dominated by one or a few species—a very different situation from the low biodiversity of northern coniferous forests. This raises some fundamental questions. First, what is it about the evolutionary history of tropical rain forests that has allowed such diversity to evolve? Part of the answer relates to the comparative stability of patches of rain forest during the ice ages. It is thought that during these periods, drought forced tropical rain forests to contract into islands in a sea of savanna (flat grassland), and these expanded and coalesced again as wetter periods returned. This would have promoted genetic isolation of populations, allowing individual species to become well established before joining other species when rain forests grew again. This phenomenon of isolation is so important for a variety of species to develop. We may also ask why it is that among the diversity of species in tropical rain forests, a few have not dominated and suppressed the rest in a struggle for existence. At least part of the answer is that populations of specialized pathogens and herbivores develop near mature trees and attack new recruits of the same species nearby. Thus, the chance that a new seedling will survive can be expected to increase with its distance from a mature tree of the same

species, reducing the likelihood of dominance by one or a few species in the forest.

The diversity of rain forest trees provides for a corresponding diversity of resources for herbivores. A variety of fresh, young leaves are available throughout the year, and a constant procession of seed and fruit production provides reliable food for specialists such as fruit-eating bats. Moreover, a diversity of flowers, such as epiphytic orchids with their specialized pollinating mechanisms, requires a parallel, specialized diversity of pollinating insects. Rain forests are the center of diversity for ants—43 species have been recorded on a single tree in a Peruvian rain forest. And there is even more diversity among the beetles; Erwin (1982) estimated that there are 18,000 species of beetles in 1 hectare (10,000 square meters) of Panamanian rain forest compared with only 24,000 in the whole of the United States and Canada.

There is intense biological activity in the soil of tropical rain forests. Leaves that fall to the ground decompose faster than in any other ecological community. As a result, the soil surface is almost bare. The mineral nutrients in fallen leaves in rain forests are rapidly released and are unlikely to be removed by natural processes. In other ecological communities, rains may carry nutrients well below the upper soil levels at which roots can recover them. By contrast, the mineral nutrients in a rain forest are safe from loss to the lower soil layers and almost all are in fact held by the plants themselves.

100-2: Economic Reasoning

Economics begins with the assumption that human actions are rational—intended for maximum self-benefit. For example, people pay more for a larger box of the same cereal than for a smaller one. Yet people sometimes seem to act irrationally. When a celebrity endorses a product, sales increase, although the endorsement appears to convey no information about product quality. Nonetheless, economists stubbornly insist on finding rational explanations for all behavior. Why? Imagine a physicist who encounters his first helium-filled balloon, which flies up rather than down when released near the ground, a blatant challenge to the laws of gravity. Two courses are open to him. He can say, "Well, the laws of gravity have exceptions." Or he can say, "Let me see if there is any way to explain this strange phenomenon without abandoning basic principles of physics." If he takes the latter course, and if he is sufficiently clever, he will eventually discover the properties of objects that are lighter than air and recognize that their behavior is in perfect harmony with existing theories of gravity. In the process, he will not only learn about helium-filled balloons; he will also come to a deeper understanding of how gravity works.

Now, it might very well be that there are real exceptions to the laws of gravity, and that our physicist will one day encounter one. If he insists on looking for a good explanation without abandoning his theories, he will fail. If there are enough such failures, new theories will eventually arise to supplant the existing ones. Nevertheless, the wise course of action, at least initially, is to see whether surprising facts can be reconciled with existing theories. The attempt itself is a good mental exercise for the scientist, and there are sometimes surprising successes. Moreover, if we are too quick to abandon our most successful theories, we will soon be left with nothing at all.

Much primitive agriculture shares a strange common feature. There are very few large plots of land; instead, each farmer owns several small plots scattered around the village. (This pattern was endemic in medieval England and exists today in less developed parts of the world.) Historians have long debated the reasons for this scattering, which is believed to be the source of much inefficiency. Perhaps it arises from inheritance and marriage: At each generation, the family plot is subdivided among the heirs, so that plots become tiny; marriages then bring widely scattered plots into the same family. This explanation suffers because it seems to assume a form of irrationality: Why don't the villagers periodically exchange plots among themselves to consolidate their holdings?

Inevitably, this problem attracted the attention of the economist and historian Deirdre McCloskey, whose instinct for constructing ingenious economic explanations is unsurpassed. Instead of asking "What social institutions led to such irrational behavior?" McCloskey asked, "Why is this behavior rational?" Careful study led her to conclude that it is rational because it is a form of insurance. A farmer with one large plot is liable to be completely ruined in the event of a localized flood. By scattering his holdings, the farmer gives up some potential income in exchange for a guarantee that he will not be wiped out by a local disaster. This behavior is not even exotic. Every modern insured homeowner does the same thing. One way to test McCloskey's theory is to ask whether the insurance "premiums" (that is, the amount of production that is sacrificed by scattering) are commensurate with the amount of protection being "purchased," using as a standard, the premiums that people are willing to pay in more conventional insurance markets, and by this measure, it holds up well. A serious criticism is this: If medieval peasants wanted insurance, why didn't they buy and sell insurance policies, as we do today? A simple answer is that nobody had yet figured out how to do it. But some economists demand more exact answers to this objection, and they are right. Theories should be tested to their limits.

101-1: Facial Expressions

Even without the more elaborate forms of nonverbal communication like sign language, gestures, or miming, humans can actually communicate a great deal using only their bodies. Much of this, such as the posture of a person who is sad, is unintentional, but we can give deliberate signals with our bodies as well. Invading someone's space can be a way of intentionally signaling aggression. Other species also use body language, which was probably one of the earliest forms of communication between animals. However, unlike many other animals, humans also have an impressive ability to communicate things using only our faces. Once again, this can be intentional or unintentional. Our facial expression is almost constantly advertising our current emotional state and can even betray us by revealing feelings that we attempt to hide. We can also add important nuance to verbal communication using our faces. Given the right context, a simple glance at a friend can often communicate a great deal.

Many of the cues and signals that come from our facial expressions are universal among all of the people in our species. This indicates that this form of communication is much more innate (inborn rather than learned) than the recent innovation of spoken language. For example, the languages spoken among the hunter-gatherer tribes of New Guinea have absolutely nothing in common with English or any other Indo-European language. (Most of them have nothing in common with each other!) Yet, a smile means exactly the same in the rain forests of New Guinea as it does in the streets of New York City. This speaks to the unconscious, genetic, and inborn nature of facial expressions because if facial expressions were learned, like language, we would expect divergence among the cultures of the world. Instead, we see striking universality. Further proof of this is that people born blind display the same facial expressions as their sighted neighbors. They could not have learned the expressions by seeing them in others. It seems clear that facial expressions and their meanings are almost completely built into the human brain.

Many animal species also communicate with each other using facial expressions. This appears largely limited to mammals for two reasons. First, invertebrates, fish, amphibians, and reptiles, even if they live in communities, show only the simplest hints of interactive social dynamics and cooperation. When it comes to cooperation and communication in the animal world, birds and mammals are the stars of the show. Second, facial expressions require elaborate overlapping musculature in the facial region, which birds do not really have. There are forty-three muscles in the average human face.

Many of them are strange, as skeletal muscles go, because they do not connect two bones in order to allow skeletal locomotion. Instead, they are loosely connected to the skull on one end and skin tissue on the other. In other words, their only purpose is to squeeze and stretch the skin of the face. This sort of thing does not exist anywhere else in the body. Much of our face musculature exists purely so that we can make facial expressions.

This is why mammals have the monopoly on facial expressions. We are the only animal group that has both the social-cooperative nature to use communication in the first place and the necessary muscles to accomplish that communication using our faces. One might be tempted to think that mammals evolved our elaborate facial muscles for the purpose of communication, since that seems like pretty much all we use them for, but it turns out that our face muscles perform an even more basic and essential function: suckling (drinking milk from the mother's breast). By studying very old mammal fossils, paleontologists have discovered that a muscle group that was originally located in the throat was co-opted and relocated to the face of early mammals and resulted in much-improved suckling from the then-recent mammalian innovation: the breast. With the new facial muscles and the highly social context of nursing, the needed components were in place for the invention of facial expressions as a means of silent communication by early mammals, who must have lived in constant fear of predation by the dominant animals of their day: the dinosaurs.



101-2: Greece Emerges from the Dark Ages

The Mycenaean Greek civilization, which featured cities and a thriving economy, dominated the Aegean and parts of the Mediterranean beginning about 1600 B.C.E. But around 1100 B.C.E., Greek society entered a period of decline known as the Dark Ages that lasted for roughly three centuries. A new Greek culture emerged in the eighth century—the conventional start date is 776 B.C.E.—marking the beginning of the Archaic Age, which brought a revival of Greek trade and culture as the Greeks again became one of the major economic and political forces in the Mediterranean world. This happened for several reasons. For one thing, the Dark Ages had seen a constant population increase, and Greece, with its great expanses of rocky and hilly soil, simply could not grow enough agricultural staples to support a large population. Methods for dealing with the extra mouths were needed.

One response was to import food. As a consequence, Greek trade and manufacturing expanded. In exchange for olive oil, fine pottery, silver, and slaves, the Greeks acquired grain in Egypt and in the lands on the Black Sea coast. The expansion of Greek commerce brought the Greeks into conflict with the current Mediterranean trading power, the Phoenicians. The large, 30-oared Greek warships, known as triakonters, of the ninth and eighth centuries B.C.E. were replaced by fleets of iron-beaked ships known as pentakonters, which were 90 feet long and had 50 oars. The clumsy, two-decked Phoenician warships were no match for the more maneuverable Greek ships, and the Greeks soon wrested control of much of Mediterranean trade from their Phoenician competitors, who at this time also were under attack by the Assyrians.

The revival of Greek trade also had other consequences. The Greeks brought back culture as well as grain, and as a result Greek culture evolved based largely on borrowings from the much more ancient and sophisticated cultures of the Near East. From the Phoenicians

they borrowed the alphabet, for they now had just as great a need to keep records. In order to adapt the Phoenician alphabet, which had been designed to write Semitic languages, to write their own language, the Greeks added new letters representing vowels to make it more clear which word was meant.

In the late seventh century B.C.E., the Greeks of Ionia adopted coinage from the Lydians who lived in Turkey. In the early sixth century, coinage spread to the mainland, with the island city of Aegina issuing staters (coins of a standard weight) bearing a turtle on one side and a simple punch mark on the other. Around 570 B.C.E., Corinth issued coins bearing an image of Pegasus, the winged horse thought to have alighted at Corinth to drink from a sacred well, and about 525 B.C.E. Athens began issuing its famous "Owls," which bore the head of the goddess Athena on one side and Athena's owl on the other. A multitude of other Greek cities also issued coins as a means of asserting their status and independence. The coins of important trading cities such as Aegina, Corinth, and Athens became standard currency in the Mediterranean world because of their extensive circulation. One problem, however, was that the Greeks could not agree on standard weights for all their coins, and since trading cities attempted to carve out their own spheres of economic influence by favoring the use of coins with their own weights, this resulted in a complicated mathematical calculation if, say, someone wanted to use Corinthian coins to pay for something with a price expressed in Athenian coins.

The Greeks also assimilated ideas about artistic expression from the East. During the Dark Ages, artistic designs had consisted of simple geometric patterns. The Archaic Age introduced a multitude of Eastern styles. Greek sculpture assumed a very Egyptian look, and, in the Orientalizing style, at its height around 750 B.C.E., Greek pottery depicted many Eastern motifs, such as lions, bulls, and sphinxes. But Greek artists soon developed their own idiosyncratic style by incorporating into their designs scenes drawn from their own myths and legends. Greek potters competed with each other to perfect different manufacturing techniques and artistic styles, and they soon were the acknowledged masters of pottery making in the Mediterranean.

102-1: Animal Communication

Communication among animals has risks, as when an animal that makes a noise reveals itself to a predator. But it also has important advantages. Communication serves animals' needs to survive and reproduce and their ability to project their genes into future generations by aiding close relatives. When a male cowbird sings his song, he is attending to his need to attract a female cowbird and get his genes into the next generation. Worker honeybees, which are honeybees that do not reproduce, perform dances to communicate to other bees in the hive the location of a food source, with specific paths, speeds, and angles indicating the distance and direction of the food. By imparting this information, a worker bee is aiding the survival of her sisters in the hive and the further reproductive efforts of her mother (this is the only way worker bees can help spread their genes).

Animals do not only communicate with their relatives. Prey also communicate with predators, as when a gazelle stots (jumps up suddenly in the air) to communicate to the lion stalking it that he has been spotted and thus is unlikely to catch the gazelle. They should both save themselves the bother of a pointless chase that will just waste their time and energy. Sometimes predators listen in on the communications of their prey—as when dolphins detect the sounds made by fish. In discussing animal communication, it is customary to identify a signaler and the recipient of a signal. This does not imply that either individual is consciously aware or even unconsciously intends to send or receive the signal. An older view construed communication as a cooperative activity in which a sender and receiver collaborated to get a message across. It is now understood that the picture can be much more complex. Receivers may eavesdrop on signals that the signaler would rather they did not pick up. Bats, for example, listen in on the mating calls of male frogs in order to locate and attack them.

Some species are sensitive to the presence of potential eavesdroppers and modify their signals accordingly. Monkeys that raid farmers' crops have quiet alarm calls, known as "conspiratorial whispers," that they use to warn each other of the arrival of the farmer. These calls are clear enough that the monkeys hear each other but quiet enough that the farmer does not detect them. In other situations, signalers compete to get their message across with ever louder, larger, and brighter signals. It is thought that the large bright tail of the peacock evolved in this way as a form of competition among male birds to impress the females.

The tail of the peacock is an example of an honest signal—one that is intended to provide accurate information. The size and bright coloration of an impressive male's feathers indicate health, requiring adequate nutritious food, energy and preening (grooming) to maintain. Peacocks do not have the resources to fake their tails. Furthermore, having bright, flashy feathers or a long train that might slow you down can come at the cost of being more susceptible to predators. A male that is fit enough to survive despite this potential handicap must surely be a prize for a lucky female.

The deep notes of frogs' and toads' mating calls are also honest signals of their owner's size. The laws of physics impose a necessary relationship between the depth of a tone and the size of the body that produces it: a big body means a deep call. While this is generally true, a recent study showed that males of one species of green frog, *Rana clamitans*, may sometimes lower the dominant frequency of their calls during territorial contests in order to make the calls sound deeper and the frog seem larger than normal. In other words, not all signals are honest. Examples of dishonest signals include the behavior of mimics, which copy the behavior of another individual. For example, two species of fireflies emit similar patterns of flashes of light. Females of one species, the *Photinus* species, use their characteristic flashing pattern to attract mates. Females of the other species, the *Photuris* species, mimic the flashing pattern of *Photinus* females so that they can lure *Photinus* males and eat them.

102-2: The Choreography of Merce Cunningham

Choreographer Merce Cunningham was an experimentalist who, in his final years, was almost routinely hailed as the world's greatest living choreographer. Cunningham productions—many of them collaborations with composer John Cage—aroused much controversy. Three aspects of Cunningham's choreographic theories were especially provocative: his use of chance and indeterminacy (not precisely determining the completion of his works but leaving some choices up to the performers); his treatment of stage space; and his tendency to regard the components of a dance production as independent entities.

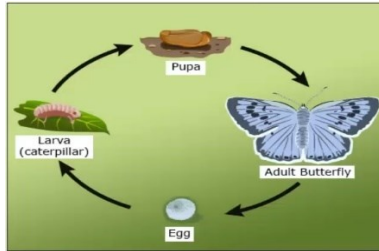
Wishing his dances to possess some of the unpredictability of life itself, Cunningham began to make choreographic use of chance. For Cunningham, chance did not mean chaos, and his dances were not improvisations or free-for-alls. Typically, he prepared in advance a multitude of movement possibilities—more than he may actually have needed in a work—and then decided which sequences (groups of movements) he would use by some simple device, such as flipping a coin. Or he would choreograph works in which the episodes might be performed in any order. Since Cunningham prepared so much, one might ask why he bothered with chance at all. He would reply that chance can reveal to a choreographer ways of combining movements that the rational, conscious mind might not have otherwise thought of on its own. All people are, to some extent, prisoners of their mental habits. But Cunningham believed that through chance, choreographers could free themselves from habit and discover attractive new movement sequences.

A theatrical preoccupation with indeterminacy is perhaps to be taken for granted in an age of mechanical formats, such as films, recordings, and television. One characteristic of the mechanical is fixity: once something is on film, it will stay that way forever—or at least until the film decays. But even when live performers are trying to speak the same words, sing the same music, or dance the same steps, every live performance differs, if only slightly, from every other live performance. Therefore, it can be argued that in his concern for indeterminacy, Cunningham was calling attention to the indeterminacy inherent in all live theater.

A second characteristic of Cunningham's choreographic approach was his treatment of stage space. Whereas many dances are structured around a central focus—a ballerina or danseur, or a hero or heroine, who may be framed by an ensemble—Cunningham gave equal importance to all parts of the stage. Corners and sides were as important as center stage, and many things happened simultaneously in different stage areas. Instead of being choreographically guided toward a single point, the spectator's eye was left free to roam as it pleased across a field of activity.

Finally, Cunningham regarded the components of a production—movement, music, decor—as coexisting independent entities. The dancers' steps were not phrased to coincide with the musical phrases; the scenery did not illustrate the choreography. Dance, music, and scenery simply occurred in the same space and time. The most radical manifestations of Cunningham's belief in the independence of theatrical elements occurred in what he termed "Events." These 90-minute productions, performed without intermission, consisted of choreographic sequences from dances already in the Cunningham repertoire. Yet the sequences were not presented to the audience as detached excerpts. Instead, they were woven together and performed to musical scores other than those for the works from which they derived. Thus, Events presented old movements in a new context, and the change of context often drastically altered the effect of the movements. Through his Events, Cunningham emphasized that everything is subject to change.

Each of Cunningham's dances managed to have its own special mood. Although there is nothing really tropical about *Rainforest*, the choreography is sensuous, and Andy Warhol's decor of floating silver pillows somehow seems appropriate for such a lush creation. *Quartet*, which, despite its title, is a dance for five—one outsider keeps trying in vain to join a group of four people—can be interpreted as an image of social exile or the gulf between generations. Cunningham's dances consist of many different elements, but like objects in a landscape, they may cohere to produce an unmistakable atmosphere.



103-1: The Decline of the Large Blue Butterfly in England

The large blue butterfly undergoes a four-phase life cycle: egg, larva, pupa, and adult. Adults lay eggs on plants that provide food for the larvae, known as caterpillars. When the caterpillar is fully grown, it enters the pupa stage, from which it emerges as a winged adult. The population of the large blue declined greatly in southern England in the late nineteenth century; this was attributed to poor weather. Final extinction of the species in England in 1979 followed two successive hot, dry breeding seasons. Since the beautiful butterfly is highly sought after by collectors, excessive collecting was presumed to have caused the long-term decline that made the species vulnerable to deteriorating climate.

In an attempt to protect the butterfly and its habitat, a reserve was established in the 1930s that excluded both collectors and domestic livestock, but this did not stop the decline. Evidently, habitat had changed through time, including a reduction in wild thyme, which provides the food for the early development period of the large blue's caterpillar. Moreover, the loss of grass-eating rabbits (through disease) and the exclusion of grass-eating cattle and sheep from the reserve harmed short-turf grasses, which require periodic grazing to flourish. Short woody plants (shrubs) replaced the short-turf grasses that had provided an optimal environment for the large blue. Thyme survived; however, the butterflies continued to decline to extinction in Britain.

A more complex story has since been revealed by research associated with the reintroduction of the large blue to England from continental Europe. The larvae of the large blue butterfly in England and on the European continent prey on colonies of *Myrmica* ant species. Larval large blues must enter a *Myrmica* nest to feed on larval ants. Similar predatory behavior and/or tricking ants into feeding them as if they were the ants' own offspring are features of the natural history of various butterfly groups. After hatching from an egg laid on the larval plant food, the large blue's caterpillar feeds on thyme flowers until it reaches the final larval phase, around August. At dusk, the caterpillar drops to the ground from its birth plant, where it waits motionless until a

Myrmica ant finds it. The worker ant observes the larva for an extended period, perhaps more than an hour, during which the ant eats a sugary substance secreted from the caterpillar's nectar organs. At some point the caterpillar becomes swollen and adopts a posture that seems to convince the ant that it is dealing with an escaped young ant, and the caterpillar is carried into the ant nest. Until this stage, the caterpillar's growth has been modest, but in the ant nest the caterpillar becomes predatory on ant eggs and larvae and grows rapidly. The caterpillar spends the winter in the nest and, nine to ten months after entering the nest, pupates (passes through the pupal stage) in early summer of the following year. The caterpillar requires an average of 230 immature ants for successful pupation. It apparently escapes predation by the ants by secreting surface chemicals that mimic those of the ant offspring, and probably receives special treatment in the colony by producing sounds that mimic those of the queen ant. The adult butterfly emerges in the summer and departs rapidly from the nest before the ants can identify it as an intruder.

Adoption and incorporation into the ant colony turns out to be the critical stage in the life history. The complex system involves the "correct" ant, *Myrmica sabuleti*, being present, and this in turn depends on the appropriate microclimate associated with short turf grass. Longer grass causes cooler near-soil microclimate conditions, favoring other *Myrmica* species, including *Myrmica scabrinodes*, which may displace *Myrmica sabuleti*. Although caterpillars associate apparently indiscriminately with any *Myrmica* species, caterpillar survivorship differs dramatically; with *M. sabuleti* approximately 15 percent survive, but an unsustainable reduction to less than 2 percent survivorship occurs with *M. scabrinodes*. For large blue populations to succeed, more than 50 percent of the adoption by ants must be by *M. sabuleti*. Other factors affecting survivorship are that the ant colony must provide enough larvae to feed the caterpillar well and that it must lie within 2 meters of the birth thyme plant. Such nests are associated with newly burnt grasslands, which are rapidly colonized by *M. sabuleti*.

103-2: The Nile River Valley

Though the verdant green course of the Nile Valley appears to be the very antithesis of the bleached Saharan sands through which it flows, these two contrasting features have together constituted a primary force in the development of agriculture and settled human society in Africa. The Sahara acted as a pump, drawing people from surrounding regions into its watered environments during the good times and driving them out again as conditions deteriorated (though not necessarily returning them to their point of origin). The Nile, for its part, was a refuge for people retreating from the desert and then a reservoir from which the desert region was repopulated as conditions improved.

The Nile is commonly believed to have been a conduit along which the principles of food-crop cultivation moved from Egypt into more southern regions of Africa along with the wheat, barley, peas, and lentils initially domesticated some 9,000 years ago in Southwest Asia. It is true that the ancient Egyptian civilizations founded on the Nile over 5,000 years ago were sustained by the exceptional productivity of those crops, but they were a comparatively recent introduction, arriving long after the cultivation of indigenous African plants had begun farther south.

Contrary to expectations, the earliest evidence of a shift toward a critical dependence on food production as opposed to food gathering comes not from the floodplain of the Nile but from sites in what is now the empty and waterless Sahara. The development was complex, involving the domestication of plants and livestock, technological innovation, the establishment of villages, and an increasing level of social interdependence.

The Sahara Desert had been widely inhabited until the last glacial maximum, when conditions of increasing aridity drove people (and practically all animals) out of its previously productive wooded grassland and savanna environments. Essentially nomadic—though exhibiting a tendency to settle at lakesides and other sources of food and water—the groups moving to the north and south continued to follow their established hunting and gathering way of life. The groups that moved east into the Nile Valley, however, adopted a distinctly sedentary lifestyle. Indeed, they had no choice. The narrow strips of green floodplain flanking the Nile were bounded by waterless desert, and movement along the riverside plains was restricted by the presence of competing groups.

The Nile along which people congregated 18,000 years ago was a quite different river from that which flows through Egypt today. It was much smaller and flowed more slowly through a tangle of braided channels across a wide elevated floodplain rather than along a single massive stream. The river carried a heavy sediment load, and the silts deposited along its length steadily raised the level of the river and its floodplain.

In effect, the Nile Valley at that time was an elongated oasis: a sharply defined area of inhabitable territory beyond the boundaries of which human survival was impossible. The oasis extended all the way from the Sudan to what is now Cairo—a distance of over 800 kilometers—but it was nowhere more than a few kilometers wide. Its food resources were varied and nutritious. Quantities and availability, however, were always subject to seasonal variation and the vagaries of the Nile's annual flood. During the critical two to four weeks when the flood was at its height each year, people and animals were driven out of the Nile Valley and crowded into the narrow band of inhabitable land that lay between the floodwaters and the arid desert beyond.

People hunted large mammals for meat and hides, but environmental and territorial constraints meant that only small numbers of a limited range of species were available. Indeed, large mammals constitute a very small proportion of the faunal remains found at archaeological sites, and only three species are represented: the hartebeest, the dorcas gazelle, and a type of wild cattle called the aurochs. Birds—particularly coots and migratory geese and ducks—were caught regularly. Fish, however, were by far the most important source of protein—primarily catfish. For carbohydrates, the Nile Valley offered its inhabitants a considerable variety and seasonal abundance. Twenty-five different seeds, fruits, and soft vegetable tissues have been distinguished among archaeological remains.

104-1: Cloth Manufacturing in England

By the fourth century, the European cloth industry, which had previously been centered mainly in Flanders (northern Belgium), Italy, and Holland, transferred to England, where fewer property taxes and market restrictions made greater profits for cloth makers. In England, the industry's production methods yielded improved cloth and changed the economy.

A technique called fulling was widely used in England and caused manufacturers to relocate from urban areas of England to different rural villages all over the country. The process of fulling involves submerging a cloth in water (usually containing a natural clay detergent called fuller's earth) and beating it vigorously. Proper fulling will shrink the cloth, making the fabric tighter and stronger, and make the surface smoother and softer. Three traditional methods of fulling were used: a submerged cloth was beaten with the feet, with the hands, or with clubs. One of these is depicted in an ancient wall painting in Pompeii of a man standing in a trough containing water and pounding a cloth with his feet. These traditional methods were still used in Flanders, Italy, and, for a time, England. But then, in the eleventh or twelfth century (historians disagree on the precise date), a new method was introduced: two wooden hammers were attached to a drum and a crank was turned to raise and drop the hammers on the cloth. The real breakthrough came when this device was hooked to a water mill (one probably constructed to grind grain). As a result, a single operator overseeing a series of hammers could perform the work that had previously required a crew of fullers—and he could do it much more quickly, too.

This is why the historian Eleanora Carus-Wilson observed that the invention of the fulling mill "was as decisive an event [for the woolen industry] as were the mechanization of spinning and weaving in the eighteenth century." Whether fulling mills developed in the eleventh or twelfth century, they were so common in the thirteenth century that they revolutionized the English industry, leaving the Continent (the rest of Europe) far behind. Fulling gave English cloth a significant advantage on the international market. With many fewer fulling mills, cloth makers on the Continent could full only some of their cloths, which entailed a large sacrifice in quality.

The ascendance of the fulling mill helps explain the English woolen industry's marked preference for villages and rural areas on good streams. Such locations had several additional advantages. Moving water was useful for dyers, who needed to rinse excess dye from their cloths. Moreover, locating in rural areas permitted firms to escape the repressive regulations imposed by guilds (professional organizations of practitioners of

the trades), to avoid the higher taxes of towns and cities, and to pay lower wages (the cost of living was lower in rural areas than in cities).

Why didn't the woolen industry on the Continent similarly disperse to small towns and villages? Because in Continental Europe only the cities provided enough freedom and property rights to sustain industry. On the European Continent, the rule of the nobility prevailed in the countryside, and everyone had to fear the local lord's greed. But in England, freedom and security prevailed throughout the country, and medieval English industrialists did not need to concentrate in crowded expensive, dirty cities—many devoid of water power—as their counterparts in Flanders, in Holland, along the Rhine, and in Italy were forced to do. As a result, the English woolen industry was remarkably decentralized.

Finally, dispersion and relatively unimpeded capitalism may have contributed to the international dominance of English woolens by encouraging the production of more fashionable and attractive products. As the historian AR Bridbury put it, to explain the success of the English woolen industry, it is not enough to cite better wools or lower prices; what should be stressed is "art and skill... the exotic dyeing of these cloths and... the subtle blending of design and color in their creation... the search for making cloth which would be more fashionable internationally." In textile centers on the Continent, the guilds—very traditional in their outlook—often limited experimentation with colors and designs, and originality nearly always suffers when creative people are crowded together and fully aware of one another's work. England's dispersed woolen industry produces greater variations in styles and quality.

104-2: Origins of the Deciduous Forests

Forests dominated by deciduous trees (trees with leaves that fall off at the end of the growing season) are found around the world. But they were not always so widespread, and there are unanswered questions about their evolution. The ancestors of deciduous trees can be traced back to the Cretaceous Period, 113-65 million years ago, when dinosaurs ruled the land and the continents occupied different positions from today. At this time, extensive deciduous forests existed only at very high latitudes, near the north and south poles.

Alaska was closer to the North Pole during the late Cretaceous than it is today, but the climate was milder, reflecting higher temperatures for the entire planet. This can be deduced from the shape of fossil leaves. There is a remarkably consistent relationship between leaf shape and mean annual temperature for modern plant species. For example, the proportion of species with toothed leaves consistently increases as one moves from warm to cold climates. Based on statistical analyses of a large number of such leaf characteristics, temperatures in the Cretaceous Arctic generally hovered above freezing during the winter. This is confirmed by the lack of geological evidence of permafrost, a layer of earth that is permanently frozen. However, fossilized tree trunks at Ellesmere Island in northern Canada display damaged sections (growth interruptions) consistent with damage caused by severe frost indicating that occasional frosts occurred. Also, although the winters were generally mild, they were—like modern Arctic winters—very dark. Because of the tilt of Earth's axis, very little sunlight reaches high northern latitudes for several months each year, so trees and shrubs of the ancient Arctic could not photosynthesize (produce chemical energy from sunlight) for several weeks or even months depending on the latitude. This is apparent from growth rings from fossilized wood of the period; the annual growth rings end abruptly, evidence of a rapid interruption of photosynthesis and plant growth with the onset of winter darkness.

The combination of mild temperatures and dim light during the winter may explain why so many Cretaceous trees shed their leaves at the end of the growing season. This would help reduce water loss and energy use during an extended period of darkness. Because the winters were relatively warm, leaves on an evergreen tree, which does not lose its leaves, would continue to undergo respiration (breathing), resulting in a steady loss of energy during a period when photosynthesis was not possible and energy stores could not be replenished. Thus, the food reserves of the plant would be expended to keep the

leaves alive. In contrast, deciduous trees dropped their leaves and entered a period of deep dormancy during which they lost relatively little energy. When the Sun reappeared in the spring, new leaves sprang out to take advantage of the short—but almost continuously sunny—growing season.

This explanation for the prevalence of deciduous trees in the ancient Arctic is so logical that it was not rigorously tested for many years. In the early 2000s, however, this hypothesis was undermined by experiments with both evergreen and deciduous species belonging to plant groups that grew in the Arctic or Antarctic forests of the Cretaceous. A team of researchers compared growth rates of young trees for two types of deciduous trees—ginkgo and dawn redwood—and two types of evergreen trees—southern beech and coast redwood—in greenhouses with a simulated (imitated) Cretaceous Arctic winter (dim light and mild temperatures). Surprisingly, the deciduous trees lost more of their food reserves (as measured by the amount of carbon in the trees) during the simulated winter than did the evergreen trees. This is due to the immediate and massive loss of carbon as leaves are shed, which offsets the carbon saved by winter dormancy. If these modern trees responded in a similar way to their ancient relatives, then it is perplexing that evergreen trees did not predominate in Arctic forests. Deciduous trees may have an advantage during the summer rather than during the winter, however, because their leaves could grow and photosynthesize more rapidly than evergreen leaves during the brief growing season.

105-1: Proto-industrialization in Europe

The wealthy merchants of the early modern era (approximately 1500-1800) took commodities (products) made by others and shipped them to distant markets. Or they supervised manufacturing enterprises, especially textiles, which worked on the domestic or "putting-out" system. In this system, the merchant acquired raw materials, distributed them to country workers who labored in their own cottages, and collected the finished goods for sale. Employing men, women, and children, this system flourished into the eighteenth century. It constituted a "proto-industrial" phase of European manufacturing, featuring rural production and city-directed exchange.

Technological innovation also contributed to the growth of the European economy. By 1500, Europe already had put machines to work in productive ways. Waterwheels and windmills helped grind grain, brew ale, pump water, make paper, saw wood, and treat textiles. The circulation of goods was facilitated by the building of canal networks, which in turn was made possible by the invention of locks (gates on rivers or canals that change the water level, allowing boats to pass obstacles). Over the next centuries, other new machines and industrial processes improved textile and glass manufacturing, coal mining, and iron production.

All this growth, however, had a disadvantage: rapid inflation. The injection of new quantities of bullion (ingots or gold coins) into the European economy, the deliberate devaluation of existing currencies, and the steady climb of the European population caused an explosion of values and prices. Prices rose for goods, while wages and rents lagged, with regional variations: in Spain, prices more than tripled in the century before 1600; in England, the prices of basic goods rose nearly sixfold during the same period. The pattern of steep price increases affected different groups differently—it allowed merchants to accumulate capital for investment in more manufacture and trade, while it had a very negative impact on peasants and urban workers.

The forward wave of commercial economy favored population increase, which in turn supported the economic boom. The European population generally increased until about 1600, when it surpassed for the first time the pre-bubonic plague level of 73 million. During the seventeenth century, population growth slowed again, owing in part to the disastrous effect of the Thirty Years' War, which reduced the population of central Europe

by 30-40 percent. The marriage age increased in many regions, showing that people were adopting a familiar strategy to reduce the birth rate. Spain and Italy fell behind the north Atlantic nations, and France's rate of growth fell behind England's. In the eighteenth century, the population began again to increase as mortality rates declined. After a last outbreak of the bubonic plague in Marseilles in 1720, the bubonic plague left Europe (perhaps because of unfavorable shifts in the ecology of the black rat, the critical carrier of the deadly infectious flea). The cycle of hunger receded too, and the proto-industrial cottage industries, which allowed young persons to make a profit from their labor without delaying marriage, drove the numbers upward.

Despite economic growth, the steady upward pressure on prices and the increasing population brought hardship, especially to the rural poor. Grain prices rose by factors of three, four, even six, while the average price of manufactured goods merely doubled. Bread was expensive or scarce, while landowners strove to extract more labor from their workers. The real wages of agricultural workers fell catastrophically, and the force of hunger, briefly subdued in the aftermath of the bubonic plague, returned. The numbers of the landless increased, and beggars proliferated.

With economic growth came urbanization. Merchants concentrated in cities, where goods changed hands, where banks held funds, where artisans labored, and where consumers purchased. In the early modern centuries, towns became cities, and cities expanded. For the first time, many European cities grew to exceed the largest cities previously known in the Western world (London, Paris, Naples, and Milan). Not only did large cities grow larger, but a greater proportion of Europeans lived in cities and towns; even country workers often migrated to the city for part of their lives before returning to their villages.

105-2: Mate Choice in Birds

Male birds are often strikingly different in plumage and size from their female counterparts. Charles Darwin concluded that exaggerated sexual differences such as the tail of a peacock or the displays of the wild turkey evolve as a result of what he called sexual selection—namely, contests among males for mates and female preferences for particular males. As potential male reproductive success increases, so does the value of the characteristics—large size, fancy plumage, intricate songs, and striking displays—that are responsible for the success. The resulting evolutionary process of sexual selection leads to differences between the sexes in size and ornamentation. Darwin's insights into the evolutionary role of sexual selection are now largely confirmed, but the effects of competition among males, female choice, and resources other than mates connect in even more intricate ways than Darwin proposed.

Mutual assessment of prospective partners is a vital aspect of the early stages of courtship and pair formation. The ornaments and displays favored and maintained by sexual selection are those that reliably reflect the superior condition of certain males, enabling females to select the best possible mates. For example, house finch females prefer brightly colored males, which have better survival rates and are better family providers. The familiar flight displays of male bobolinks over lush fields advertise their condition. Females favor males that display longer. Such males have larger fat reserves and consequently, raise more young than their neighbors do.

Why do females choose males with more elaborate plumage or displays? The good-genes hypotheses propose that exaggerated male plumage and courtship displays truthfully signal genetic or physiological superiority. Females should recognize superior males and select them to sire offspring. What aspects of genetic or physiological superiority might exaggerated courtship displays serve to indicate? One possibility would be a male's superior survival skill. For example, the enormous tail of a peacock might actually be a handicap during flight or escape. So would be bright colors that attract predators. Males that survive to display such handicaps would have superior stamina or abilities to escape predators. Evolution would tend to favor bigger and bolder badges of this so-called handicap superiority if females preferred to mate with the males that bore such badges.

Another application of the good-genes hypotheses asserts that ornamented plumage provides an index to a male's health, particularly its resistance to pathogens and parasites. Females could detect disease-prone males by the lower quality of their display plumage or by their reduced display stamina. Strong evidence now exists for this hypothesis. Among their many effects, parasites reduce the brightness of ultraviolet coloration of bird feathers. The glittering blue plumage of male satin bowerbirds, for example, has a single wavelength peak in the ultraviolet. The visual intensity of this peak predicts the male's level of infection by blood parasites, because ultraviolet shine decreases with increasing infection.

Other evidence for the use of a bird's appearance as an index to its health comes from studies of red junglefowl, which are the ancestors of domestic chickens. Marlene Zuk and her colleagues first established that hens of the red junglefowl mated more quickly with roosters bearing large, fleshy, red combs on their heads. The hens use comb size as an index to the health of a potential mate. Comb size is strongly affected by the level of the hormone testosterone, which, in turn, affects the bird's physical condition. Intestinal worms reduce comb size, with the result that hens prefer roosters without worms over those that are infected.

The large sizes and conspicuous plumage favored in reproductive displays may be disadvantages in other regards. Large size itself requires greater energy expenditure. There is some evidence that large male red-winged blackbirds are at a disadvantage because they must sacrifice display time for feeding. Among species of North American blackbirds, males that are much larger than females tend to suffer greater mortality as nestlings. Similarly, because they grow twice as fast as females to reach their full adult size by the end of their first summer, male western capercaillie, a huge species of European grouse, are more vulnerable than females to starvation when food is scarce. And the same bright colors that announce a male's presence to potential rivals or mates may also attract predators.

106-1: Theatrical Staging

Staging, the most obvious function of a play's director, essentially involves positioning the actors on the stage or set and moving them about in a theatrically effective manner. The basic architecture of staging is called "blocking," which refers to the timing and placement of a character's entrances, exits, rises, stage crosses, and other major movements of all sorts. The blocking pattern that results from the interaction of characters in motion provides the framework of an overall staging; it is also the physical foundation of an actor's performance—many actors have difficulty memorizing their lines until they know the blocking that will be associated with them.

The director may block a play either by preplanning the movements ("preblocking") on paper or by allowing the actors to improvise movement (or create it on their own without preparation) on a rehearsal set and then settling on a blocking pattern sometime before the first performance before an audience. Often a combination of these methods is employed, with a director favoring one method or the other depending on the specific demands of the play, the rehearsal schedule, rapport with the acting company, or the extent of the director's own preparation. Complex or stylized plays and settings and short rehearsal periods usually dictate a great deal of preblocking; simple domestic plays and experienced acting ensembles are often accorded more room for improvisation. Each method can produce highly commendable results in the right hands and at the right time. Both, however, can present serious problems if misapplied or ineptly handled.

For the most part, the blocking of a play is hidden in the play's action; it tends to be effective insofar as it is not noticed and insofar as it simply brings other values into play and focuses the audience's attention on significant aspects of the drama. By physically enhancing the dramatic action and lending variety to the play's visual presentation, a good blocking pattern can play a large role in creating theatrical life and excitement. But beyond this, there are moments when inspired blocking choices can create astonishing theatrical effects—effects that are not hidden at all but are so surprising and shocking that they compel intense consideration of specific dramatic moments and their implications. Such a dramatic effect was achieved, for example, by director Peter Brook in his celebrated 1962 production of *King Lear*, when Paul Scofield, as Lear, suddenly rose to his feet and, with one violent sweep of his arm, overturned the huge oak dining table at which he had been seated, sending pewter mugs crashing to the floor as he raged at his daughter Goneril's treachery. This stunning action led to a reevaluation of the

characters of both Lear and Goneril and of the relationship between this tempestuous and sporadically vulgar father and his socially ambitious daughter.

Some plays require specialized blocking for certain scenes—for duels, for example, or dances. Such scenes demand more than basic blocking and are frequently directed by specialists, such as dueling masters or choreographers, working with the director. These specialized situations are not at all rare in the theater—almost every Western play that was written before the nineteenth century includes a duel or a dance or both—and the ability to stage an effective fight scene or choreographic interlude (or at least to supervise the staging of one) is certainly a requisite for any director who aspires to work beyond the strictly realistic theater.

"Business" is a theater term that refers to the small-scale movements a character performs within the larger pattern of entrances and crosses and exits. Mixing a drink, answering a telephone, adjusting a tie, shaking hands, fiddling with a pencil, winking an eye, and drumming on a tabletop are all "bits of business" that can lend a character credibility, depth, and fascination. Much of the stage business in a performance is originated by the actor—usually spontaneously over the course of rehearsal—although it may be stimulated by a directorial suggestion or request. The director ultimately must select from among the rehearsal inventions and determine what business will become a part of the finished performance; when this determination is made, bits of business become part of the blocking plan.

106-2: Pollen Gathering by Animals

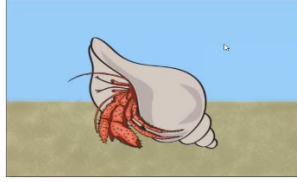
Pollen contains a seed plant's male reproductive cells, and in flowering plants it is produced by anthers, part of the male reproductive organs. Pollination in flowering plants is the process whereby pollen is transported from a flower's anthers to the female reproductive organ (stigma), enabling reproduction. This is sometimes accomplished by wind or water, but pollen is more commonly transported by animals that visit flowers, especially bees, butterflies, moths, and other insects, but also including some birds, reptiles, and mammals. It is generally assumed that most pollen is transferred passively (without deliberate effort) onto a visiting animal's body when physical contact is made. Pollen adheres to the surfaces of visitors because of its naturally sticky properties. Its adhesion is often enhanced by the characteristics of the animal's surfaces: branched hairs for bees, scaled surfaces for butterflies and moths, and fur or feathers in vertebrates.

There are some indications that what is termed passive gathering can be augmented by electrostatic attraction between pollen grains and animals' bodies. Plants are inherently slightly negatively charged in warm, still air, the magnitude of the charge field depending on the plant's composition and architecture and on weather conditions. The charge is greater at sharp tips and is therefore often higher in flowers than elsewhere. Hence it is possible that pollen grains acquire negative charges within the flower. In contrast, insects, and especially hairy bees, tend to acquire a positive surface charge as they fly. The charge difference might aid the transfer of pollen onto visiting insects. In short, even without direct contact, pollen grains could be attracted onto a close-up bee. Pollen might also be unloaded from bees onto dry stigmas in reverse fashion (the pollen having acquired a positive charge while on the flying insect), especially where stigmas stick out from flowers and thus concentrate charge. Small electrostatic charges could also lead a visiting insect to leave an electrical "footprint" on flowers, detectable to subsequent visitors for at least several minutes and thus potentially allowing avoidance of recently depleted flowers.

Rather few kinds of pollinators eat pollen and so gather it deliberately, but those that do are important and numerically dominant kinds. Syrphids (a type of fly that gathers in high numbers) may eat pollen directly from the anthers. A few bees also gather pollen directly with their mouthparts, eating some immediately and carrying the rest back to the nest in their crop (a part of the foregut that functions like an expandable sack). In such cases, only the pollen that is accidentally scattered during consumption (or that passively adheres to other parts of the insect) is of any use for pollinating. Most bees normally do not eat pollen at the flower but instead collect it—with their legs and mouthparts—by

rubbing the underside of their body across a flower, or by using various comblike or rake-like structures. In these cases, the pollen is then packed away on or into specific collecting areas of the insect's body, often with the addition of some moistening flower nectar, a sugary liquid. Hence there may be substantial interruptions to food gathering while a bee lands, grooms (removes pollen from its hair), and packages its pollen, although more advanced bees require only a brief hover between flower probings to achieve grooming and packaging.

Although active pollen gathering may be highly beneficial to a flower-visiting animal, there are problems from the plant's point of view. First, much of the pollen is hidden away in sites on the visitor where it will never contact a stigma. Therefore, plant evolution might act to place as much pollen as possible in sites on the visitor that are hard to groom. With suitable arrangement of floral parts, there is then a good chance that the stigma of another flower of the same species will receive some pollen from the insect. Second, many insect pollinators, especially ants, have special glands that produce small quantities of substances that protect their nests against fungal and bacterial growth but that can also kill pollen. Third, when saliva and nectar are added to the pollen gathered by bees, the pollen may suffer damaging biochemical change, since bees do not "want" the pollen to germinate (develop) once deposited in the nest. For such reasons, usually the longer pollen adheres to a visitor, the more its viability is compromised.



107-1: Hermit Crabs in Snail Shells

Hermit crabs depend on a properly fitting snail shell for protection from predators. Crabs in shells that are too small grow more slowly, are less tolerant of air exposure and drying out, and are more likely to be eaten by a predator than those that can withdraw completely into their shell. When a crab has outgrown its snail shell, it has to look for a bigger one that fits properly. They locate shells by smell, either when the preceding owner (the snail) dies and begins to decay or by detecting calcium, the major component of snail shells. The crabs do not generally kill snails to find new shells, although they may fight with other hermit crabs over empty shells, especially when empty shells are scarce. After locating a new shell, the crab investigates its surface and internal size by rolling it over and exploring it with its claws and walking legs. If it looks like a good fit, the crab withdraws its abdomen from its old shell and inserts it into the new one so quickly that it can be difficult to see. Switching shells is risky—the crab could be attacked by a predator or lose one or both shells to other hermit crabs.

Crabs rarely abandon their current shells without a new home lined up. However, if a crab has been buried in the sand during a storm or flooding event, it is likely to abandon its shell to return to the surface. Although abandoning its shell increases its chances of surviving such a burial, this behavior also increases the risk of predation and of being buried or injured in another flooding event.

Particular species of hermits tend to prefer certain species of snail shells, but size is the critical factor. Occupying larger and heavier shells gives them greater protection, but it takes more energy to lug a large house around. Shells often become covered with other organisms, such as barnacles or algae, that make them heavier and more difficult to carry. Some hermit species live in shells that get entirely covered by bryozoan colonies that eventually dissolve the snail shell, leaving the crab with a house composed entirely of the bryozoans. As the bryozoan colony grows, so too do the crabs, so these crabs have no need to change their homes.

Hermit crabs with poor-fitting shells are chemically attracted to dying gastropods (snails and related animals) and other hermit crabs, where a shell may become available. In the Mediterranean species *Clibanarius erythropus*, gastropod predation sites attract dozens of hermit crabs. Researcher Elena Tricarico and colleagues observed that these aggregations function as shell exchange markets: the first crab to arrive takes the empty shell, and a chain of shell exchanges among the crabs follows. They found that simulated snail predation sites quickly attracted a larger number of hermit crabs than other types of sites; therefore, aggregation is the most efficient tactic for this species to acquire new shells. In Belize, the land hermit *Coenobita clypeatus* has an even more organized shell exchange in what is called a "synchronous vacancy chain" by Randi Rotjan and colleagues, who discovered this unique behavior. When a large, empty shell becomes available, many crabs gather around it, which can take hours. As they gather, the crabs arrange themselves into a line of decreasing size, starting with the largest crab holding onto the empty shell. As though choreographed, the crabs begin shell swapping, one after the other, a smaller crab climbing into a new shell right after it is vacated by the slightly larger crab ahead of it. What makes the synchronous chain possible is that smaller crabs linger near a too-large shell, perhaps attracting others, waiting until a bigger crab comes along, which increases their chances of getting a good-fitting hand-me-down.

Most hermit crabs and most snails are right-handed and spiral clockwise. Some species, however, are left-handed and spiral counterclockwise, and they need to find snail shells that are coiled that way. Since the majority of snail species are right-handed, left-handed hermits may have trouble finding left-handed snails. Adults of the left-handed *Petrochirus diogenes*, the largest hermit crab in the Caribbean, are often found in shells of the queen conch *Eustrombus gigas*. As an exception to the general rule, they will attack and eat the conch, thus obtaining both a meal and a shell.



107-2: Impressionists and Their Paints

By the mid-nineteenth century, the provision of artists' materials in Europe had become a regular trade with its own well-defined distribution channels. Previously, pigments (substances that give paints their color) tended to be categorized either with imported spices or with drugs (since some artists' materials also had pharmaceutical uses), and so they were sold by grocers who dealt also in foods and medications. But as paint manufacturing became increasingly a matter of chemical synthesis (production) rather than the grinding of naturally-occurring pigment, it was transformed into an industry, and most retailers did little more than package the ready-made paints from manufacturers.

Another major innovation in paint supplies was the collapsible metal tube, invented in 1841 by an American portrait painter named John Rand. The tin tubes replaced containers made from the bladders (hollow organs) of pigs, in which oil paints had previously been stored, and made paints far less liable to dry out in their packaging. This was particularly significant for the Impressionist painters—artists who represented the effect of light on objects with short brush strokes of color and preferred working outdoors rather than in the studio; Auguste Renoir, a painter and one of the founding members of the Impressionist school, remarked that "without paints in tubes, there would have been no Cezanne, no Monet, no Sisley or Pissarro, nothing of what the journalists were later to call Impressionism."

This commercialization of artists' supplies contributed to the distancing of the painter from his or her materials—the beginnings of that problematic attitude toward the primary substance of paint that was ultimately to impel some artists of the twentieth century to experiment with the use of house paints. J.F.L. Merimee, a French chemist and painter, complained in 1830 that painters were no longer able to distinguish good materials from bad. And there certainly were bad materials around. To the paint merchant, the bottom line was profit, and sellers often cared little for the quality or long-term stability of their colors. These people, said Merimee, "had a stronger feeling toward

their own immediate profit than any regard to the preservation of pictures." They used binding materials that gave the paints a long shelf life, but at the cost of creating poor drying properties on the canvas. Because the pigment was the most expensive component of the paint, the color-makers would be inclined to minimize the amounts used. But as all oils turn slightly yellow as they dry, paints with a high oil-to-pigment ratio discolor more markedly. And to keep the paint stiff while lowering the pigment content, some manufacturers added wax, which resulted in a sticky material that was liable to crack. Some retailers diluted their pigments with extenders—inert materials such as chalk or gypsum that merely made the colored material go further. There were even cases of deliberate falsification, of one pigment being passed off as another more expensive variety or surreptitiously mixed with it. The ambiguous relation of some paint names to the pigments they contained only heightened the temptation for duplicity.

As a protection against such practices, many artists strove to establish a good relationship with a particular color merchant, relying on him to control paint quality. Impressionist painters Pissarro and Cezanne, and later Vincent van Gogh, used Julien Tanguy, who ran a small shop on the rue Clauzel in Montmartre from 1874. The Danish painter Johan Rohde described it as a "paltry little shop, poorer than the most miserable second-hand dealers in Adelgade [in Copenhagen]. There were stacks of pictures—no doubt payment for materials—and very valuable things among them," which drew many visitors to the shop. The wild van Gogh was befriended by Tanguy when the artist came to Paris in 1886, and he twice painted the merchant's portrait, despite complaining on several occasions about his wares. Tanguy ground his own colors, and one of the advantages of such a supplier was that he could provide materials to order: van Gogh, for instance, made specific requests for coarsely ground pigments. Among the Impressionists, only Edgar Degas showed much curiosity about what was actually in his colors. His notebooks contain many chemical recipes and technical notes that show a very limited knowledge of chemistry.